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ABSTRACT

A quantum computer must be capable of manipulating quantum information while at the same time protecting it from error and loss of quantum coherence due to interactions with the environment. Topological quantum computation (TQC) offers a particularly elegant way to achieve this. In TQC, quantum information is stored in exotic states of matter which are intrinsically protected from decoherence, and quantum computation is carried out by dragging particle-like excitations (quasiparticles) around one another in two space dimensions. The resulting quasiparticle trajectories define world-lines in three-dimensional space-time, and the corresponding computation depends only on the topology of the braids formed by the world-lines.

Quasiparticles that can be used for TQC are expected to exist in a variety of fractional quantum Hall states, among them the so-called Fibonacci anyons. These quasiparticles are conjectured to exist in the $\nu = 12/5$ fractional quantum Hall state which has been observed in experiments. It has been shown that qubits can be encoded using three or four Fibonacci anyons and single-qubit gates can be carried out by braiding quasiparticles within each qubit. Braids that approximate single-qubit gates can be found through brute force searching and the result can be systematically improved, to any desired accuracy, by applying the Solovay-Kitaev algorithm in $SU(2)$.

Two-qubit gates are significantly harder to implement, mostly due to the following two reasons. First, the Hilbert space of the quasiparticles forming two qubits is considerably larger than the Hilbert space of the quasiparticles of a single qubit. Therefore, performing a brute force search to find braids that approximate two-qubit gates, as well as the implementation of the Solovay-Kitaev algorithm for subsequent improvements are prohibitively more difficult. Second, to construct any entangling two-qubit gate, one needs to braid some

of the quasiparticles from one qubit around quasiparticles of the other qubit. This process will inevitably lead to leakage errors, i.e. transitions from the qubit space to other available states in the Hilbert space.

In this thesis, I will present several efficient methods to construct two-qubit gates using a specific class of quasiparticles. In particular, I show that the problem of finding braids that correspond to two-qubit gates can be reduced to a series of smaller problems which involve braiding only three objects at a time. The required computational power for finding these braids is equivalent to that needed to find single-qubit gates, therefore, these braids can be found with the same high degree of accuracy and efficiency. The end result of this work is an efficient procedure for translating (or “compiling”) arbitrary quantum algorithms into specific braiding patterns for Fibonacci anyons, as well as quasiparticles of certain other fractional quantum Hall states that can be used for TQC.

CHAPTER 1

Introduction

During the past two decades, the theory of *quantum information processing* has evolved into a major branch of research both in condensed matter physics and in computer science. Progress in this field started when it was realized that a generalization of the classical theories of information processing to include quantum mechanical resources can lead to profoundly new paradigms in our ability to process information. This new possibility is manifested in the idea of the *quantum computer*.

Quantum computers are hypothetical devices that are capable of performing tasks beyond the reach of any conceivable classical computer. Research in quantum information processing is important for at least two reasons. From a practical point of view, with our ever growing demand for computers that are faster and more powerful, the potential applications of quantum computers are highly desired. Nevertheless, the study of quantum information processing is also important from a conceptual point of view as it may provide answers to fundamental questions such as: What are the ultimate limitations that Nature (as we understand it) can impose on our ability to perform efficient computations?

Among the many proposals for realizing a quantum computer, *Topological Quantum Computing* is particularly appealing. This method of quantum computing obtains its uniqueness from the fact that it takes advantage of states of matter with exceptional properties — as if Nature has tailored unique systems for us to use as reliable quantum computers. These systems are the *topological* states of matter.^a Topological quantum computing is a method of quantum computing which exploits the exceptional features of topological phases of matter to give rise to computers that are essentially error-free. This Thesis is concerned with this method of quantum computing.

^aSee Sec. 1.2.

Topological quantum computing is in the intersection of the theory of quantum computing and the physics of topological phases of matter. Each of these two fields can, in fact, be considered as interdisciplinary in their own rights — the theory of quantum computation is a field with roots in computer science and quantum physics, and the theory of topological phases of matter borrows many of its ideas from topology, quantum field theory, physics of strongly correlated many-body systems and the theory of quantum phase transitions, among others. Given the diversity of topics and the vast literature available in each of these fields, providing a comprehensive, yet compact introduction to the field is a difficult task. Nonetheless, in this Chapter I will try to provide a brief survey of these topics, with emphasis on those aspects that are directly related to the problems addressed in Chapters [2](#) and [3](#).

1.1 Quantum Computing Basics

For the most part, today’s classical computers are constructed based on classical physics: the physics of Newton and Maxwell. Even though some quantum mechanical effects are taken into account in the engineering of certain devices used in the hardware of computers, the computer as a whole is still a classical machine. With the progress in technology, as device sizes get smaller and reach subatomic scales, quantum mechanical effects become more and more pronounced. Therefore, for future generations of computers dealing with quantum mechanics seems to be inevitable.

Interestingly, this seeming difficulty, might in fact, open new doors to our ability to perform computations. Theoretical work in the past two decades shows that quantum computers, i.e. computers that are built based on principles of quantum mechanics, can achieve performances beyond the reach of any conceivable classical computer. In this Section, I will review some of the basic properties of quantum computers, with the goal of providing the reader with the minimal background necessary to understand the main results of this Thesis.

1.1.1 History and Introduction

In 1982, in an article Richard Feynman pointed out that there seems to be a fundamental problem with the ability of classical computers to efficiently simulate quantum systems [\[1\]](#). Yet, this does not seem to be a problem in Nature; after all, real quantum systems are

constantly simulating themselves! Feynman then suggested that it might in fact be possible to harness the power of a quantum system in a “quantum mechanical computer” to efficiently simulate other quantum systems. Later, David Deutsch, in his quest for a “universal model of computation”, showed that a quantum computer is capable of efficiently simulating any physical process, while a universal classical computer cannot efficiently simulate a quantum computer [2]. Therefore, quantum computers, if they can be built, are fundamentally more powerful than classical computers.

But exactly how the power of quantum systems can be utilized in a quantum computer is a subtle issue. Perhaps the reason many people started believing in the possibility of using a quantum system to perform computation was the discovery of *quantum algorithms*. Quantum algorithms are algorithms that can run efficiently on a quantum computer and often have no efficient classical counterparts. Here, “efficient” means the time it takes for the algorithm to execute grows no faster than polynomially with the size of the problem it is intended to solve. Perhaps the most famous manifestation of the superior abilities of quantum computers is Shor’s factoring algorithm, discovered in 1995 [3]. This algorithm provides a procedure for factoring large numbers, N , in $O((\log N)^3)$ steps. This efficiency should be contrasted with the best classical algorithms which can carry out the task in $O(N^{1/3})$ steps. Therefore, Shor’s algorithm offers a remarkable exponential speedup over classical algorithms. Over the years, a few other quantum algorithms have been proposed. An example is Grover’s search algorithm which offers a quadratic speedup over the best classical search algorithms [4]. Compared to Shor’s algorithm, this polynomial speedup is modest, however, considering the vast applications of search algorithms, even a quadratic speedup can prove to be very effective.

As for other applications, in the field of information theory, it has been shown that using quantum resources can improve classical methods of communication. For example, the possibility of *quantum cryptography* [5] greatly enhances the security of our cryptographic methods, or *superdense coding* [6] can improve the capacity of channels used in transmitting data. For physicists however, the most exciting application of quantum computers remains to be Feynman’s original idea, i.e. the possibility of efficiently simulating quantum systems. It should also be mentioned that aside from possible applications, theoretical study of quantum computation and information is important from a conceptual point of view. One hopes that this field would, one day, provide answers to fundamental questions that address the

ultimate limitations that the physical world can impose on our ability to perform efficient computations (see for example [7]).

Despite the fact that there are no known laws of physics that prohibit us from building a quantum computer, it is still a very difficult task. The main difficulty, as will be described in more detail later, is that compared to classical computers, quantum computers are more susceptible to errors and these errors can potentially destroy any computation. If the problem of errors is not treated properly the implementation of a quantum computer in a realistic setting is practically impossible.

The breakthrough came in the mid 1990's, with the discovery of *quantum error correcting codes* [8, 9, 10] and the possibility of *fault-tolerant* quantum computing [11]. The main result of the theory of fault-tolerant quantum computing is the *threshold theorem*. This theorem states that if the error rate (i.e. the probability of an error occurring) per *quantum gate*^b is below a certain threshold, then it is always possible to reduce the effects of errors by using certain procedures, known as quantum error correcting codes, to carry out arbitrarily long quantum computation with any desired accuracy.

Since in general it is not possible to completely isolate a quantum system from the environment, in all models of quantum computation the occurrence of errors is, indeed, inevitable. Therefore, as was mentioned above, the discovery of quantum error correcting codes and fault-tolerant procedures was an essential step towards realizing a quantum computer. In 1997, Alexie Kitaev proposed a different approach for dealing with the problem of errors in quantum computers [12]. This approach which is known as topological quantum computing, is based on the idea that quantum information can be stored in certain states of matter that are intrinsically protected from the environment and therefore are essentially decoherence-free.^c The subject of this Thesis is related to this particular approach which will be described in more detail in Sec. 1.2.3.

1.1.2 Models of Quantum Computation

In a classical computer, information is stored in binary bits: 0's and 1's. The information then is processed by applying circuits consisting of Boolean gates: AND, OR, NOT, ... and

^bSee the next Section.

^cStrictly speaking, even in these models random errors which lead to decoherence of quantum states do occur, but under certain conditions, as will be described in the following Sections, the error rate in these systems is exponentially small.

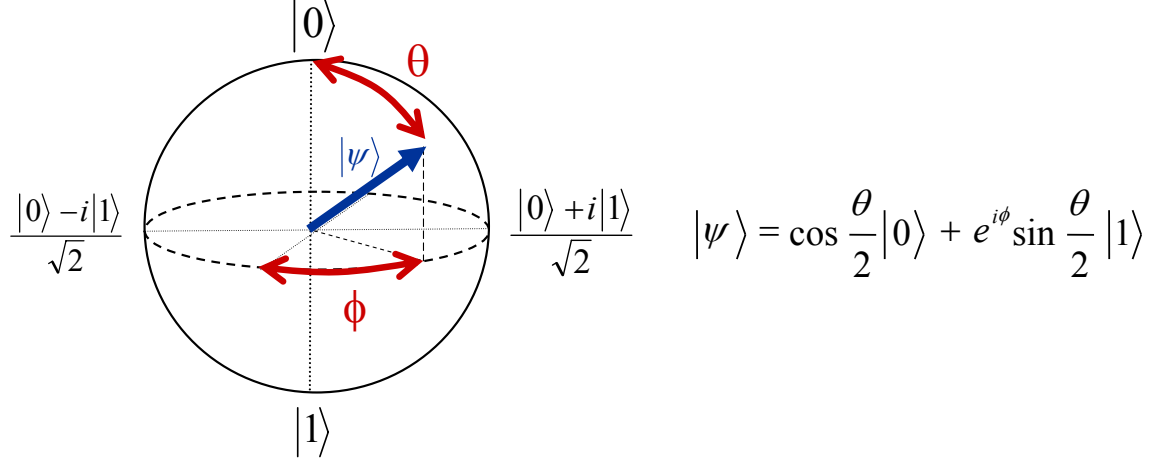


Figure 1.1: Representation of a qubit in the Bloch sphere. $|0\rangle$ and $|1\rangle$ (with unit length) which correspond to classical states of a qubit, provide a basis for an arbitrary state of a qubit, $|\psi\rangle$ to be expressed in terms of the two angles, θ and ϕ .

their combinations. Similar ideas can be applied to quantum computers in what is known as the *quantum circuit model* [13]. In this model, the building blocks of a quantum computer are *quantum bits* (qubits, for short) which are two-level quantum systems. For example, a particle with spin 1/2 can serve as a qubit in which, spin up can correspond to the state $|0\rangle$ and spin down can represent $|1\rangle$. The main difference between qubits and classical bits is that qubits can also exist in a state which is a quantum superposition of $|0\rangle$ and $|1\rangle$. In general, $|0\rangle$ and $|1\rangle$ are orthogonal states that span the Hilbert space of a qubit and the state of a qubit, $|\Psi\rangle$ can be written as

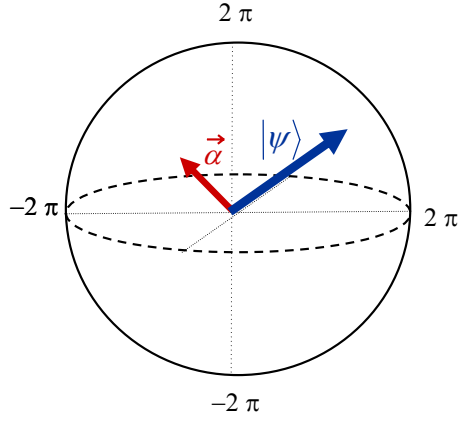
$$|\Psi\rangle = a|0\rangle + b|1\rangle. \quad (1.1)$$

Here, a and b are complex numbers and $|a|^2 + |b|^2 = 1$. As is shown in Fig. 1.1, states $|\Psi\rangle$, can be represented by vectors in the *Bloch sphere*.

Since each qubit lives in a 2-dimensional Hilbert space, when N qubits are present, the corresponding Hilbert space is 2^N -dimensional. A quantum state in this Hilbert space then can exist in a quantum superposition state of the form,

$$|\Psi\rangle = a_1|00 \dots 0\rangle + a_2|00 \dots 1\rangle + \dots + a_{2^N}|11 \dots 1\rangle, \quad (1.2)$$

where $\sum_i |a_i|^2 = 1$. A consequence of this possibility is that qubits can also exist in an



$$|\psi\rangle \xrightarrow{U_{\vec{\alpha}}} U_{\vec{\alpha}} |\psi\rangle$$

$$U_{\vec{\alpha}} = e^{i\vec{\alpha} \cdot \vec{\sigma} / 2}$$

Figure 1.2: Single-qubit gates as rotations. Single-qubit gates, $U_{\vec{\alpha}}$, can be represented by a vector, $\vec{\alpha}$, in a solid sphere of radius 2π . the direction of $\vec{\alpha}$ represents the axis of rotation, and its magnitude determines the rotation angle.

entangled state — a state which cannot be decomposed into individual qubit states. A famous example is the singlet state,

$$|\Psi\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \quad (1.3)$$

Entangled states are important tools in quantum computation and quantum information and for a large part, are responsible for the power of quantum computers.

Quantum analogs of classical Boolean gates are *quantum gates*. These gates are essentially rotations (up to phases) in the space of qubits (see Fig. 1.2) and can be represented by unitary operators, forming a continuum. For example, a phase gate of the form,

$$P(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \quad (1.4)$$

rotates a single qubit about the $\hat{z}$ axis by an angle of θ . Typically, in any real implementation of a quantum computer one does not have access to a continuum of single-qubit rotations and usually only a few such gates can be constructed exactly. Nonetheless, as long as the available gates can generate the unitary group $SU(2)$ it is always possible to find excellent approximations to any desired single-qubit gate. For example, a Hadamard gate,

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (1.5)$$

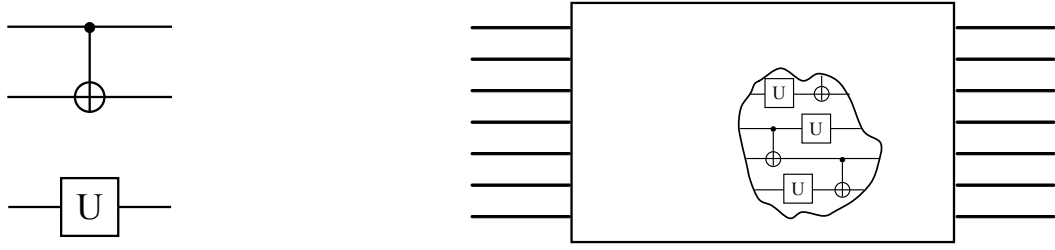


Figure 1.3: A universal set of quantum gates. Any quantum computation can be carried out by applying gates from a universal set, consisting of single-qubit gates, U , and a controlled-NOT gate.

and a phase gate of the form,

$$P(\pi/4) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}, \quad (1.6)$$

together with their inverses can be used to generate arbitrary single-qubit gates.

It has been shown that any quantum computation (i.e. any unitary operation in $SU(N)$ where N is the dimensionality of the Hilbert space) can be carried out by using what is known as a *universal set of quantum gates*. This set of gates consists of single-qubit gates and at least one entangling two-qubit gate [13, 14]. The most famous two-qubit gate is the controlled-NOT (CNOT) gate. This gate acts on two qubits, a control and a target, in such way that if the control qubit is in the state $|1\rangle$, CNOT flips the target, but if the control qubit is in the state $|0\rangle$, CNOT does nothing. If the states of control and target qubits are denoted by $|ct\rangle$, in a basis that is labeled by $|ct\rangle = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the unitary operation corresponding to a CNOT gate has the form,

$$U_{CNOT} = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array} \right). \quad (1.7)$$

As was mentioned above, the main difficulty in building a quantum computer is the existence of errors which, essentially, come in two varieties. One type of error is due to

imperfect implementation of the gates. As was described above, most of the gates used in any quantum circuit cannot be carried out exactly and involve approximations, therefore, each approximated gate introduces a fixed amount of error to the computation. Another kind of error is known as *decoherence*. Decoherence occurs because, in general, quantum states of the form $\psi = \alpha|0\rangle + \beta|1\rangle$ are very fragile and easily couple with the environment. The interactions with the environment will effectively *measure* such quantum states, resulting in the loss of their quantum coherence.

In principle, the effects of the first type of errors can be reduced by constructing more accurate gates (by using longer sequences of the available exact gates), and the effects of the second type can be reduced by better isolating the quantum system from the environment. However, when considering large scale computations carried out over many qubits, using many more quantum gates, these solutions are not practical. As was pointed out earlier, the breakthrough was the discovery of quantum error correcting codes and fault-tolerant quantum computation. Quantum error correcting codes are procedures for encoding quantum information associated with one qubit, into the entangled state of several qubits. The idea is to spread the valuable quantum information over many qubits, thus, storing the quantum information *globally* so that if a few of the qubits are affected by *local* errors, the original information can still be recovered.

To manipulate quantum information coherently, one must apply fault-tolerant quantum gates. These gates are designed in such way that when applied on encoded data, they perform the same operation even if the implementation of the gates is not perfect. As was mentioned above, the theory of fault-tolerant quantum computing then states that if the error rate of each quantum gate is below a certain threshold, it is always possible to use quantum error correcting codes to carry out arbitrarily long quantum computation with arbitrary reliability.

We summarize by mentioning a set of criteria (known as DiVincenzo criteria) for building a quantum computer. For implementing a quantum computer, we must, (1) have access to a scalable system of qubits i.e., two-level quantum systems, and (2) have the means to initialize the qubits to pure states (e.g. $|0\rangle$). Then, (3) we must be able to carry out single-qubit gates, and entangle qubits by carrying out two-qubit gates. These gates must be carried out with sufficient accuracy so that fault-tolerant quantum computation can be applied. At the end, (4) we must be able to individually address the qubits and measure the corresponding states, and of course, (5) all this must be carried out in such way that the decoherence of

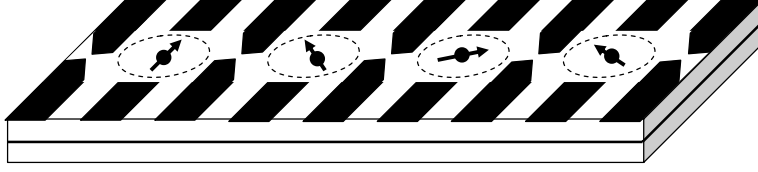


Figure 1.4: The Loss-DiVincenzo model of a quantum computer. It consists of an array of quantum dots (dashed ovals) each with one electron trapped in it. Spin of each trapped electron plays the role of a qubit.

quantum states does not interfere with the outcome of the computation. This means the time it takes to initialize the qubits, perform a computation and read out the result, must be shorter than the decoherence time of the quantum states.

We also mention in passing that the quantum circuit model, described above, is not the only model for quantum computation. Other models include *adiabatic quantum computation* [15] and *measurement-based quantum computation* (see for example [16] and the references therein). Despite their differences, all these models are based on the idea of preparing a quantum system in some initial state, then evolving it into a desired final state in a controlled way, and finally performing the readout. Note that these models can all efficiently simulate each other, therefore, they are essentially equivalent. In this Thesis we will not refer to these other models and will focus on the quantum circuit model.

1.1.3 Models for Physical Realization

A qubit can be realized by a two-level quantum system and a natural proposal for implementing a qubit is to use the spin of electron. In 1998, Loss and DiVincenzo [17] proposed a model for physical implementation of a quantum computer in which, qubits are spins of single electrons trapped in quantum dots (see Fig. 1.4). In this model, quantum gates can be carried out by controlling external magnetic and electric fields and by turning on and off the Heisenberg exchange interaction ($\mathbf{S}_i \cdot \mathbf{S}_j$) between the spins of trapped electrons. Note that when considering single electrons as qubits, the Heisenberg interaction on its own is not enough to carry out arbitrary quantum computation, however, as was shown in [18], if *logical* qubits are encoded using several electrons, then it is possible to carry out arbitrary quantum gates just by using the exchange interaction.

There are many other proposals for implementing qubits, for example, nuclear spin [19], photon polarization [20], etc. The common feature of all these proposals is that they store information in a *local* degree of freedom, be it in real space or momentum space. The dependence of quantum states to local degrees of freedom is the reason behind their vulnerability to decoherence — the environment will eventually couple to these local degrees of freedom and cause the quantum states to collapse into classical states, hence, losing their quantum coherence.

As was mentioned in Sec. 1.1.2, one solution to the problem of errors is to use quantum error correcting schemes. In this approach, one can protect the quantum information from the environment by spreading it over the global state of many entangled qubits. Alternatively, Kitaev’s idea for topological fault-tolerance was to store quantum information in global (topological) degrees of freedom of certain two-dimensional quantum systems. These global degrees of freedom are, by definition, immune to local disturbances and decoherence. Thus, if quantum error correction is the treatment of errors at the *software* level, Kitaev’s alternative approach is to treat the problem of errors at the *hardware* level. This method of quantum computing is called *topological quantum computing*. To understand this model some background on topological phases of matter is necessary which will be provided in the next Section.

1.2 Non-Abelian Phases of Matter

Topological quantum computing (TQC) is a method of quantum computing which is intrinsically fault-tolerant and, as was described above, this fault-tolerance is obtained at the hardware level. The same way that quantum error-correcting codes store quantum information in a highly entangled state of many qubits, in TQC, quantum information is stored in a highly entangled state of matter — the so-called *topologically ordered* phases of matter.

Theoretical work indicates that topological phases can occur in two space dimensions. These states are often characterized by degenerate ground states that are separated from the excited states by large energy gaps. The ground states of these systems have the property that the corresponding degeneracy depends only on the topology of the two-dimensional system. Furthermore, localized particle-like excitations in these systems possess exotic

properties. For instance, these excitations can carry fractional charge and obey fractional statistics. Quasiparticles that obey fractional statistics are known as *anyons* [21]. In this Section, I will review the basic features of topologically ordered phases of matter and their anyon excitations, and describe how TQC can be carried out.

1.2.1 Topologically Ordered States

In condensed matter physics, various phases of matter are often characterized by their symmetries and long range correlations: different symmetries correspond to different phases and a particular symmetry is the characteristic feature of a certain phase of matter. For example, liquids have translational symmetry, while in crystals (solids) this symmetry is (partially) broken. In most cases, broken symmetries give rise to the concept of *order* and different kinds of order can be classified using symmetry groups. For example, all possible orders in crystals arise from the different ways the translational symmetry can be broken and they can be classified using group theory.

Still, there are some phases of matter that do not fit in this classification. For example, in fractional quantum Hall systems *different* phases of matter can exist which all have the *same* symmetry. Thus, these systems cannot be classified using order parameters arising from broken symmetries. Nevertheless, these systems possess a different kind of order which, to some extent, can be characterized by the degeneracy of their ground states and the properties of their quasiparticle excitations.

In 1990, Wen and Niu showed that the ground state degeneracy of certain two-dimensional systems depends *only* on the topology of the surface the system is defined on [22]. For example, in the Fractional quantum Hall effect, if the underlying two-dimensional electron gas is wrapped around a surface with genus g ,^d then the ground state of the system at filling fraction $\nu = 1/3$, is 3^g -fold degenerate (see Secs. 1.3.2, 1.3.3 for more details on fractional quantum Hall effect). Furthermore, this degeneracy is robust — local perturbations, as long as they do not change the topology of the entire system, will not affect the ground state degeneracy. Thus, ground state degeneracy is a universal property that characterizes a new phase — a *topologically ordered* phase.^e

^dLoosely speaking, genus is the number of handles on a connected orientable surface. So for example, genus of a sphere is equal to 0 while a torus has genus 1.

^eTo add to the excitement, note that the topological degeneracy of the ground state is closely related to the fact that the quasiparticle excitations in these systems carry fractional charge and obey fractional

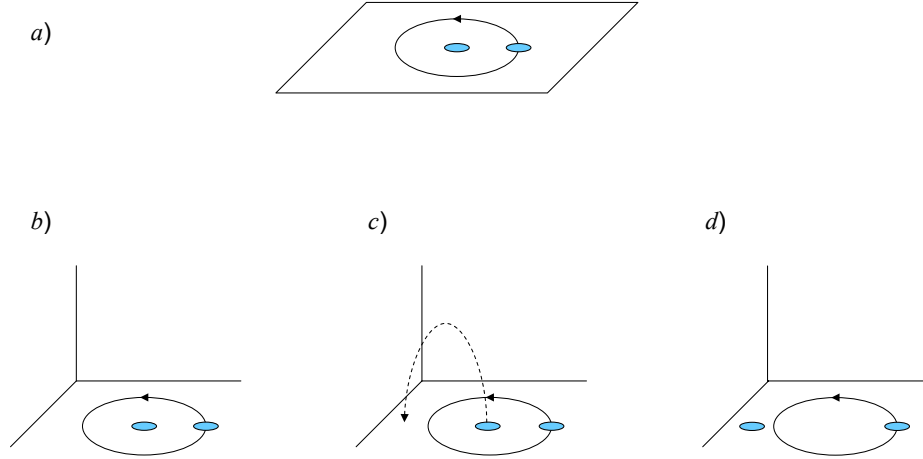


Figure 1.5: Statistics of identical particles in two versus three dimensional space. (a) Taking one particle around another, in two dimensions, cannot be smoothly deformed into the identity, therefore the corresponding unitary operation is non-trivial. (b)-(d) Taking one particle around another, in three dimensions can be smoothly deformed into the identity, shown in (d) without one particle having to cut through the trajectory of the other particle. Therefore the corresponding unitary operation is trivial.

In 1991, Moore and Read showed that certain fractional quantum Hall states can have particularly exotic properties [23]. In these states, when localized particle-like excitations (quasiparticles) are present and their positions are fixed, as long as they are sufficiently far apart,^f the corresponding Hilbert space is degenerate and its degeneracy grows exponentially with the number of quasiparticles present. Furthermore, this degeneracy is topological, i.e., different states cannot be distinguished by local measurements and only global measurements which measure the topology of the system can distinguish between these states. Thus, these systems provide an exponentially large Hilbert space, with states that are protected from the environment — a perfect place to hide quantum information!

In 1997 Kitaev proposed that ground states of topologically ordered phases can be used for constructing a robust memory for storing quantum information [12]. In this model, quantum information is stored in terms of certain quantum numbers carried by quasiparticle excitations. These quantum numbers give rise to global degrees of freedom that are spread

statistics, as will be described in great detail in the following Sections.

^fIn principle, for the ground state to be degenerate, the quasiparticles must be infinitely far apart.

over the entire system and cannot be measured locally. Quantum computation is then carried out by moving quasiparticle excitations around each other in two space dimensions. To understand how this can be carried out I should describe another characteristic of topologically-ordered phases (which is directly related to the degeneracy of the ground state) i.e. the fact that phases of matter that are topologically ordered can have quasiparticle excitations that carry fractional charge and obey fractional statistics. In the next Section I will review the concept of statistics for identical particles and describe the general circumstances under which fractional statistics can arise.

1.2.2 Anyons

In our usual three-dimensional quantum world, there are two different classes of identical particles: bosons and fermions. These two classes of identical particles arise from two kinds of statistics obeyed by these particles — bosons obey Bose statistics and fermions obey Fermi statistics. Why there are only two possibilities? Consider two identical particles in the three dimensional space. What happens when we (adiabatically) exchange these particles twice? If we exchange two identical particles in two dimensions twice, the result is topologically equivalent to the process in which, we keep the position of one particle fixed while moving the other particle around it. In three dimensions this procedure can always be smoothly deformed into the identity i.e. two exchanges are equivalent to the identity (see Fig. 1.5 (b)-(d)). Therefore, the final state of the system (after two exchanges) is the same as the initial state. As a consequence, if we exchange two identical particles only once, the state of the system obtains an overall phase of $+1$ or -1 corresponding to bosons or fermions, respectively. This means, in three dimensions, when many identical particles are present, the state of the system depends *only* on the positions of particles and thus, the statistics of identical particles in three space dimensions can be described by a one-dimensional representation of the *permutation group*, i.e. ± 1 .

Surprisingly, if we restrict our physical space to two dimensions, it is possible to have richer varieties of identical particles with more interesting statistics. The reason is that, unlike three dimension, in two dimensions moving one particle around another particle is a nontrivial process: it cannot be smoothly deformed to the identity without one particle cutting through the trajectory of the other particle (see Fig. 1.5 (a)). Therefore, when considering two space dimensions, exchanging two identical particles is no longer equivalent

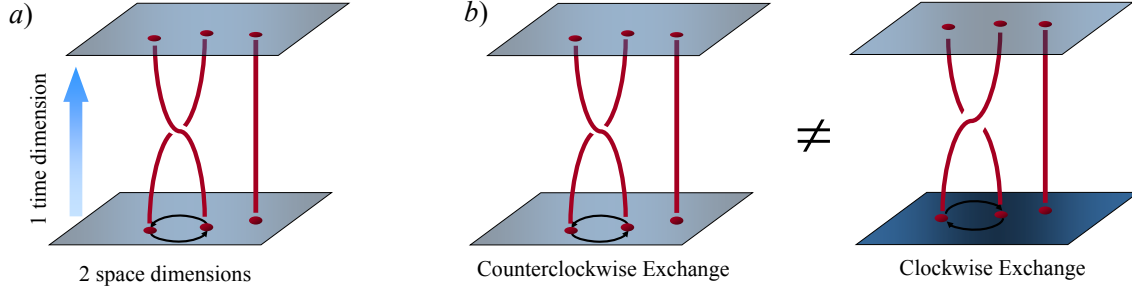


Figure 1.6: Exchanging particles in 2+1 dimensions. (a) Exchanging particles in a two-dimensional space corresponds to braiding their world-lines in three-dimensional space-time. (b) Exchanging particles in a clockwise manner leads to a braid which is topologically distinct from the braid resulting from exchanging particles in a counterclockwise manner.

to the identity. Instead, in general (when the underlying state of the system can be described by a scalar wavefunction) the state of the system acquires an overall phase of $e^{i\phi}$, where ϕ can be “any”^g phase — hence, identical particles in two dimensions are called “anyons” [21].

We see that in two dimensions, exchanging two anyons twice, does not return the system to its original state. Therefore, the final state of the system depends not only on the final positions of the particles, but also on *how* the exchanges took place. To see this more clearly, consider time flowing in a direction perpendicular to the plane of particles as in Fig. 1.6 (a). In this picture, exchanging particles in two-dimensional space corresponds to braiding *world-lines* of particles in (2+1)-dimensional space-time. As is shown in Fig. 1.6 (b), in two dimensions, there is a topological distinction between exchanging particles in a clockwise or a counterclockwise manner (the two cannot be smoothly deformed into one another). Therefore, the operation associated with exchanging anyons, in general, can no longer be described by a representation of the permutation group; instead, these exchanges are described by a representation of the *braid group* (see Fig. 1.7).

This observation can have profound consequences. Note, for instance, that in general a braid group can have multidimensional irreducible representations i.e. braid generators can be represented by matrices. As was described in Sec. 1.2.1, for certain topologically ordered phases, when quasiparticle excitations are present, the corresponding Hilbert space is degenerate and the state of the system is described by a vector. In these systems,

^gStrictly speaking, for anyons to be stable, ϕ is restricted to be a rational multiple of 2π (see, e.g. [24]).

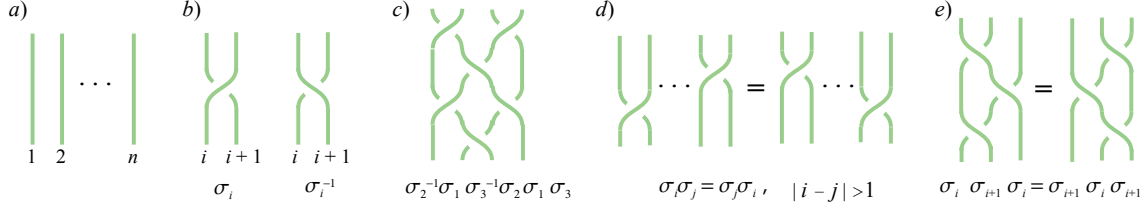


Figure 1.7: The Braid group. The braid group, B_n , can be generated using a set of braid generators $\sigma_1, \sigma_2, \dots, \sigma_{n-1}$, acting on n strands (a). (b) σ_i corresponds to exchanging strands i and $i + 1$ in a clockwise manner while a counterclockwise exchange corresponds to σ_i^{-1} . (c) An example of a group element. The multiplication corresponds to combining braid generators. (d), (e) The defining conditions of a braid group: $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ for all i , and $\sigma_i \sigma_j = \sigma_j \sigma_i$, for all $|i - j| > 1$.

when identical quasiparticles are exchanged, the corresponding unitary operation can be represented by a matrix — an irreducible representation of the braid group. In general, different matrices associated with exchanging different pairs of particles do not commute, hence, they form a non-Abelian representation of the braid group. Quasiparticles that obey this kind of *non-Abelian statistics* are called *non-Abelian anyons* or sometimes *nonabelions* [23]. In what follows, we will refer to “topologically ordered phases with non-Abelian quasiparticle excitations”, simply as “non-Abelian phases”. Also note that in what follows, when considering fractional statistics we often use the words particle, quasiparticle, quasihole and anyon interchangeably.

1.2.3 Topological Quantum Computing

We have seen that the evolution of a system of well-separated non-Abelian anyons due to adiabatically moving quasiparticles around each other, is described by multidimensional unitary operations. A remarkable fact about these operations is that they depend *only* on the topology of the paths used to carry out the braid and not the details of the quasiparticle space-time trajectories. In other words, wiggles (due to local interactions with environment) in the path of the quasiparticles around each other, as long as they do not change the topology of the braid, will not affect the resulting unitary operation (see Fig. 1.8). Therefore, unitary operations corresponding to braids are robust. In topological quantum computing, the idea is that the braid operations can, in principle, simulate quantum computation. For

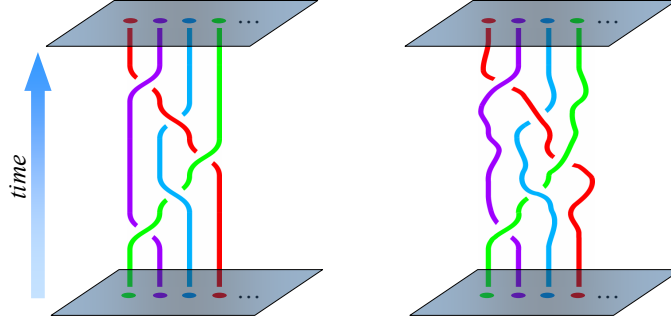


Figure 1.8: Topological robustness. The unitary operation corresponding to exchanging anyons depends only on the topology of the braid and not on the details of how the exchanges took place. The two patterns on the right and on the left, produce the same unitary operation.

example, in the quantum circuit model each gate can be simulated by a braid. Since the braid operations are robust, the resulting quantum computation is immune to errors caused by local interactions with the environment, leading to intrinsically fault-tolerant quantum computation.

To summarize, in topological quantum computing quantum information is stored in global degrees of freedom, carried by localized quasiparticle excitations. If the quasiparticles are kept sufficiently far apart, the associated degrees of freedom are spread over the entire system and local disturbances, including small interactions with the environment will not affect the stored quantum information. Quantum computation then is carried out by adiabatically moving quasiparticle excitations around each other in two space dimensions, or equivalently, by braiding their world-lines in the three-dimensional space-time. Because of the topological robustness of braid operations, the quantum computation carried out this way is inherently fault-tolerant.

One question that naturally arises is that whether these non-Abelian phases are rich enough to give rise to arbitrary quantum computation. In 2000, Freedman, Larsen and Wang showed that for a class of non-Abelian anyons described by the so-called $SU(2)_k$ ^h *Chern-Simons-Witten (CSW) theories*,ⁱ for certain values of the parameter k , arbitrary

^hRead $SU(2)$ “level” k .

ⁱCSW theory is an effective topological quantum field theory that describes the low energy limit of quantum Hall systems.

quantum computation can be carried out just by braiding anyons around each other [25, 26].^j Furthermore, it has been shown that a universal quantum computer can simulate any possible braid operation [27]. Therefore, quantum computers that are based on braiding non-Abelian anyons are, in effect, equivalent to other existing proposals for implementation of a quantum computer. However, from a practical point of view, it is the inherent robustness of this method of quantum computing which may prove to be worth struggling with the technological challenges that must be overcome before this method of quantum computing can ever be realized.

1.3 The Quantum Hall Effect

The theory of topological quantum computing is based on the hypothesis that states of matter with non-Abelian properties can be found in Nature. Throughout the years, many theoretical models have been proposed that predict the existence of such states. These models include a number of quantum systems, for example, rotating Bose gases [28], frustrated spin systems [29, 30, 31], Josephson junction arrays [32] and quantum Hall systems [22]. Among these theoretical models, the only candidates that are backed up with experimental evidence are the fractional quantum Hall (FQH) states. In this Section, I will briefly review the basics of quantum Hall effect and the properties of the ground states and excitations (i.e. quasiparticles and quasiholes). After this quick review, I will describe how certain FQH wavefunctions with exotic properties are constructed and how these special properties can be understood.

1.3.1 The Integer Effect

The integer quantum Hall effect (IQHE) was discovered by von Klitzing *et al* in 1980 [34]. In their experiment, a two-dimensional gas of electrons (2DEG) was formed in the interface of Silicon and Silicon Oxide in a metal-oxide-semiconductor field-effect transistor (MOSFET) and was subject to a strong perpendicular magnetic field (see Fig.1.9). The observation was that the Hall resistance $R_H = V_H/I$, as a function of the magnetic field B , displays a plateau behavior, in contrast to the classical Hall effect in which the Hall resistance grows linearly

^jSee Sec. 1.4.

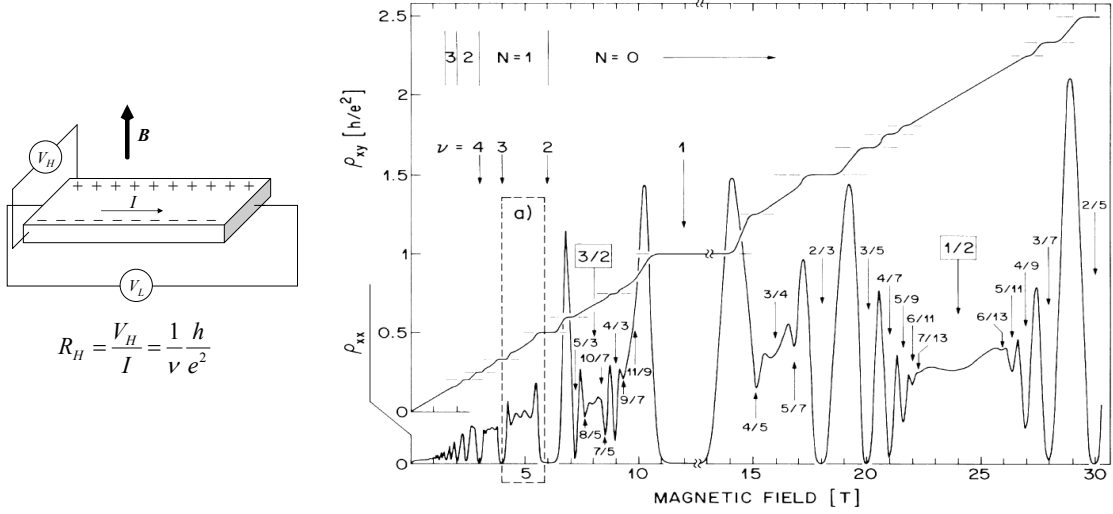


Figure 1.9: The quantum Hall effect. Left: The apparatus consists of a two-dimensional gas of electrons subject to a strong magnetic field, at low temperatures. Right: Diagonal resistivity, ρ_{xx} and Hall resistance, ρ_{xy} , as a function of the magnetic field. Data taken from [33].

with the magnetic field,

$$R_H = \frac{B}{ne}, \quad (1.8)$$

where n is the density of electrons. In IQHE, the plateaus form at $R_H = h/(\nu e^2)$, with ν being an integer. In addition, the longitudinal resistance R_L at the corresponding values of the magnetic field vanishes which implies the flow of a dissipationless current.

This effect can be explained by considering a model of non-interacting electrons confined to two dimensions subject to an external perpendicular magnetic field [35] as described by the following Hamiltonian,

$$H = \frac{1}{2m_e} \sum_i (-i\hbar \nabla_i - e\vec{A})^2. \quad (1.9)$$

In this model, electrons undergo a circular motion with quantized energies, $E_n = \hbar\omega_c(n + 1/2)$, and cyclotron frequency $\omega_c = eB/m$. The associated energy levels (i.e. the *Landau levels*) possess a finite degeneracy which is equal to the number of flux quanta piercing the sample, i.e. $N_\phi = AB/\Phi_0$ where A is the area of the sample and $\Phi_0 = h/e$ is the magnetic

flux quantum. The ratio, $\nu = N/N_\phi$, where N is the number of electrons, is called the *filling fraction*.

As the magnetic field increases, electrons fill in the lowest available Landau levels. Assuming that the magnetic field polarizes the spin of the electrons (and so there is no spin degeneracy) each Landau level can contain up to N_ϕ electrons. At those values of the magnetic field where the filling fraction is an integer ($B = n\Phi_0/\nu$) and an integer number of Landau levels are exactly full, the Hall resistance is,

$$R_H = \frac{1}{\nu} \frac{h}{e^2} \quad (1.10)$$

while the longitudinal resistance vanishes,

$$R_L = 0. \quad (1.11)$$

The fact that R_L vanishes can be explained by noting that dissipation (hence the resistance) occurs when electrons can scatter to other available energy levels. When the lowest Landau level (LLL) is completely full, the closest available states are in the next energy level which is separated from the LLL by a large energy gap of magnitude $\hbar\omega_c$. At low temperatures these states are essentially unavailable, therefore, electrons cannot scatter and the current is dissipationless.

The formation of broad plateaus in R_H and wide minima in R_L can be attributed to the existence of residual disorder in the system [36]. In brief, residual dirt and disorder in the system will cause some of the states in the Landau levels to *localize*, and shifts their energies. Electrons trapped in these localized states become isolated and no longer contribute to the electrical conduction. These localized states do not affect the measurements of carrier densities in the system since the remaining extended states in the Landau levels make up for this loss. Therefore, as long as electrons are filling these localized states, R_H and R_L do not change, giving rise to flat regions in R_H and R_L . As the last word in closing this section on IQHE, it should definitely be mentioned that what really makes IQHE interesting is the precision it provides for the quantized values of R_H while there is dirt and disorder in the system. This was explained by Laughlin, based on a gauge invariance argument which I will not explain here and the reader is referred to [36] and [37].

1.3.2 The Fractional Effect

The fractional quantum Hall effect (FQHE) was first discovered by Tsui *et al* in 1982 [38]. The observation was that at sufficiently High magnetic fields and low temperatures, the Hall resistance R_H , exhibits plateau behavior at filling fractions that are rational fractions of the form $\nu = p/q$ where p and q are coprimes and q (in most cases) is odd.

This observation came as a surprise since in the single-particle model used to describe IQHE, the presence of plateaus was attributed to the large energy gaps that exist between the Landau levels. If the same model is to be used for FQHE, there should be gaps *within* the Landau levels, which is not the case. Therefore, this effect cannot be explained by a simple model of noninteracting particles and the Coulomb interaction between the particles must be taken into consideration.

The Hamiltonian of a system of interacting electrons in two dimensions has the following form,

$$H = \frac{1}{2m_e} \sum_i (-i\hbar \nabla_i - e\vec{A})^2 + \frac{1}{2} \frac{e^2}{4\pi\epsilon} \sum_{i \neq j} \frac{1}{|\vec{x}_i - \vec{x}_j|}. \quad (1.12)$$

For a class of filling fractions of the form $\nu = 1/M$ where M is an odd integer, Laughlin proposed the following variational wavefunction,

$$\psi_L = \prod_{i < j} (z_i - z_j)^M e^{-\sum_i \frac{|z_i|^2}{4l_0^2}}, \quad (1.13)$$

where z_i are the complex coordinates of electrons in two dimensions and $l_0 = \sqrt{\hbar/eB}$ is the magnetic length.

This wavefunction is carefully designed to satisfy a number of desired properties: (1) the factor $|z_i - z_j|$ (the so-called *Jastrow form*) keeps the electrons apart, (2) since M is odd, the Laughlin wavefunction is totally antisymmetric which implies that the underlying particles giving rise to FQHE, (in 2DEG systems) are electrons (fermions), (3) it is a holomorphic function (there are no $\bar{z}$ dependences) which is the defining property of electron wavefunctions in the LLL, and (4) it is an eigenfunction of total angular momentum. Finally (and perhaps most importantly), the Laughlin wavefunction 1.13, has a very good overlap with the numerical solutions of the Hamiltonian in Eq. 1.12, even though it is not the exact ground state of this Hamiltonian.^k

^kThe Laughlin wavefunction is in fact the exact ground state of a short range two-body Hamiltonian proposed by Haldane in 1983 [39].

Note that the filling fraction $\nu = 1/M$ can also be read off from the Laughlin wavefunction, Eq. 1.13. In the first Landau level, the last (highest) occupied state corresponds to the highest angular momentum. The highest angular momentum in turn, corresponds to the highest power of z in the Laughlin wavefunction, i.e. $M(N-1)$ (note that the Laughlin wavefunction is an eigenfunction of the total angular momentum with an eigenvalue which is proportional to the highest power of z). So in the limit of large N , the LLL contains $N_\phi = MN$ states, which can be occupied by N electrons. Therefore, the filling fraction which is defined as the number of electrons per flux quantum is simply $\nu = 1/M$.

When the magnetic field is in the vicinity of $B = Mn\Phi_0$, the system is in its lowest energy and the density of electrons is *pinned* to a certain value, therefore, these electrons form an *incompressible* fluid. The ground state of the system is separated from the excited states by a large energy gap. As a result, small changes to the magnetic field or the number of electrons do not change the density of the quantum Hall fluid. Instead, these perturbations can introduce particle-like excitations to the system. In particular, presence or absence of additional flux quanta in the system can be interpreted as existence of quasiholes or quasiparticles, respectively, which are free to roam around the system. These excitations are expected to possess exotic properties, for example, they can carry fractional charge and obey fractional statistics. The wavefunction of a Laughlin quasihole at position w is of the form,

$$\psi_{L,qh}(w) = \prod_i (z_i - w) \psi_L, \quad (1.14)$$

where ψ_L is the Laughlin wavefunction defined in 1.13. The wavefunction of a Laughlin quasiparticle has the following more complicated form,

$$\psi_{L,qp} = \prod_i \left(\partial_{z_i} - \frac{w}{l_0^2} \right) \psi_L. \quad (1.15)$$

Laughlin showed that these quasiholes and quasiparticles carry electric charge of $-e/M$ and $+e/M$ respectively.

The fact that fractional quantum Hall states support localized excitations with fractional charge follows from the following very general argument [36]. Imagine piercing the sample with an infinitely thin solenoid and adiabatically passing one flux quantum through it. Due to Faraday's law we have,

$$\frac{\partial \phi}{\partial t} = \oint \vec{E} \cdot d\vec{l} = E(2\pi r) = 2\pi r \frac{1}{\nu} \frac{h}{e^2} J = \frac{1}{\nu} \frac{h}{e^2} \frac{dQ}{dt}. \quad (1.16)$$

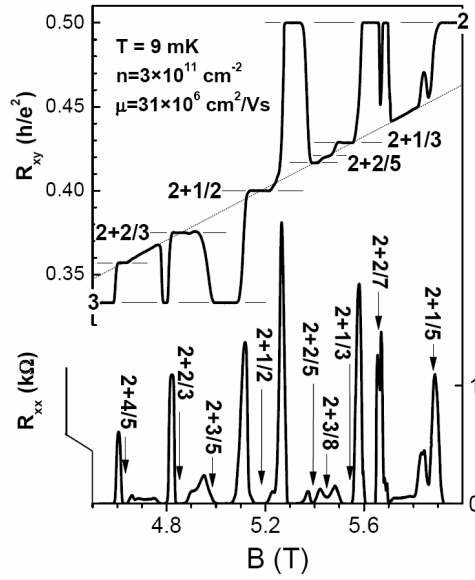


Figure 1.10: The first experimental evidence for the formation of a plateau at filling fraction $\nu = 12/5$. Data taken from [43].

Note that the induced electric field, $\vec{E}$ (which is tangential) will cause a radial current to flow away from the thin solenoid, therefore causing the background charge to deplete, leaving a hole of charge Q behind. Therefore,

$$Q = \nu \frac{h^2}{e} \int_0^{\Phi_0} d\phi = \nu e. \quad (1.17)$$

This fractional charge has been directly observed in experiments [40].

Perhaps even more interestingly, it has been suggested that these quasiholes and quasiparticles might obey fractional statistics. In 1984, Arovas, Schrieffer and Wilczek showed explicitly that this is indeed the case and exchanging two Laughlin quasiparticles or quasiholes will result in an overall phase of $e^{i\pi/M}$ [41]. Therefore, the notion of anyons discussed in Sec.1.2.2, at least in the Abelian form, can be realized as quasiparticle and quasihole excitations of the Laughlin states. Some initial experiments for observing fractional statistics in these states have been carried out [42].

Throughout the years, many experiments on FQH effect have been carried out and

many plateaus corresponding to many different filling fractions have been observed (see for example, Fig. 1.9). After the success of Laughlin's trial wavefunction for describing FQH states corresponding to $\nu = 1/M$, other theories such as Jain's composite fermion theory [44] and the hierarchy picture of Haldane and Halperin [39, 45] were proposed to describe the observed plateaus. The common characteristics of all these states was that the corresponding filling fraction had an *odd* denominator. In the above mentioned theories, the odd denominator in the filling fraction was attributed to the fact that in the quantum Hall effect the underlying particles are electrons and therefore obey Fermi statistics.

The saga continued when in 1987 a new plateau at filling fraction $\nu = 5/2$ was observed [33]. As is shown in Fig. 1.10, this plateau has the same characteristics as other observed plateaus except that it has an *even* denominator, and therefore, it could not be explained by any of the previous theories. Using analogies with *conformal field theory* (CFT), in 1991, Moore and Read [23] proposed a wavefunction to describe the state corresponding to the filling fraction $\nu = 5/2$.¹ This wavefunction has the form,

$$\psi_{MR} = \text{Pf}\left(\frac{1}{z_i - z_j}\right) \prod_{i < j} (z_i - z_j)^2 e^{-\sum_i \frac{|z_i|^2}{4l_0^2}}, \quad (1.18)$$

where, the Pfaffian factor (Pf) is an antisymmetrized sum over all pairs of particles, $\text{Pf}\left(\frac{1}{z_i - z_j}\right) = \mathcal{A}\left(\frac{1}{z_1 - z_2} \frac{1}{z_3 - z_4} \cdots\right)$. Roughly speaking, in this picture, the even denominator in the filling fraction is related to the assumption that in the Moore-Read state electrons form pairs (similar to Cooper pairs in the BCS theory) to lower their energies. I will describe this in more detail when considering clustered states in the next Section (see also [46, 47, 48]).

A remarkable prediction of the Moore-Read state is the fact that the corresponding quasiparticle excitations, possess particularly exotic properties. Similar to quasiparticle excitations of the Laughlin states, the Moore-Read state also has quasiparticle excitations that carry fractional charge and obey fractional statistics. However, unlike Laughlin states when quasiparticle excitations of the Moore-Read state are present, the corresponding Hilbert space is degenerate and its degeneracy grows exponentially with the number of quasiparticles present (see the next Section). Therefore, the state of the system in the presence of quasiparticle excitations at fixed positions can no longer be described by a scalar wavefunction, instead, it has a vector form. Consequently, the operators that describe the

¹Strictly speaking, the Moore-Read state corresponds to a state at filling fraction $\nu = 1/2$ which is analogous to the $\nu = 5/2$ state in the lowest Landau level.

evolution of the system when quasiparticles move around one other will be represented by matrices. Therefore, these quasiparticles can in principle, realize the notion of non-Abelian anyons described in Sec. 1.2.2.

Though a full understanding of non-Abelian states requires a number of advanced mathematical techniques (e.g. topological quantum field theory, conformal field theory, etc.) many of their basic properties, including the multidimensionality of the Hilbert space in the presence of quasiparticles, can be understood at the “wavefunction” level, similar to Laughlin’s original approach. In the following Section, I will describe some of the properties of non-Abelian anyons of the Moore-Read state, as well as the so-called Read-Rezayi states in more detail.

1.3.3 Non-Abelian States in FQHE

As was discussed in Sec. 1.3.2, Laughlin states are excellent trial wavefunctions that describe FQH states at filling fraction $\nu = 1/M$ where M is an odd integer. For other filling fractions with odd denominator, theories such as Jain’s composite fermion picture [44], or the hierarchy picture of Haldane and Halperin [39, 45] can be used. As was mentioned above, the observation of $\nu = 5/2$ state in 1987 came as a surprise since the odd denominator in FQH states was thought to be related to the fermionic nature of the underlying electrons. To solve the mystery, Moore and Read proposed to use a *paired* state to explain the state corresponding to this filling fraction [23]. Numerical work indicates that the Moore-Read state has a good overlap with the ground state of the Hamiltonian in Eq. 1.12 at filling fraction $\nu = 5/2$ [49, 50].

The idea of paired states was later extended by Read and Rezayi to include *clustered* states [51]. These states describe incompressible quantum Hall fluids at filling fractions $\nu = k/(km + 2)$, where k (called the “level”) is the number of particles in each cluster (for example, for the paired state of Moore and Read $k = 2$) and m is a nonnegative integer which determines the nature of the underlying particles (see below). It has been shown that the Read-Rezayi state at $k = 3$, corresponding to the filling fraction $\nu = 3/5$, has a good overlap with the numerically calculated ground state of the Hamiltonian 1.12 [46, 52]. This state is particularly important since the particle-hole conjugate of this state in the second Landau level at filling fraction $\nu = 12/5$ has been observed in experiments [43].

It should be noted that, even though our discussion of FQH states, so far, has been focused

on the realization of these states in two-dimensional electron gases (2DEG), in principle, these states can also exist in bosonic systems, such as rotating Bose gases (RBG), although these states have not yet been observed experimentally [28]. In the expression for the filling fraction, $\nu = k/(km+2)$, an odd m corresponds to a system of fermions (such as 2DEG) and an even m corresponds to a system of bosons (e.g. RBG). So for example, for a fermionic Laughlin state $\nu = 1/3$ while for a bosonic Laughlin state $\nu = 1/2$. Likewise, for a fermionic Moore-Read state $\nu = 1/2$ while for a bosonic Moore-Read state $\nu = 1$.

In this Section, I will describe a particularly transparent way to construct the Read-Rezayi sequence of non-Abelian states, due originally to Cappelli *et al* [53] (see also [54]). The advantage of this construction is that it provides a simple way to visualize charge fractionalization, as well as the multidimensionality of the Hilbert space of FQH states in the presence of quasiparticle excitations.

In presenting this construction, it will also be useful at times to consider the so-called *thin cylinder* limit. The usefulness of this limit, for both Abelian and non-Abelian states, has recently been emphasized by a number of authors [55, 56, 57, 58, 59, 60, 61, 62, 63]. In this limit, we imagine that the two-dimensional electron gas is rolled up tightly into a cylinder, with circumference $L_y \ll l_0$ in the $\hat{y}$ direction and extends to infinity in the $\hat{x}$ direction (see Fig. 1.11 (a)). In this limit, if one solves the Hamiltonian 1.9 in the Landau gauge, $\mathbf{A} = (0, xB, 0)$ (see, for example [35]) then the LLL wavefunctions take the form of localized Gaussians in the $\hat{x}$ direction and plane waves in the $\hat{y}$ direction,

$$\psi(x, y) = A e^{-\frac{(x-x_q)^2}{2l_0^2}} e^{iqy}. \quad (1.19)$$

Here, because of the periodic boundary conditions in the $\hat{y}$ direction, the allowed q values are,

$$q = \frac{2\pi}{L_y} n, \quad n = 0, \pm 1, \pm 2, \dots \quad (1.20)$$

For a given q value, the Gaussian part of the wavefunction is then centered at the point,

$$x_q = \frac{2\pi l_0^2}{L_y} n, \quad n = 0, \pm 1, \pm 2, \dots \quad (1.21)$$

Thus we see that the spacing between neighboring Gaussians, as we move along the $\hat{x}$ direction, is $2\pi l_0^2/L_y$ which, in the thin cylinder limit is much greater than the width of

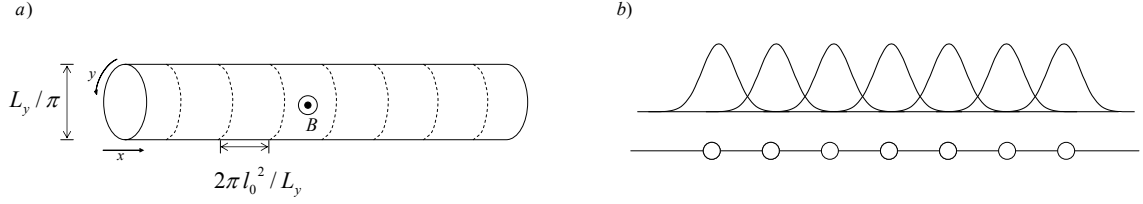


Figure 1.11: The thin cylinder limit. (a) The two-dimensional electron gas, wrapped around a cylinder. Dashed lines mark the locations of the Gaussians. (b) In the limit of very thin cylinder, $L_y \ll l_0$, the Gaussians are well separated and the system is essentially equivalent to a one-dimensional chain.

each Gaussian ($\sim l_0$), and so the Gaussians barely overlap. In this limit of well separated Gaussians, we can represent each Gaussian as a thin strip encircling the cylinder (see Fig. 1.11).

Note that in this limit, the LLL Hilbert space essentially becomes that of a one-dimensional *chain* (see Fig. 1.11 (b)). We can then represent states in the many-particle LLL Hilbert space using an occupation number representation. For spin-polarized fermions, the occupation of each site (i.e. Gaussian) on the chain can be either 0 or 1. The LLL many-particle Hilbert space is therefore spanned by states which can be represented as strings of 0's and 1's, corresponding to the Gaussian occupation numbers as we move along the cylinder, e.g.,

$$\cdots 101000101010101 \cdots . \quad (1.22)$$

For bosons, any occupancy number is allowed, and so the many-particle Hilbert space is spanned by states which can be expressed in the form,

$$\cdots 35675143059674 \cdots . \quad (1.23)$$

Note that in what follows, for simplicity, we mostly focus on the bosonic case which turns out to be more transparent. For the fermionic case, the reader is referred to Tables 1.1 and 1.2.

To see the usefulness of the thin cylinder limit, let us consider first the Laughlin

wavefunction (1.13). Writing this wavefunction with the Gaussian part suppressed we have,^m

$$\psi_L = \prod_{i<j}^N (z_i - z_j)^M. \quad (1.24)$$

This wavefunction (which describes FQH states at filling fraction $\nu = 1/M$) can be written in the form

$$\psi_L = \prod_{i<j}^N (z_i - z_j)^2 \prod_{i<j}^N (z_i - z_j)^m, \quad (1.25)$$

where $m = M - 2$ is the quantity m introduced above. In the bosonic case ($m = 0$) the wavefunction 1.25 becomes,

$$\Phi_L = \prod_{i<j}^N (z_i - z_j)^2. \quad (1.26)$$

In the limit where the spacing between the Landau levels is much greater than the interaction energy between the electrons, i.e. $\hbar\omega_c \gg e^2/(4\pi\epsilon l_0)$, the constraint that the wavefunction must live entirely in the LLL becomes exact. As shown by Haldane [39], within this restricted Hilbert space, the wavefunction Φ_L is then an exact, zero energy ground state of the following 2-body interaction potential,

$$V = \sum_{i<j} \delta^2(z_i - z_j), \quad (1.27)$$

where $\delta^2(z_i - z_j)$ is the two-dimensional delta function. This is easily seen by noting that V is nonzero only when two or more bosons occupy the same position, and Φ_L vanishes whenever two bosons are brought to the same position. Thus, $V\Phi_L = 0$.

We can now consider what the Laughlin state evolves into in the thin cylinder limit. As described above, in this limit the set of one-particle basis states for this system in the LLL becomes effectively a one-dimensional “chain” of well-separated Gaussians (Fig. 1.11 (b)). The short range interaction potential, 1.27, then implies that in the ground state, no two bosons can sit in the same Gaussian (if they did there would be a nonzero amplitude to find the two bosons in the same point in space and thus a contribution from V to the energy). By

^mNote that the Laughlin wavefunction in Eq. 1.13 is written in the symmetric gauge and the suppressed Gaussian factor in the Laughlin wavefunction should not be confused with the Gaussian part of the wavefunction in Eq. 1.19

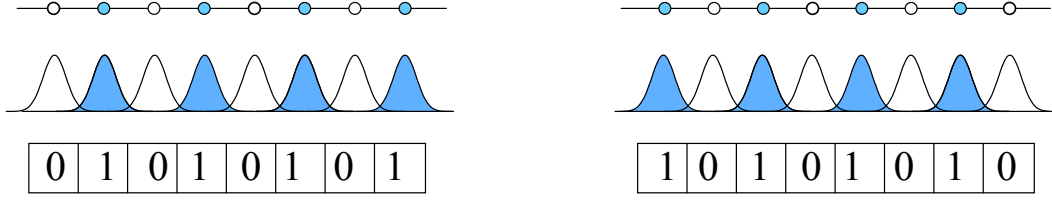


Figure 1.12: Two ground states of the bosonic Laughlin state in the thin cylinder limit (see also [63]).

similar reasoning, it will also cost energy if two bosons are occupying neighboring Gaussians due to their finite (albeit exponentially small) overlap. Once these constraints are taken into account, one can then safely ignore the contributions from V due to the overlap of the next-nearest-neighbor Gaussians, since these will be smaller than the nearest neighbor contribution by yet another exponential factor. This is all we will need to know to proceed here. For a rigorous discussion of this limit see [56, 57, 58, 59, 60, 61, 62, 63].

From the above reasoning, we see that in the limit of a thin cylinder, no more than one boson can sit on any two neighboring positions. Using the occupation number representation described above, it is easy to see that this gives rise to the following two degenerate ground states (see Fig. 1.12),

$$\begin{array}{l} 10101010101010 \\ 01010101010101 \end{array} \quad (1.28)$$

Note that in this limit, the two degenerate ground states can clearly be distinguished by local measurements. Thus the topological degeneracy of the Laughlin state evolved into a simpler degeneracy associated with a locally observable broken translational symmetry in the thin cylinder limit. For fermions ($m = 1$) the degeneracy of the ground state on a cylinder is 3-fold. The associated ground states are given in Table 1.1.

Now consider introducing excitation to the system. If a quasi-hole is inserted at position w , as was shown in 1.14, the corresponding wavefunction will have the form,

$$\psi_{L,qh}(w) = \prod_i^N (z_i - w) \psi_L \quad (1.29)$$

which describes quasiholes of charge $Q = +e\nu = +e/(m + 2)$. Note that when $m = 0$, this wavefunction is also, a zero energy eigenstate of the potential given in 1.27. It was shown by [58, 59] that in the limit of a thin cylinder, these excitations can be shown as *domain walls* between two different ground states. To see this, consider one of the bosonic ground states in 1.29, for example

$$1010101010101010. \quad (1.30)$$

If we remove one electron, we effectively create a hole with charge $+e$ in the position previously occupied by the electron,

$$10101010|0101010. \quad (1.31)$$

In this notation, ‘|’ indicates a domain wall. Note that we can insert the other ground state from 1.29 in the position of this domain wall,

$$101010|0101010|01010 \quad (1.32)$$

without changing the energy of the system. Since the net charge of the system is still $+e$, (as in 1.31), the new domain walls in 1.32 must each carry a charge of $+e/2$. Note that $+e/2$ is the smallest charge possible for quasiholes of a bosonic Laughlin state ($m = 0$) as given in Eq. 1.29. In the thin cylinder limit, this charge corresponds to the domain wall between the two different ground states given in 1.29. For the fermionic case ($m = 1$), the smallest possible charge for a Laughlin quasihole is $+e/3$. An example of the corresponding domain wall is shown in Table 1.2.

Having reviewed the properties of the Laughlin state and the usefulness of the thin cylinder limit, we next turn to the Moore-Read state. Following Cappelli *et al.* [53] consider the Laughlin wavefunction in Eq. 1.25. To derive the Moore-Read state, we modify this wavefunction by first, dividing the electrons into two groups, $A = \{1, \dots, N/2\}$ and $B = \{N/2 + 1, \dots, N\}$ and writing down a product of Laughlin wavefunctions — one for the A particles and one for the B particles,

$$\psi_L \rightarrow \prod_{i < j \in A} (z_i - z_j)^2 \prod_{i < j \in B} (z_i - z_j)^2 \prod_{i < j}^N (z_i - z_j)^m \quad (1.33)$$

For this expression to be an acceptable wavefunction for identical particles (i.e. a wavefunction which is completely symmetric or antisymmetric) it is necessary to symmetrize over all

the particles from group A and group B so they will be indistinguishable. The result is the Moore-Read wavefunction,

$$\psi_{MR} = \mathcal{S} \left[\prod_{i < j \in A} (z_i - z_j)^2 \prod_{i < j \in B} (z_i - z_j)^2 \right] \prod_{i < j}^N (z_i - z_j)^m, \quad (1.34)$$

where $\mathcal{S}$ is the symmetrizer symbol. As was described in Sec. 1.3.2, the filling fraction corresponding to this wavefunction can be read off by counting the number of flux quanta, N_ϕ . This quantity is equivalent to the highest power z_i in the wavefunction 1.34, i.e., $N_\phi = 2(N/2 - 1) + m(N - 1)$. Therefore, in the thermodynamic limit, ψ_{MR} as given in 1.34, describes the ground state of a FQH state at filling fraction $\nu = 1/(m + 1)$.

When $m = 0$, the bosonic part of this wavefunction has the form

$$\Phi_{MR} = \mathcal{S} \left[\prod_{i < j \in A} (z_i - z_j)^2 \prod_{i < j \in B} (z_i - z_j)^2 \right]. \quad (1.35)$$

Similar to the Laughlin wavefunction, this wavefunction is the exact, zero energy, ground state of the following three-body potential,

$$V = \sum_{i < j < k} \delta^2(z_i - z_j) \delta^2(z_j - z_k). \quad (1.36)$$

This can be seen by noting that on the one hand, V vanishes unless at least three bosons occupy the same position. On the other hand, if three or more particles are brought to the same position, Φ_{MR} vanishes, therefore, $V\Phi_{MR} = 0$. Note that if two particles are brought to the same position, the product $V\Phi_{MR}$ vanishes, however, due to the symmetrizing factor in the expression for Φ_{MR} , there would always be at least one term in Φ_{MR} that does not vanish. Thus, in the Moore-Read state, it is possible for two bosons to occupy the same position.

Now we again turn to the thin cylinder limit. Following similar reasoning as we did for the Laughlin case, the interaction, 1.36, implies that in the limit of a thin cylinder, up to two bosons can sit on any two neighboring positions. Therefore, bosons can occupy the Gaussians in two different patterns: (1) each Gaussian is occupied by a single boson. This will give rise to the following ground state

$$1111111111111111. \quad (1.37)$$

(2) Every other Gaussian is occupied by two bosons, as in the following two ground states.

$$\begin{array}{l} 20202020202020 \\ 02020202020202. \end{array} \quad (1.38)$$

Therefore, in the bosonic case, the highest density ground states are three-fold degenerate. The fermionic case, corresponding to $m = 1$, is shown in Table 1.1. In this case, the wavefunction 1.34 describes a fermionic system at filling fraction $\nu = 2/4$.ⁿ This means, no more than two particles can occupy any four neighboring orbitals which leads to a 6-fold degeneracy in the ground state as is shown in Table 1.1 (for more on the fermionic case see e.g. [63]). It should also be noted that, in general, the ground state degeneracy depends on the geometry of the surface the system is defined on, as well as the nature and parity of the underlying particles. Here we have focused on systems defined on a cylinder (disc). For a detailed discussion of the ground state degeneracy of Moore-Read states the reader is referred to [64].

Now consider inserting one quasihole excitation at position w . Similar to Eq. 1.29 we have,

$$\psi_{MR,qh}(w) = \prod_i (z_i - w) \psi_{MR}. \quad (1.39)$$

The quasihole described by this wavefunction carries the charge $Q = +e\nu$ which, in this case, is $+e/(m+1)$. Note that when $m = 0$, this wavefunction is also a zero energy eigenstate of the potential in 1.36.

Up to this point everything is very similar to the Laughlin quasiholes, however, the Moore-Read states have a much more complex structure. This can become clear by noting that in Eq. 1.39 we can take the $\prod_i (z_i - w)$ factor inside the symmetrizer in Eq. 1.34 to get,

$$\begin{aligned} \psi_{MR,qh}(w) &= \mathcal{S} \left[\prod_{i \in A} (z_i - w) \prod_{i \in B} (z_i - w) \prod_{i < j \in A} (z_i - z_j)^2 \prod_{i < j \in B} (z_i - z_j)^2 \right] \\ &\times \prod_{i < j} (z_i - z_j)^m. \end{aligned} \quad (1.40)$$

Once the Laughlin quasihole operator is inside the symmetrizer, we can *break* apart the quasihole at position w and adiabatically separate the two parts by taking them to positions

ⁿThe reason for writing the filling fraction as $\nu = 2/4$ instead of $\nu = 1/2$ will be more obvious when we introduce cluster states later in this Section.

w_1 and w_2 . The result is the following wavefunction,

$$\begin{aligned} \psi_{MR,qh}(w_1, w_2) &= \mathcal{S}[\prod_{i \in A} (z_i - w_1) \prod_{i \in B} (z_i - w_2) \prod_{i < j \in A} (z_i - z_j)^2 \prod_{i < j \in B} (z_i - z_j)^2] \quad (1.41) \\ &\times \prod_{i < j} (z_i - z_j)^m. \end{aligned}$$

This *further* fractionalization of the quasiholes is characteristic of non-Abelian states (e.g. it is *not* possible in the simple Laughlin state). Since the total charge of the quasihole in 1.39 was $+e/(m+1)$, the wavefunction in Eq. 1.42 describes two quasiholes at positions w_1 and w_2 , each carrying the charge $Q = +e/(2m+2)$.

In the limit of a thin cylinder, again quasiholes (and quasiparticles) can be represented as domain walls between different ground states. For example in the bosonic case ($m=0$), the following sequence,

$$02020|1111111|02020 \quad (1.42)$$

represents one way to create two quasiholes, each carrying the charge $+e/2$.

To see the multidimensionality of the Hilbert space in the presence of quasihole excitations, consider introducing two quasiholes to the system, each with charge $+e/(m+1)$. If we create these quasiholes at positions w and w' , the corresponding wavefunction would be,

$$\begin{aligned} \psi_{MR,qh}(w, w') &= \prod_i (z_i - w)(z_i - w') \mathcal{S}[\prod_{i < j \in A} (z_i - z_j)^2 \prod_{i < j \in B} (z_i - z_j)^2] \quad (1.43) \\ &\times \prod_{i < j} (z_i - z_j)^m. \end{aligned}$$

Similar to the procedure described above, we can take the quasihole factors $\prod_i (z_i - w)(z_i - w')$ inside the symmetrizer. Each of the quasiholes then can break apart into two new quasiholes, each carrying half of the original charge. The result is a wavefunction with *four* quasiholes at positions w_1, w_2, w_3 and w_4 ,

$$\begin{aligned} \psi_{MR,qh}(w_1, w_2, w_3, w_4) &= \mathcal{S}[\prod_{i \in A} (z_i - w_1)(z_i - w_2) \prod_{i \in B} (z_i - w_3)(z_i - w_4) \quad (1.44) \\ &\times \prod_{i < j \in A} (z_i - z_j)^2 \prod_{i < j \in B} (z_i - z_j)^2 \prod_{i < j} (z_i - z_j)^m]. \end{aligned}$$

As was mentioned above, these quasiholes each carry a charge of $+e/(2m+2)$.

Note the wavefunction in 1.45 represents one way to distribute the four quasiholes among the particles of type A and type B . In general, this distribution is not unique; for example, if we denote the positions of quasiholes in Eq. 1.45 by $(12)_A(34)_B$, it is easy to see that, in principle, we can have two other wavefunctions corresponding to $(13)_A(24)_B$ and $(14)_A(23)_B$. It has been shown by Nayak and Wilczek that for the Moore-Read state, two of these three wavefunctions are linearly independent [65]. These two linearly independent wavefunctions can form a basis for describing the state of a system of four Moore-Read quasiholes. In other words, the corresponding Hilbert space is two-fold degenerate.

In the thin cylinder limit, the two degenerate states in the presence of four quasiholes can be represented by the following two sequences,

$$\begin{array}{l} 02020|11111|02020|11111|02020 \\ 02020|1111|0202020|1111|02020. \end{array} \quad (1.45)$$

Note that the main difference between these two sequences is that in the first sequence, the middle string of 0's and 2's is in phase with the two strings of 0's and 2's on the sides, while in the second sequence, the middle string is out of phase [58, 66]. The sequences corresponding to fermions ($m = 1$) are given in Table 1.2. For a rigorous discussion of the fermionic case the reader is referred to [55] or [61], for example.

Nayak and Wilczek have shown that, in general,^o the degeneracy of the Hilbert space of n Moore-Read quasiholes is $(\sqrt{2})^{n-2}$ -fold [65]. The fact that the Hilbert space of the Moore-Read state in the presence of quasiholes is degenerate implies that the wavefunctions describing the state of the system must be multidimensional (vectors) and the process of adiabatically interchanging quasiholes should be described by a multidimensional representation (matrix) of the braid group. This means elementary quasiholes of the Moore-Read state can be non-Abelian anyons (see Sec. 1.2.2).

As was mentioned in Sec. 1.3.2, the Laughlin and the Moore-Read states are special cases of a general class of states proposed by Read and Rezayi in [51]. These states are labeled by two parameters, m and k and describe particles at filling fraction $\nu = k/(km + 2)$. As before, $m = 0$ corresponds to systems of bosons and $m = 1$ corresponds to fermions. In this model, $k = 1$ corresponds to Laughlin states and $k = 2$ corresponds to Moore-Read states.

^oThe degeneracy of the Hilbert space in the presence of the Moore-Read quasiholes, in general, also depends on the geometry of the surface the system is defined on. Here for the most part we focus on a cylinder (or a disc) geometry.

For completeness, we now consider the general case of the Read-Rezayi states. Here, again we start with the Laughlin wavefunction, 1.25, then divide the electrons into k types, $A_1, \dots, A_k$, and finally, symmetrize over all particle types. The result is a wavefunction of the form,

$$\begin{aligned} \psi_{RR} = & \mathcal{S} \left[\prod_{i < j \in A_1} (z_i - z_j)^2 \prod_{i < j \in A_2} (z_i - z_j)^2 \dots \prod_{i < j \in A_k} (z_i - z_j)^2 \right] \\ & \times \prod_{i < j} (z_i - z_j)^m. \end{aligned} \quad (1.46)$$

From this wavefunction, one can see that the number of flux quanta $N_\phi = 2(N/k - 1) + m(N - 1)$, therefore, in the thermodynamic limit $\nu = k/(km + 2)$, as expected.

As before, when $m = 0$ the bosonic wavefunction,

$$\begin{aligned} \Phi_{RR} = & \mathcal{S} \left[\prod_{i < j \in A_1} (z_i - z_j)^2 \prod_{i < j \in A_2} (z_i - z_j)^2 \dots \prod_{i < j \in A_k} (z_i - z_j)^2 \right] \\ & \times \prod_{i < j} (z_i - z_j)^m, \end{aligned} \quad (1.47)$$

is the exact, zero energy, ground state of a $(k + 1)$ -body potential of the form,

$$V = \sum_{i_1 < i_2 < \dots < i_{k+1}} \delta^2(z_{i_1} - z_{i_2}) \delta^2(z_{i_2} - z_{i_3}) \dots \delta^2(z_{i_k} - z_{i_{k+1}}). \quad (1.48)$$

Again, this can be seen by noting that the potential V in 1.48 vanishes unless at least $k + 1$ particles are brought to the same position. The wavefunction Φ_{RR} , however, vanishes when $k + 1$ or more particles occupy the same position, therefore, $V\Phi_{RR} = 0$. Note that when k particles are brought together, the product $V\Phi_{RR}$ vanishes while the wavefunction, Φ_{RR} , remains nonzero. As before, this is because of the presence of the symmetrizer in the expression for Φ_{RR} — there will always be at least one term that does not vanish when k particles are brought to the same position.

This *clustering* property, in the limit of the thin cylinder, implies that up to k bosons can occupy every two neighboring orbitals, therefore, the degeneracy is $k + 1$ -fold. An example of these states is given in Table 1.1. For the fermionic states ($m = 1$), the optimal situation is when clusters of k fermions are as far apart from each other as possible. Given the filling fraction at these states, $\nu = k/(k + 2)$, the lowest energy configuration corresponds to the case in which up to k electrons occupy every $k + 2$ neighboring orbitals. Therefore,

for fermions the degeneracy of the ground state is $\binom{k+2}{k}$ -fold, as shown in Table 1.1 (for a detailed discussion of the ground state degeneracy of the Read-Rezayi states see [67]).

Now consider introducing quasihole excitations to the system. As before, the wavefunction corresponding to a quasihole at position w has the form,

$$\psi_{RR,qh}(w) = \prod_i (z_i - w) \psi_{RR}, \quad (1.49)$$

where ψ_{RR} is defined in Eq. 1.47. This quasihole which carries the charge $Q = +e\nu = +ek/(km + 2)$, can further fractionalize to give rise to k new quasiholes each with charge $+e/(km + 2)$. This can be shown by taking the quasihole operator, $\prod_i (z_i - w)$, inside the symmetrizer in Eq. 1.47. The result is a wavefunction, describing k quasiholes at positions $w_1, w_2, \dots, w_k$,

$$\begin{aligned} \psi_{RR,qh}(w_1, w_2, \dots, w_k) &= \mathcal{S} \left[\prod_{i_1 \in A_1} (z_{i_1} - w_1) \prod_{i_1 \in A_2} (z_{i_2} - w_2) \dots \prod_{i_k \in A_k} (z_{i_k} - w_k) \right. \\ &\times \prod_{i_1 < j_1 \in A_1} (z_{i_1} - z_{j_1})^2 \prod_{i_2 < j_2 \in A_2} (z_{i_2} - z_{j_2})^2 \dots \prod_{i_k < j_k \in A_k} (z_{i_k} - z_{j_k})^2 \left. \right] \\ &\times \prod_{i < j} (z_i - z_j)^m. \end{aligned} \quad (1.50)$$

In general, the degeneracy of the Hilbert space in the presence of the Read-Rezayi quasiholes depends on the geometry of the surface the system is defined on. This degeneracy can be determined from what is known as the *Bratteli diagram*. We postpone the definition of the Bratteli diagram to Secs. 2.1 and 3.1 and just mention that in the limit of large n , where n is the number of quasiholes, the degeneracy of the Hilbert space grows asymptotically as d^n where $d = 2 \cos \pi/(k + 2)$. The quantity d which determines the rate of growth of the Hilbert space, is called the *quantum dimension* of the corresponding quasiholes. Note that for Laughlin states ($k = 1$) we have $d = 1$, indicating that the Hilbert space in the presence of quasiholes is nondegenerate and for the Moore-Read states ($k = 2$), $d = \sqrt{2}$, consistent with the results of Nayak and Wilczek, described above. We close this section by mentioning that for quasiholes of the Read-Rezayi state at level $k = 3$, the quantum dimension is the golden mean ($\phi = (\sqrt{5} - 1)/2$) and the dimensionality of the Hilbert space grows as Fibonacci numbers. These quasihole excitations are closely related to the so-called *Fibonacci* anyons which are the subject of Chapter 2 of this Thesis.

Table 1.1: Ground state wavefunctions, the corresponding potential, the thin cylinder limit, ground state degeneracy on a cylinder and the corresponding filling fraction. Superscripts B and F correspond to bosons and fermions respectively. L , MR and RR_k correspond to Laughlin, Moore-Read and Read-Rezayi at level k , respectively.

	Ψ_{gr}	V	Thin Cylinder	Deg.	ν
$L - B$	$\Phi_L = \prod_{i < j} (z_i - z_j)^2$	$\sum_{i < j} \delta^2(z_i - z_j)$	0101010101 1010101010	2	$\frac{1}{2}$
$L - F$	$\psi_L = \Phi_L \prod_{i < j} (z_i - z_j)$		001001001001 010010010010 100100100100	3	$\frac{1}{3}$
$MR - B$	$\Phi_{MR} = \mathcal{S}[\prod_{i < j \in A} (z_i - z_j) \prod_{i < j \in B} (z_i - z_j)^2]$	$\sum_{i < j < k} \delta^2(z_i - z_j) \delta^2(z_j - z_k)^2$	020202020202 111111111111 202020202020	3	1
$MR - F$	$\psi_{MR} = \Phi_{MR} \prod_{i < j} (z_i - z_j)$		110011001100 011001100110 001100110011 100110011001 101010101010 010101010101	6	$\frac{1}{2}$
$RR_k - B$	$\Phi_{RR} = \mathcal{S}[\prod_{i < j \in A_1} (z_i - z_j)^2 \dots \prod_{i < j \in A_k} (z_i - z_j)^2]$	$\sum_{i_1 < \dots < i_k} \delta^2(z_{i_1} - z_{i_2}) \dots \delta^2(z_{i_k} - z_{i_{k+1}})$	$0k0k0k0k0k$ $\vdots$	$k + 1$	$\frac{k}{2}$
$RR_k - F$	$\psi_{RR} = \Phi_{RR} \prod_{i < j} (z_i - z_j)$		$\underbrace{11 \dots 1}_k \dots \underbrace{10011 \dots 1}_k 00$ $\vdots$	$\binom{k+2}{k}$	$\frac{k}{k+2}$

Table 1.2: Wavefunctions in the presence of quasiholes, the thin cylinder limit and the charge of the corresponding elementary excitations. Superscripts B and F correspond to bosons and fermions respectively. L , MR and RR_k correspond to Laughlin, Moore-Read and Read-Rezayi at level k , respectively.

	Ψ_{qh}	Thin Cylinder	Q_{qh}
$L - B$	$\Phi_{L,qh} = \prod_i (z_i - w) \prod_{i < j} (z_i - z_j)^2$	101010 010101	$+e/2$
$L - F$	$\psi_{L,qh} = \Phi_{L,qh} \prod_{i < j} (z_i - z_j)$	100100 010010	$+e/3$
$MR - B$	$\Phi_{MR,qh} = \mathcal{S}[\prod_{i \in A} (z_i - w_1) \prod_{i \in B} (z_i - w_2) \prod_{i < j \in A} (z_i - z_j)^2 \prod_{i < j \in B} (z_i - z_j)^2]$	202020 111111	$+e/2$
$MR - F$	$\psi_{MR,qh} = \Phi_{MR,qh} \prod_{i < j} (z_i - z_j)$	11001100 010101	$+e/4$
$RR_k - B$	$\Phi_{RR,qh} = \mathcal{S}[\prod_{i \in A_1} (z_i - w_1) \dots \prod_{i \in A_k} (z_i - w_k) \prod_{i < j \in A_1} (z_i - z_j)^2 \dots \prod_{i < j \in A_k} (z_i - z_j)^2]$		$+e/2$
$RR_k - F$	$\psi_{RR,qh} = \Phi_{RR,qh} \prod_{i < j} (z_i - z_j)$		$+e/(k+2)$

1.4 Quantum Computing with FQH States

In Sec. 1.1, we pointed out that to perform quantum computation, we need to have access to a large Hilbert space which is isolated from the environment (the computational space), and we must be able to operate on this space fault-tolerantly, i.e. without introducing fatal errors to our computation. In Sec. 1.2.3, we showed that, in principle, it is possible to protect quantum information from interactions with the environment, by encoding it using particular global properties of certain two-dimensional quantum systems. In Secs. 1.3.2 and 1.3.3, we argued that these quantum systems may in fact, be realized as ground states of certain fractional quantum Hall states at filling fractions that are experimentally observed.

To carry out quantum computation one must braid quasiparticle excitations around each other. To calculate the exact braiding pattern that corresponds to a particular computation, one must know the exact form of the unitary operations that result from exchanging two quasiholes in a certain basis. Furthermore, we need to have a consistent set of unitary operations that describe any change of bases in the Hilbert space of our anyons. This information, can in principle, be obtained from the wavefunctions of the Read-Rezayi states in the presence of quasiholes.

The wavefunctions of the Read-Rezayi states described in Sec. 1.3.3 are constructed based on an analogy with *conformal field theory* (CFT) (see for example [23, 51] and Sec. III.D.2 of [68]). In particular, these wavefunctions are correlators of fields in the Z_k parafermion conformal field theory [69]. Though in principle it is possible to calculate the exact form of correlators of the conformal field theory, in practice it has proven to be a hard problem [70].^P Nonetheless, it has been shown that the information about braiding quasiholes (braid matrices) can be calculated using a mathematical structure known as *quantum group* [72]. The Z_k parafermion CFT, used to construct the Read-Rezayi states is closely related to the quantum group $SU(2)_k$ (read $SU(2)$ “level” k) and the braid matrices associated with $SU(2)_k$ are calculated in [73].

OK. So we have our braid matrices in hand. Can we carry out arbitrary quantum computation? Freedman, Larsen and Wang have shown that braid generators of $SU(2)_k$, for $k \geq 3$ and $k \neq 4$, can in fact, generate a *dense cover* for $SU(N)$, where N is the

^PExplicit wave functions for these states were worked out in [70], and the non-Abelian braiding properties of the corresponding quasiparticles have been verified numerically in [71].

dimensionality of the Hilbert space [25]. This means for any arbitrary operator u in $SU(N)$, there is an operator corresponding to the product of a sequence of braid generators, which is arbitrarily close to u . Therefore, in principle it is possible to approximate all operations in $SU(N)$ (i.e. any desired quantum computation) just by using braid matrices of $SU(2)_k$ (for the above mentioned values of k) and this approximation can be carried out to any desired accuracy.

To carry out topological quantum computation, we must prepare a physical system that supports non-Abelian anyons (e.g. a two-dimensional electron gas in a fractional quantum Hall state with non-Abelian excitations). Non-Abelian anyons can be created by pulling particle-hole pairs out of the vacuum (ground state of the system) and adiabatically separating them [74]. As will be described in Chapter 2, “qubits” associated with these newly created non-Abelian anyons are readily initialized to the logical state $|0\rangle$. Quantum computation is then carried out by braiding the world-lines of non-Abelian anyons around each other in the three-dimensional space-time. After the computation is complete, pairs of non-Abelian anyons are *fused* together and the result can be read out. Since the storage and manipulation of quantum information is carried out using global degrees of freedom, local interactions with the environment cannot affect the process.

There is no question that the technological difficulties for building a quantum computer are immense. On the quest for building a quantum computer, it is not at all clear which path will eventually lead to success — topological quantum computing or the proposals that rely on error correcting codes. Regardless of the outcome and whether or not we will have a topological quantum computer one day, the fact that Nature provides us with the possibility of physical fault-tolerance is amazing and worth pursuing.

1.5 Outline of The Thesis

Suppose we have a quantum algorithm, written in terms of quantum gates in the form of a quantum circuit. Also imagine we have managed to build a topological quantum computer, in which we have control over braiding non-Abelian anyons. A natural question to ask is: How would one find braiding patterns that correspond to a particular quantum algorithm? In other words, how do we translate (compile) a quantum algorithm to the machine language of braids, understandable by a topological quantum computer?

As was mentioned above, this question was addressed in [25] where the authors offered a

constructive proof for the possibility of carrying out arbitrary quantum computation using anyons of $SU(2)_k$ with $k \geq 3$ and $k \neq 4$. The main result of this Thesis is to provide efficient methods for explicitly carrying out a universal set of quantum gates for $SU(2)_k$ anyons. The reason for focusing on this type of anyons is the possibility that some of these anyons may, in fact, exist in fractional quantum Hall states that are experimentally observed.

This Thesis is organized as follows. Chapter 2 provides a quick introduction to $SU(2)_k$ theory and the properties of its non-Abelian anyons. The rest of the Chapter (except for Sec. 2.7) focuses on anyons corresponding to $k = 3$ and provides a detailed description of how qubits are encoded and how single-qubit gates can be carried out. This will include a brief review of the *Solovay-Kitaev* algorithm, which is used to systematically improve the accuracy of braids that approximate quantum gates. Then I will describe the harder problem of finding braids for two-qubit gates and present several methods for carrying out such gates.

The reason for focusing on $k = 3$ in particular is that, first, these anyons are closely related (and for purposes of quantum computing essentially equivalent) to a particularly simple anyon model known as the *Fibonacci* model. The simplicity of this model allows us to construct several different classes of two-qubit gates which will be described in detail. The second reason for paying special attention to $SU(2)_3$ model is that, in the Read-Rezayi sequence of FQH states, one the filling fractions associated with $k = 3$, is $\nu = 12/5$ which has been observed in experiments. Therefore, if the corresponding quasiparticle excitations can be produced and their motions can be controlled, the results of this Thesis can be experimentally verified (adding to the excitement).

In Chapter 3, I will consider the problem of finding braids corresponding to single-qubit gates and two-qubit gates for anyons of $SU(2)_k$ for all $k > 3$. This Chapter starts by reviewing some of the properties of $SU(2)_k$ anyons described in Chapter 2 and provides more details on their braiding and fusion properties. Then I will quickly explain how qubits can be encoded and how single-qubit gates can be carried out. The main problem addressed in this Chapter is to find efficient methods for carrying out two-qubit gates. I will explain why the methods for constructing two-qubit gates for Fibonacci anyons described in Chapter 2 cannot be used for general $SU(2)_k$ anyons when $k > 3$, and finally in Sec. 3.3.2 I will provide the details of a new method that can be used for carrying out such gates.

CHAPTER 2

Compiling Braids for Fibonacci Anyons

The main purpose of this Chapter is to give an efficient method for determining braids which can be used to carry out a universal set of a quantum gates (i.e. single-qubit rotations and controlled-NOT gates) on encoded qubits for quasiparticles of $SU(2)_k$ at $k = 3$, thought to be physically relevant for the experimentally observed [43] $\nu = 12/5$ fractional quantum Hall state [51, 52] ($\nu = 12/5$ corresponds to $\nu = 2/5$ in the second Landau level, and this is the particle-hole conjugate of $\nu = 3/5$ corresponding to $k = 3$). We refer to the process of finding such braids as “topological quantum compiling” since these braids can then be used to translate a given quantum algorithm into the “machine code” of a topological quantum computer. This is analogous to the action of an ordinary compiler which translates instructions written in a high level programming language into the machine code of a classical computer.

It should be noted that the proof of universality for $SU(2)_3$ quasiparticles is a constructive one [25, 26], and therefore, as a matter of principle, it provides a prescription for compiling quantum gates into braids. However, in practice, for two-qubit gates (such as controlled-NOT) this prescription, if followed straightforwardly, is prohibitively difficult to carry out, primarily because it involves searching the space of braids with six or more strands. We address this difficulty by dividing our two-qubit gate constructions into a series of smaller constructions, each of which only involves searching the space of three-stranded braids (three-braids). The required three-braids then *can* be found efficiently and used to construct the desired two-qubit gates. This “divide and conquer” approach does not, in general, yield the most accurate braid of a given length which approximates a desired quantum gate. However, we believe that it does yield the most accurate (or at least among the most accurate) braids which can be obtained for a given fixed amount of classical computing power.

2.1 $SU(2)_k$ Particles: Fusion Rules and Hilbert Space

This Thesis deals entirely with a particular class of non-Abelian particles which correspond to quasiparticles of the Read-Rezayi sequence of FQH states. As was mentioned in Sec. 1.4, it is convenient to describe certain properties of these non-Abelian particles in the language of *quantum groups* by the so-called $SU(2)_k$ theory (see e.g. [72]).

Quantum groups are mathematical objects which first arose in the study of integrable models. The subject has a dense mathematical literature, but for our purposes, it is not necessary to understand quantum groups in all their mathematical complexity to apply them to topological quantum computation. The essential fact one needs to know about the quantum groups that describe $SU(2)_k$ particles, is that they can be viewed as a “deformed” version of the representation theory of ordinary $SU(2)$, i.e. the theory of ordinary spin. This analogy is useful for physicists, because much of the intuition for thinking about ordinary spin can be carried over to the quantum group case, and I will be emphasizing this analogy whenever possible. Note that in what follows I will be simply stating the properties of $SU(2)_k$ particles and not deriving them from scratch. The reader is referred to the literature (for example Sec. 3.2 in [73], or [72]) for more rigorous treatments of the mathematics of quantum groups.

In the quantum group description of an $SU(2)_k$ theory, each quasiparticle has a half-integer q -deformed spin (q -spin) quantum number. Just as for ordinary spin, q -spin can take on half-integer values, but unlike ordinary spin, there is a *maximum* possible spin of $k/2$. Thus the allowed q -spin values, s , for $SU(2)_k$ particles are

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots, \frac{k}{2}. \quad (2.1)$$

Just as for ordinary spin, there are rules for combining q -spin known as fusion rules. The fusion rules for the $SU(2)_k$ theory are similar to the usual triangle rule for adding ordinary spin, except that they are truncated so that there are no states with total q -spin $> k/2$. Specifically, the fusion rules for the level k theory are [72],

$$\begin{aligned} s_1 \otimes s_2 = & |s_1 - s_2| \oplus |s_1 - s_2| + 1 \oplus \dots \\ & \dots \oplus \min(s_1 + s_2, k - s_1 - s_2). \end{aligned} \quad (2.2)$$

Note that in the quantum group description of non-Abelian anyons, states are distinguished only by their total q -spin quantum numbers. The q -deformed analogs of the S_z quantum

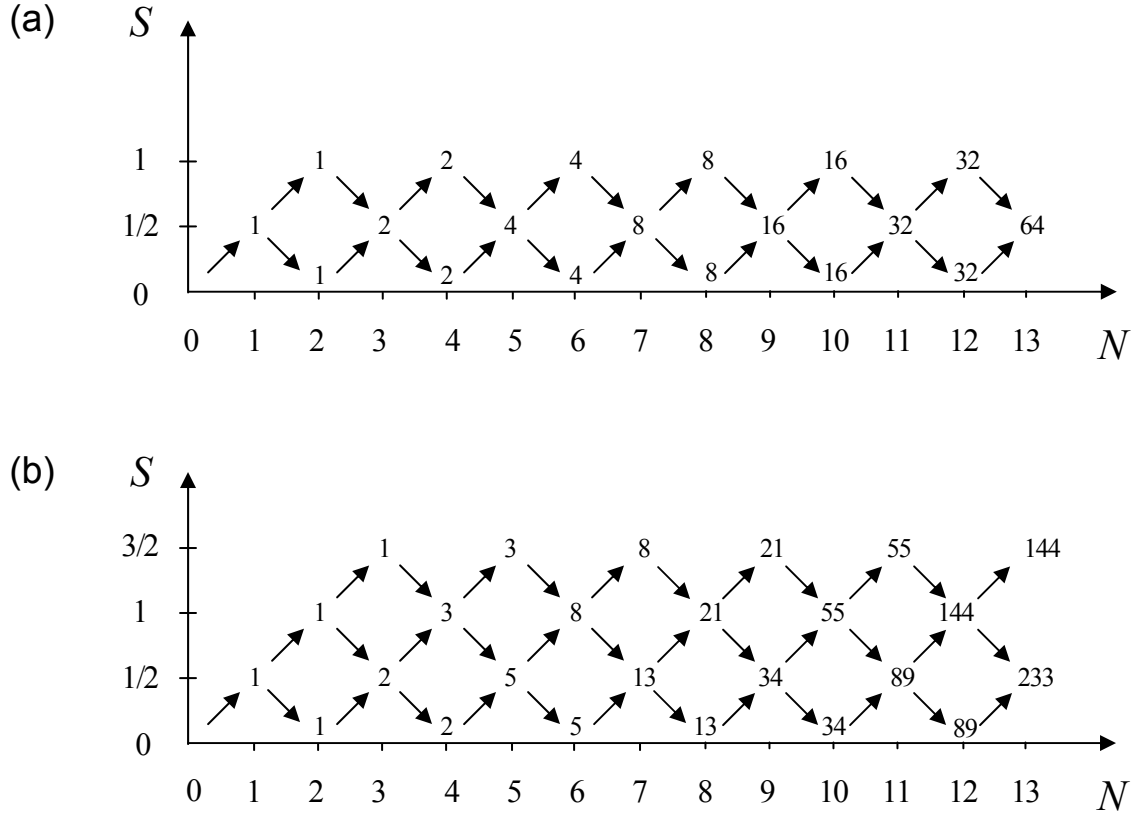


Figure 2.1: Bratteli diagrams for $SU(2)_k$ for (a) $k = 2$ and (b) $k = 3$. Here N is the number of q -spin $1/2$ quasiparticles and S is the total q -spin of those quasiparticles. The number at a given (N, S) vertex of each diagram indicates the number of paths to that vertex starting from the $(0, 0)$ point. This number gives the dimensionality of the Hilbert space of N q -spin $1/2$ quasiparticles with total q -spin S .

numbers are physically irrelevant — there is no degeneracy associated with them, and they play no role in any computation involving braiding [73]. The situation is somewhat analogous to that of a collection of ordinary spin- $1/2$ particles in which the only allowed operations, including measurement, are rotationally invariant and hence independent of S_z , as is the case in exchange-based quantum computation [18].

The fusion rules of the $SU(2)_k$ theory fix the structure of the Hilbert space of the system. For a collection of quasiparticles with q -spin $1/2$, a useful way to visualize this Hilbert space is in terms of its so-called Bratteli diagram. This diagram shows the different fusion paths

for N q -spin $1/2$ quasiparticles in which these quasiparticles are fused, one at a time, going from left to right in the diagram. Bratteli diagrams for the cases $k = 2$ and $k = 3$ are shown in Fig. 2.1.

The dimensionality of the Hilbert space for N q -spin $1/2$ quasiparticles with total q -spin S can be determined by counting the number of paths in the Bratteli diagram from the origin to the point (N, S) . The results of this path counting are also shown in Fig. 2.1, where one can see the well-known $2^{N/2-1}$ Hilbert space degeneracy for the $k = 2$ (Moore-Read) case [23, 65], and the Fibonacci degeneracy for the $k = 3$ case [51].

2.2 $SU(2)_3$ and Fibonacci Anyons

In this Chapter we will focus on the $k = 3$ case, which is the lowest k value for which $SU(2)_k$ non-Abelian anyons are universal for quantum computation [25, 26]. Before proceeding, it is convenient to introduce an important property of the $SU(2)_3$ theory, namely that the braiding properties of q -spin $1/2$ quasiparticles are the same as those with q -spin 1 (up to an overall Abelian phase which is irrelevant for topological quantum computation). This is a useful observation because the theory of q -spin 1 quasiparticles in $SU(2)_3$ is equivalent to $SO(3)_3$, a theory also known as the Fibonacci anyon theory [75, 76] — a particularly simple theory with only two possible values of q -spin, 0 and 1 , for which the fusion rules are

$$0 \otimes 0 = 0, \quad 0 \otimes 1 = 1 \otimes 0 = 1, \quad 1 \otimes 1 = 0 \oplus 1. \quad (2.3)$$

Here we give a rough proof of this equivalence. This proof is based on the fact that for $k = 3$ the fusion rules involving q -spin $3/2$ quasiparticles take the following simple form

$$\frac{3}{2} \otimes s = \frac{3}{2} - s. \quad (2.4)$$

The key observation is that since for $k = 3$ the highest possible q -spin is $3/2$, when fusing a q -spin- $3/2$ object with any other object (here we use the term object to describe either a single quasiparticle or a group of quasiparticles viewed as a single composite entity), the Hilbert space dimensionality does not grow. This implies that moving a q -spin- $3/2$ object around other objects can, at most, produce an overall Abelian phase factor. While this phase factor may be important physically, particularly in determining the outcome of interference experiments involving non-Abelian quasiparticles [77, 78, 79, 80, 81], it is irrelevant for

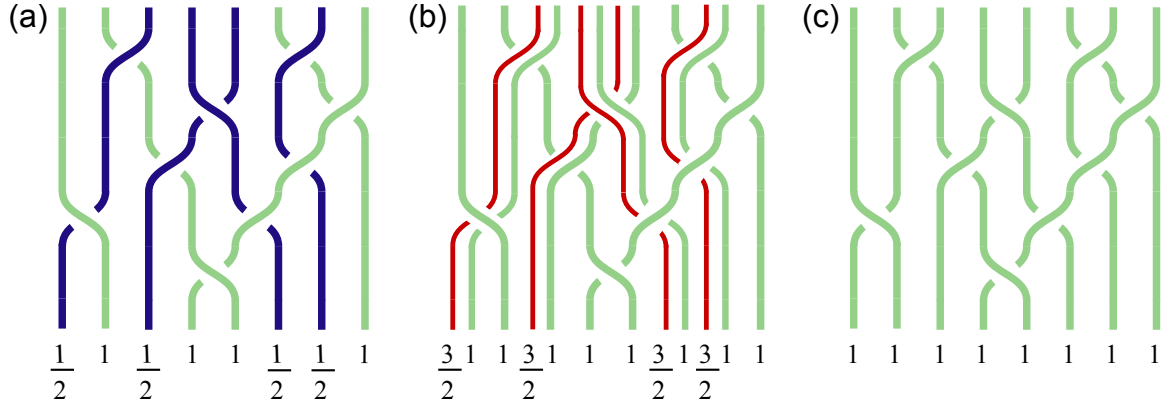


Figure 2.2: Graphical proof of the equivalence of braiding q -spin-1/2 and q -spin-1 objects for $SU(2)_3$. Part (a) shows a braiding pattern for a collection of objects, some having q -spin 1/2 and some having q -spin 1. Part (b) shows the same braiding pattern but with the q -spin-1/2 objects represented by q -spin 1 objects fused with q -spin-3/2 objects, which, for $SU(2)_3$, has a unique fusion channel. Finally, part (c) shows the same braid with the q -spin-3/2 objects removed. Because these q -spin-3/2 objects are effectively Abelian for $SU(2)_3$, removing them from the braid will only result in an overall phase factor which will be irrelevant for quantum computing.

quantum computing, and thus does not matter when determining braids which correspond to a given computation. Because 2.4 implies that a q -spin-1/2 object can be viewed as the result of fusing a q -spin-1 object with a q -spin-3/2 object, it follows that the braid matrices for q -spin-1/2 objects are the same as that for q -spin-1 objects up to an overall phase (as can be explicitly checked).

In fact, based on this argument we can make a stronger statement. Imagine a collection of $SU(2)_3$ objects which each have either q -spin 1 or q -spin 1/2. It is then possible to carry out topological quantum computation, *even if we do not know which objects have q -spin 1 and which have q -spin 1/2*. The proof is illustrated in Fig. 2.2. Figure 2.2 (a) shows a braiding pattern for a collection of objects, some of which have q -spin 1/2 and some of which have q -spin 1. Fig. 2.2 (b) then shows the same braiding pattern, but now all objects with q -spin 1/2 are represented by objects with q -spin 1 fused to objects with q -spin 3/2. Because, as noted above, the q -spin 3/2 objects have trivial (Abelian) braiding properties, the unitary transformation produced by this braid is the same, up to an overall Abelian

phase, as that produced by braiding nothing but q -spin 1 objects, as shown in Fig. 2.2 (c). It follows that provided one can measure whether the total q -spin of some object belongs to the class $1 \equiv \{1, 1/2\}$ or the class $0 \equiv \{0, 3/2\}$ — something which should, in principle, be possible by performing interference experiments as described in Refs. [80] and [82] — then quantum computation is possible, even if we do not know which objects have q -spin $1/2$ and which have q -spin 1.

2.3 Fibonacci Anyon Basics

Having reduced the problem of compiling braids for $SU(2)_3$ to compiling braids for $SO(3)_3$, i.e. Fibonacci anyons, it is useful for what follows to give more details about the mathematical structure associated with these quasiparticles. For an excellent review of this topic see [75], and for the mathematics of non-Abelian particles in general see [24].

Note that for the rest of this Chapter, except for Sec. 2.7, it should be understood that each quasiparticle is a q -spin 1 Fibonacci anyon. It should also be understood that from the point of view of their non-Abelian properties quasihole excitations are also q -spin 1 Fibonacci anyons, even though they have opposite electric charge and give opposite Abelian phase factors when braided. Because it is the non-Abelian properties which are relevant for topological quantum computation, for our purposes quasiparticles and quasiholes can be viewed as identical non-Abelian particles. Unless it is important to distinguish between the two (as when we discuss creating and fusing quasiparticles and quasiholes in Sec. 2.4) we will simply use the terms quasiparticle or Fibonacci anyon to refer to either excitation.

Figure 2.3 establishes some of the notation for representing Fibonacci anyons which will be used in the rest of this Chapter. This figure shows $SU(2)_3$ Bratteli diagrams in which the q -spin axis is labeled both by the $SU(2)_3$ q -spin quantum numbers and, in boldface, the corresponding Fibonacci q -spin quantum numbers, i.e. 0 for $\{0, 3/2\}$ and 1 for $\{1/2, 1\}$. In Fig. 2.3 (a) Bratteli diagrams showing fusion paths corresponding to two basis states spanning the two-dimensional Hilbert space of two Fibonacci anyons are shown. Beneath each Bratteli diagram an alternate representation of the corresponding state is also shown. In this representation dots correspond to Fibonacci anyons and ovals enclose collections of Fibonacci anyons which are in q -spin eigenstates whenever the oval is labeled by a total q -spin quantum number. (Note: If the oval is not labeled, it should be understood that the enclosed quasiparticles may not be in a q -spin eigenstate).

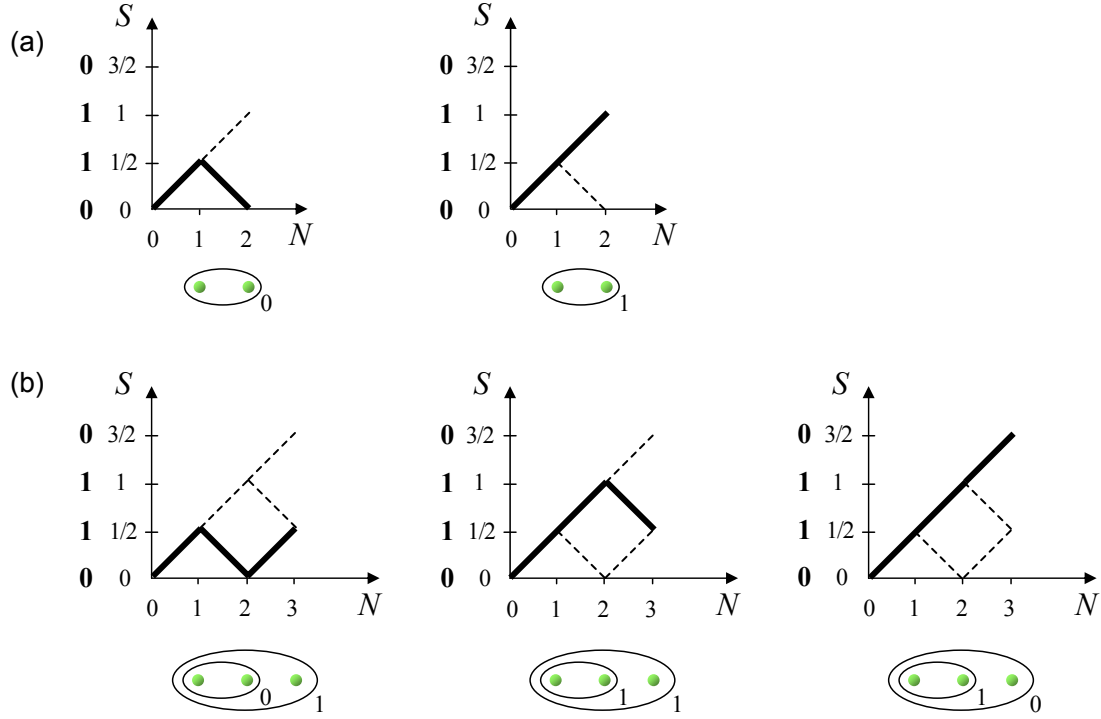


Figure 2.3: Basis states for the Hilbert space of (a) two and (b) three Fibonacci anyons. $SU(2)_3$ Bratteli diagrams showing fusion paths corresponding to the basis states for the Hilbert space of two and three q -spin $1/2$ quasiparticles are shown. The q -spin axes on these diagrams are labeled both by the $SU(2)_3$ q -spin quantum numbers $0, 1/2, 1$ and $3/2$ and, to the left of these in bold, the corresponding Fibonacci q -spin quantum numbers $0 \equiv \{0, 3/2\}$ and $1 \equiv \{1/2, 1\}$. Beneath each Bratteli diagram the same state is represented using a notation in which dots correspond to Fibonacci anyons, and groups of Fibonacci anyons enclosed in ovals labeled by q -spin quantum numbers are in the corresponding q -spin eigenstates.

In the text throughout this Chapter, we will use the notation $\bullet$ to represent a Fibonacci anyon, and the ovals will be represented by parentheses. In this notation, the two states shown in Fig. 2.3 (a) are denoted $(\bullet, \bullet)_0$ and $(\bullet, \bullet)_1$.

Fig. 2.3 (b) shows Bratteli diagram, again with both $SU(2)_3$ and Fibonacci quantum numbers, with fusion paths which this time correspond to three basis states of the three-dimensional Hilbert space of three Fibonacci anyons. Beneath these diagrams the “oval” representations of these three states are also shown, which in the text will be represented

$((\bullet, \bullet)_0, \bullet)_1$, $((\bullet, \bullet)_1, \bullet)_1$ and $((\bullet, \bullet)_1, \bullet)_0$.

In addition to fusion rules, all theories of non-Abelian anyons possess additional mathematical structure which allows one to calculate the result of any braiding operation. This structure is characterized by the F (fusion) and R (rotation) matrices [23, 75, 24]. These matrices have a particularly simple form for the case of Fibonacci anyons (due to the fact that there are only two possible values of q -spin). In Chapter 3, when we discuss topological quantum computation using $SU(2)_k$ particles with $k > 3$ we will see that the F and R matrices become more complex, or at least have more indices, but their basic mathematical properties can be fully understood for the Fibonacci anyon case.

To define the F matrix, note that the Hilbert space of three Fibonacci anyons is spanned both by the three states labeled $((\bullet, \bullet)_a, \bullet)_c$, and the three states labeled $(\bullet, (\bullet, \bullet)_b)_c$. The F matrix is the unitary transformation which maps one of these bases to the other,

$$(\bullet, (\bullet, \bullet)_a)_c = \sum_b F_{ab}^c ((\bullet, \bullet)_b, \bullet)_c, \quad (2.5)$$

and has the form

$$F = \left(\begin{array}{cc|c} \tau & \sqrt{\tau} & \\ \sqrt{\tau} & -\tau & \\ \hline & & 1 \end{array} \right), \quad (2.6)$$

where $\tau = (\sqrt{5} - 1)/2$ is the inverse of the golden mean. In this matrix the upper left 2×2 block, F_{ab}^1 , acts on the two-dimensional total q -spin 1 sector of the three-quasiparticle Hilbert space and the lower right matrix element, $F_{11}^0 = 1$, acts on the unique total q -spin 0 state. Note that this F matrix can be applied to any three objects which each have q -spin 1, where each object can consist of more than one Fibonacci anyon. Furthermore, if one considers three objects for which one or more of the objects has q -spin 0, then the state of these objects is uniquely determined by the total q -spin of all three, and in this case the F matrix is trivially the identity. Thus, for the case of Fibonacci anyons, the matrix 2.6 is all that is needed to make arbitrary basis changes for any number of Fibonacci anyons.

The R matrix gives the phase factor produced when two Fibonacci anyons are moved around one another with a certain sense. One can think of these phase factors as the q -deformed versions of the -1 or $+1$ phase factors one obtains when interchanging two ordinary spin-1/2 quasiparticles when they are in a singlet or triplet state, respectively. This phase

factor depends on the overall q -spin of the two quasiparticles involved in the exchange, so for Fibonacci anyons there are two such phase factors which are summarized in the R matrix,

$$R = \begin{pmatrix} e^{-i4\pi/5} & 0 \\ 0 & e^{i3\pi/5} \end{pmatrix}. \quad (2.7)$$

Here the upper left and lower right matrix elements are, respectively, the phase factor that two Fibonacci anyons acquire if they are interchanged in a clockwise sense when they have total q -spin 0 or q -spin 1. Again, this matrix also applies if we exchange two objects that both have total q -spin 1, even if these objects consist of more than one Fibonacci anyon. And if one or both objects has q -spin 0, the result of this interchange is the identity. Again we emphasize that in the $k = 3$ Read-Rezayi state, there will be additional Abelian phases present, which may have physical consequences for some experiments, but which will be irrelevant for topological quantum computation.

Typically the sequence of F and R matrices used to compute the unitary operation produced by a given braid is not unique. To guarantee that the result of any such computation is independent of this sequence, the F and R matrices must satisfy certain consistency conditions. These consistency conditions, the so-called pentagon and hexagon equations [75, 24, 83], are highly restrictive, and, in fact, for the case of Fibonacci anyons essentially fix the F and R matrices to have the forms given above (up to a choice of chirality, and Abelian phase factors which are again irrelevant to our purposes here) [75].

Finally, we point out an obvious, but important, consequence of the structure of the F and R matrices. When interchanging any two quasiparticles which are part of a larger set of quasiparticles with a well-defined total q -spin quantum number, this total q -spin quantum number will not change.

2.4 Qubit Encoding and General Computation Scheme

Before proceeding, it will be useful to have a specific scheme in mind for how one might actually carry out topological quantum computation with Fibonacci anyons. Here we follow the scheme outlined in [74], which, for completeness, we briefly review below.

The computer can be initialized by pulling quasiparticle-quasihole pairs out of the “vacuum”, (by vacuum we mean the ground state of the $k = 3$ Read-Rezayi state or any other state which supports Fibonacci anyon excitations). Each such pair will consist of two

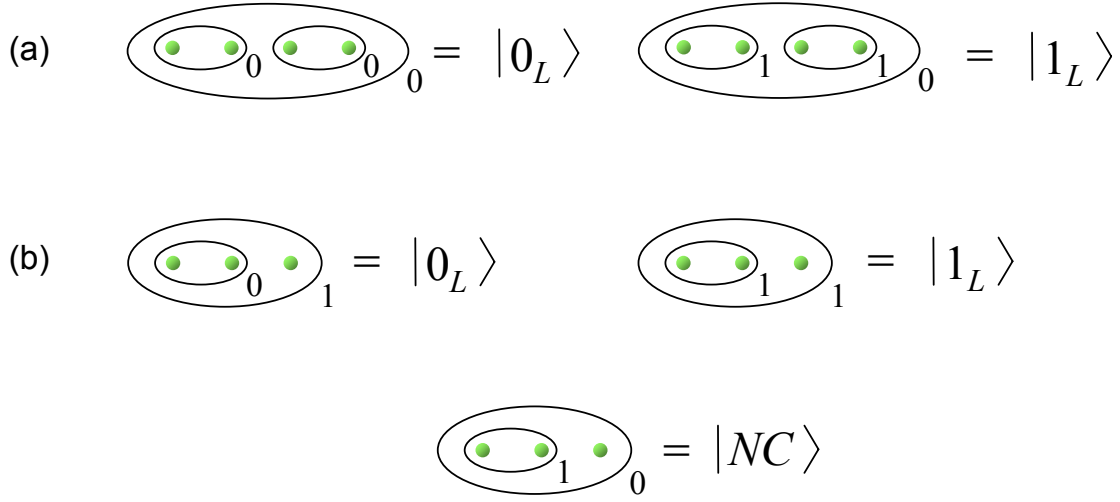


Figure 2.4: (a) Four-quasiparticle and (b) three-quasiparticle qubit encodings for Fibonacci anyons. Part (a) shows two states which span the Hilbert space of four quasiparticles with total q -spin 0 which can be used as the logical $|0_L\rangle$ and $|1_L\rangle$ states of a qubit. Part (b) shows two states spanning the Hilbert space of three quasiparticles with total q -spin 1 which can also be used as logical qubit states $|0_L\rangle$ and $|1_L\rangle$. This three-quasiparticle qubit can be obtained by removing the rightmost quasiparticle from the two states shown in (a). The third state shown in Part (b), labeled $|NC\rangle$ for “noncomputational”, is the unique state of three quasiparticles which has total q -spin 0.

q -spin 1 excitations in a state with total q -spin 0, i.e. the state $(\bullet, \bullet)_0$. In principle, this pair can also exist in a state with total q -spin 1, provided there are other quasiparticles present to ensure the total q -spin of the system is 0, so one can imagine using this pair as a qubit. However, it is impossible to carry out arbitrary single-qubit operations by braiding only the two quasiparticles forming such a qubit — this braiding never changes the total q -spin of the pair, and so only generates rotations about the $\hat{z}$ axis in the qubit space.

For this reason it is convenient to encode qubits using more than two Fibonacci anyons. Thus, to create a qubit, *two* quasiparticle-quasihole pairs can be pulled out of the vacuum. The resulting state is then $((\bullet, \bullet)_0, (\bullet, \bullet)_0)_0$ which again has total q -spin 0. The Hilbert space of four Fibonacci anyons with total q -spin 0 is two dimensional, with basis states, which we can take as logical qubit states, $|0_L\rangle = ((\bullet, \bullet)_0, (\bullet, \bullet)_0)_0$ and $|1_L\rangle = ((\bullet, \bullet)_1, (\bullet, \bullet)_1)_0$, (see Fig 2.4 (a)). The state of such a four-quasiparticle qubit is determined by the total q -spin of

either the rightmost or leftmost pair of quasiparticles. Note that the fusion rules (2.3) imply that the total q -spin of these two pairs must be the same because the total q -spin of all four quasiparticles is 0.

For this encoding, in addition to the two-dimensional computational qubit space of four quasiparticles with total q -spin 0, there is a three-dimensional *noncomputational* Hilbert space of states with total q -spin 1 spanned by the states $((\bullet, \bullet)_0, (\bullet, \bullet)_1)_1$, $((\bullet, \bullet)_1, (\bullet, \bullet)_0)_1$ and $((\bullet, \bullet)_1, (\bullet, \bullet)_1)_1$. When carrying out topological quantum computation it is crucial to avoid transitions into this noncomputational space.

Fortunately, single-qubit rotations can be carried out by braiding quasiparticles within a given qubit and, as discussed in Sec. 2.3, such operations will not change the total q -spin of the four quasiparticles involved. Single-qubit operations can therefore be carried out without any undesirable transitions out of the encoded computational qubit space.

Two-qubit gates, however, will require braiding quasiparticles from different qubits around one another. This will in general lead to transitions out of the encoded qubit space. Nevertheless, given the so-called "density" result of [26] it is known that, as a matter of principle, one can always find two-qubit braiding patterns which will entangle the two qubits, and also stay within the computational space to whatever accuracy is required for a given computation. The main purpose of this Chapter is to show how such braiding patterns can be efficiently found.

Note that the action of braiding the two leftmost quasiparticles in a four-quasiparticle qubit (referring to Fig. 2.4 (a)) is equivalent to that of braiding the two rightmost quasiparticles with the same sense. This is because as long as we are in the computational qubit space both the leftmost and rightmost quasiparticle pairs must have the same total q -spin, and so interchanging either pair will result in the same phase factor from the R matrix. It is therefore not necessary to braid all four quasiparticles to carry out single-qubit rotations — one need only braid three.

In fact, one may consider qubits encoded using only three quasiparticles with total q -spin 1, as originally proposed in [25]. Such qubits can be initialized by first creating a four-quasiparticle qubit in the state $|0_L\rangle$, as outlined above, and then simply removing one of the quasiparticles. In this three-quasiparticle encoding, shown in Fig. 2.4 (b), the logical qubit states can be taken to be $|0_L\rangle = ((\bullet, \bullet)_0, \bullet)_1$ and $|1_L\rangle = ((\bullet, \bullet)_1, \bullet)_1$. For this encoding there is just a single noncomputational state $|NC\rangle = ((\bullet, \bullet)_1, \bullet)_0$, also shown in Fig. 2.4 (b). As for



Figure 2.5: Space-time paths corresponding to the initialization, manipulation through braiding, and measurement of an encoded qubit. Two quasiparticle-quasihole pairs are pulled out of the vacuum, with each pair having total q -spin 0. The resulting state corresponds to a four-quasiparticle qubit in the state $|0_L\rangle$ (see Fig. 2.4 (a)). After some braiding, the qubit is measured by trying to fuse the bottommost pair (in this case a quasiparticle-quasihole pair). If they fuse back into the vacuum the result of the measurement is $|0_L\rangle$, otherwise it is $|1_L\rangle$. Because only the three lower quasiparticles are braided, the encoded qubit can also be viewed as a three-quasiparticle qubit (see Fig. 2.4 (b)) which is initialized in the state $|0_L\rangle$.

the four-quasiparticle qubit, when carrying out single-qubit rotations by braiding within a three-quasiparticle qubit the total q -spin of the qubit, in this case 1, remains unchanged and there are no transitions from the computational qubit space into the state $|NC\rangle$. However, just as for four-quasiparticle qubits, when carrying out two-qubit gates these transitions will in general occur and we must work hard to avoid them. Henceforth we will refer to these unwanted transitions as leakage errors.

Note that, because each three-quasiparticle qubit has total q -spin 1, when more than one of these qubits is present the state of the system is not entirely characterized by the “internal” q -spin quantum numbers which determine the computational qubit states. It is also necessary to specify the state of what we will refer to as the “external fusion space” — the Hilbert space associated with fusing the total q -spin 1 quantum numbers of each qubit. When compiling braids for three-quasiparticle qubits it is crucial that the operations on the computational qubit space not depend on the state of this external fusion space — if they did, these two spaces would become entangled with one another leading to errors. Fortunately, we will see that it is indeed possible to find braids which do not lead to such errors.

For the rest of this Chapter (except Secs. 2.6.1 and 2.7) we will use this three-quasiparticle qubit encoding. It should be noted that any braid which carries out a desired operation on

the computational space for three-quasiparticle qubits will carry out the same operation on the computational space of four-quasiparticle qubits, with one quasiparticle in each qubit acting as a spectator. The braids we find here can therefore be used for either encoding.

We can now describe how topological quantum computation might actually proceed, again following [74]. A quantum circuit consisting of a sequence of one- and two-qubit gates which carries out a particular quantum algorithm would first be translated (or “compiled”) into a braid by compiling each individual gate to whatever accuracy is required. Qubits would then be initialized by pulling quasiparticle-quasihole pairs out of the “vacuum”. These localized excitations would then be adiabatically dragged around one another so that their world-lines trace out a braid in three-dimensional space-time which is topologically equivalent to the braid compiled from the quantum algorithm. Finally, individual qubits would be measured by trying to fuse either the two rightmost or two leftmost excitations within them (referring to Fig. 2.4 (a)) for four-quasiparticle qubits, or just the two leftmost excitations (referring to Fig. 2.4 (b)) for three-quasiparticle qubits. If this pair of excitations consists of a quasiparticle and a quasihole (and it will always be possible to arrange this), then, if the total q -spin of the pair is 0, it will be possible for them to fuse back into the “vacuum”. However, if the total q -spin is 1 this will not be possible. The resulting difference in the charge distribution of the final state would then be measured to determine if the qubit was in the state $|0_L\rangle$ or $|1_L\rangle$. Alternatively, as already mentioned in Sec. 2.1, interference experiments [80, 82] could be used to initialize and read out encoded qubits.

As a simple illustration, Fig. 2.5 shows a “computation” in which a four-quasiparticle qubit (which can also be viewed as a three-quasiparticle qubit if the top quasiparticle is ignored) is initialized by pulling quasiparticle-quasihole pairs out of the vacuum, a single-qubit operation is carried out by braiding within the qubit, and the final state of the qubit is measured by fusing a quasiparticle and quasihole together and observing the outcome.

2.5 Compiling Three-Braids and Single-Qubit Gates

We now focus on the problem of finding braids for three Fibonacci anyons (three-braids) which approximate any allowed unitary transformation on the Hilbert space of these quasiparticles. This is important not only because it allows one to find braids which carry out arbitrary single-qubit rotations [25], but also because, as will be shown in Sec. 2.6, it is possible to reduce the problem of constructing braids which carry out two-qubit gates to

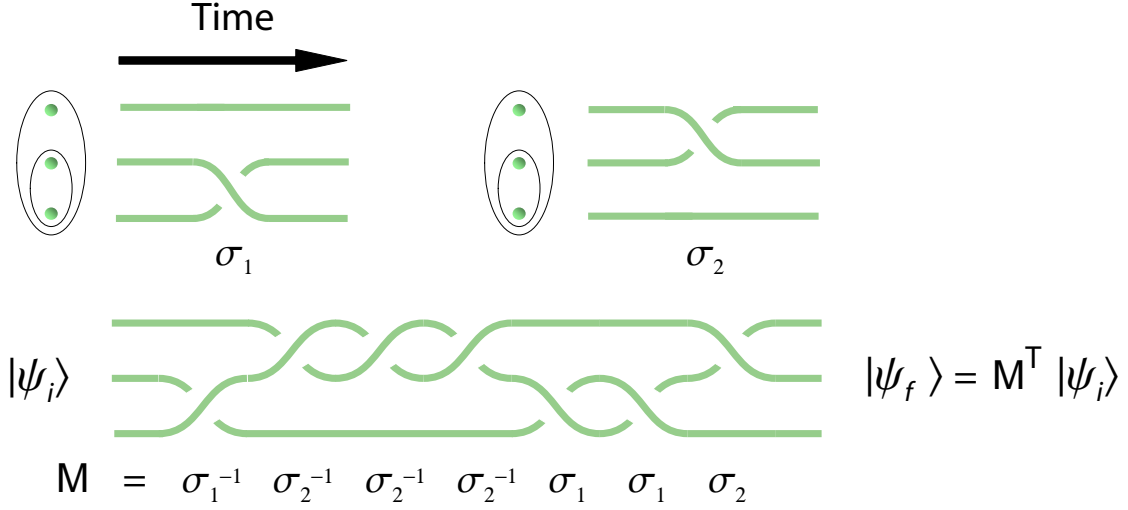


Figure 2.6: Elementary three-braids and the decomposition of a general three-braid into a series of elementary braids. The unitary operation produced by this braid is computed by multiplying the corresponding sequence of elementary braid matrices, σ_1 and σ_2 (see text) and their inverses, as shown. Here the (unlabeled) ovals represent a particular basis choice for the three-quasiparticle Hilbert space, consistent with that used in the text. In this and all subsequent figures which show braids, quasiparticles are aligned vertically, and we adopt the convention that reading from bottom to top in the figures corresponds to reading from left to right in expressions such as $((\bullet, \bullet)_a, \bullet)_c$ in the text. It should be noted that these figures are only meant to represent the topology of a given braid. In any actual implementation of topological quantum computation, quasiparticles will certainly not be arranged in a straight line, and they will have to be kept sufficiently far apart while being braided to avoid lifting the topological degeneracy.

that of finding a series of three-braids approximating specific operations.

2.5.1 Elementary Braid Matrices

Using the F and R matrices, it is straightforward to determine the elementary braiding matrices that act on the three-dimensional Hilbert space of three Fibonacci anyons. If, as in Fig. 2.6, we take the basis states for the three-quasiparticle Hilbert space to be the states labeled $((\bullet, \bullet)_a, \bullet)_c$ then, in the $ac = \{01, 11, 10\}$ basis, the matrix σ_1 corresponding to a clockwise interchange of the two bottommost quasiparticles in the figure (or leftmost in the

$((\bullet, \bullet)_a, \bullet)_c$ representation) is

$$\sigma_1 = \left(\begin{array}{cc|c} e^{-i4\pi/5} & 0 & \\ 0 & e^{i3\pi/5} & \\ \hline & & e^{i3\pi/5} \end{array} \right), \quad (2.8)$$

where the upper left 2×2 block acts on the total q -spin 1 sector ($|0\rangle_L$ and $|1\rangle_L$) of the three quasiparticles, and the lower right matrix element is a phase factor acquired by the q -spin 0 state ($|NC\rangle$). This matrix is easily read off from the R matrix, since the total q -spin of the two quasiparticles being exchanged is well defined in this basis.

To find the matrix σ_2 corresponding to a clockwise interchange of the two topmost (or rightmost in the $((\bullet, \bullet)_a, \bullet)_c$ representation) quasiparticles, we must first use the F matrix to change bases to one in which the total q -spin of these quasiparticles is well defined. In this basis, the braiding matrix is simply σ_1 , and so, after changing back to the original basis, we find

$$\sigma_2 = F^{-1} \sigma_1 F = \left(\begin{array}{cc|c} -\tau e^{-i\pi/5} & \sqrt{\tau} e^{-i3\pi/5} & \\ \sqrt{\tau} e^{-i3\pi/5} & -\tau & \\ \hline & & e^{i3\pi/5} \end{array} \right). \quad (2.9)$$

The unitary transformation corresponding to a given three-braid can now be computed by representing it as a sequence of elementary braid operations and multiplying the corresponding sequence of σ_1 and σ_2 matrices and their inverses, as shown in Fig. 2.6.

If we are only concerned with single-qubit rotations, then we only care about the action of these matrices on the encoded qubit space with total q -spin 1, and not the total q -spin 0 sector corresponding to the noncomputational state. However, in our two-qubit gate constructions, various three-braids will be embedded into the braiding patterns of six quasiparticles, and in this case the action on the full three-dimensional Hilbert space does matter.

To understand this action note that σ_1 can be written

$$\sigma_1 = \left(\begin{array}{c|c} \pm e^{-i\pi/10} \begin{pmatrix} \pm e^{-i7\pi/10} & 0 \\ 0 & \pm e^{i7\pi/10} \end{pmatrix} & \\ \hline & e^{i3\pi/5} \end{array} \right), \quad (2.10)$$

where the upper 2×2 block acting on the total q -spin 1 sector is an $SU(2)$ matrix, (i.e., a 2×2 unitary matrix with determinant 1), multiplied by a phase factor of either $+e^{-i\pi/10}$ or $-e^{-i\pi/10}$, and the lower right matrix element, $e^{i3\pi/5}$, is the phase acquired by the total q -spin 0 state. The phase factor pulled out of the upper 2×2 block is only defined up to ± 1 because any $SU(2)$ matrix multiplied by -1 is also an $SU(2)$ matrix.

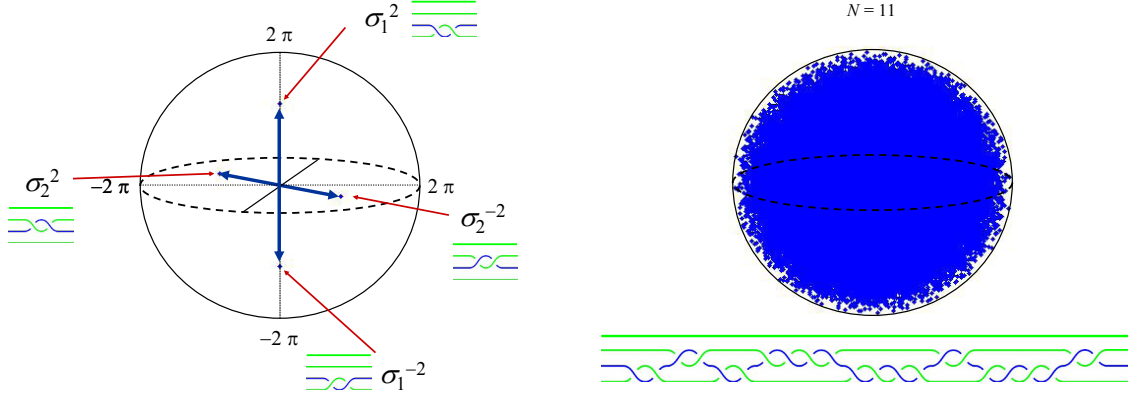


Figure 2.7: Left: Rotations corresponding to elementary braid operations. Note that since we are interested in weaves, elementary braid operations correspond to taking one particle (shown in blue) one complete round around another particle. Right: All possible rotations corresponding to braids of length $L = 22$ and a representative braid of this length.

From 2.9 it follows that σ_2 can be written in a similar fashion, with the same phase factors. Each clockwise braiding operation then corresponds to applying an $SU(2)$ operation multiplied by a phase factor of $\pm e^{-i\pi/10}$ to the q -spin 1 sector, while at the same time multiplying the q -spin 0 sector by a phase factor of $e^{i3\pi/5}$. Likewise, each counterclockwise braiding operation corresponds to applying an $SU(2)$ operation multiplied by a phase factor of $\pm e^{+i\pi/10}$ to the q -spin 1 sector and a phase factor of $e^{-i3\pi/5}$ to the q -spin 0 sector.

We define the winding, $W(B)$, of a given three-braid B , to be the total number of clockwise interchanges minus the total number of counterclockwise interchanges. It then follows that the unitary operation corresponding to an arbitrary braid B can always be expressed

$$U(B) = \left(\frac{\pm e^{-iW(B)\pi/10} [SU(2)]}{e^{i3W(B)\pi/5}} \right), \quad (2.11)$$

where $[SU(2)]$ indicates an $SU(2)$ matrix. Thus, for a given three-braid, the phase relation between the total q -spin 1 and total q -spin 0 sectors of the corresponding unitary operation is determined by the winding of the braid. We will refer to 2.11 often in what follows. It tells us precisely what unitary operations can be approximated by three-braids, and places useful restrictions on their winding.

2.5.2 Weaving and Brute Force Search

At this point it is convenient to restrict ourselves to a subclass of braids which we will refer to as weaves. A weave is any braid which is topologically equivalent to the space-time paths of some number of quasiparticles in which only a single quasiparticle moves. It was shown in [84] that this restricted class of braids is universal for quantum computation, provided the unitary representation of the braid group is dense in the space of all unitary transformations on the relevant Hilbert space, which is the case for Fibonacci anyons.

Following [84] we will borrow some weaving terminology and refer to the mobile quasiparticle (or collection of quasiparticles) as the “weft” quasiparticle(s) and the static quasiparticles as the “warp” quasiparticles.

One reason for focusing on weaves is that weaving will likely be easier to accomplish technologically than general braiding. This is true even if the full computation involves not just weaving a single quasiparticle, as was proposed in [84], but possibly weaving several quasiparticles at the same time in different regions of the computer — carrying out quantum gates on different qubits in parallel.

Considering weaves has the added (and more immediate) benefit of simplifying the problem of numerically searching for three-braids which approximate desired gates. For the full braid group, even on just three strands, there is a great deal of redundancy since braids which are topologically equivalent will yield the same unitary operation. Weaves, however, naturally provide a unique representation in which the warp strands are straight, and the weft weaves around them. There is therefore no trivial “double counting” of topologically equivalent weaves when one does a brute force numerical search of weaves up to some given length.

The unitary operations performed by weaving three quasiparticles in which the weft quasiparticle starts and ends in the middle position, will always have the form

$$U_{\text{weave}}(\{n_i\}) = \sigma_1^{n_m} \sigma_2^{n_{m-1}} \cdots \sigma_1^{n_3} \sigma_2^{n_2} \sigma_1^{n_1}. \quad (2.12)$$

Here the sequence of exponents $n_2, n_3 \cdots n_{m-1}$ all take their values from $\{\pm 2, \pm 4\}$, and n_1 and n_m can take the values $\{0, \pm 2, \pm 4\}$. Because these exponents are all even, each factor in this sequence takes the weft quasiparticle all the way around one of the two warp quasiparticles either once or twice with either a clockwise or counterclockwise sense, returning it to the middle position. We allow n_1 and n_m to be 0 to account for the possibility that the initial

or final weaving operations could each be either σ_1^n or σ_2^n with $n = \pm 2$ or ± 4 . Note that we need only consider exponents n_i up to ± 4 (i.e., moving the weft quasiparticle at most two times around a warp quasiparticle) because of the fact that $\sigma_i^{10} = 1$ for Fibonacci anyons, implying, e.g., $\sigma_i^6 = \sigma_i^{-4}$. We define the length L of such weaves to be equal to the total number of elementary crossings, thus $L = \sum_{i=1}^m |n_i|$.

We will also consider weaves in which the weft quasiparticle begins and/or ends at a position other than the middle. These possibilities can easily be taken into account by multiplying $U_{\text{weave}}(\{n_i\})$, as defined in 2.12, by the appropriate factors of σ_1 or σ_2 on the right and/or left. Thus, for example, the unitary operation produced by a weave in which the weft quasiparticle starts in the top position and ends in the middle position can be written $U_{\text{weave}}(\{n_i\})\sigma_2$, where, because of the extra factor of σ_2 , the first braiding operations carried out by this weave will be σ_2^n where n is an odd power, $n = \pm 1, \pm 3$ or 5 . This will weave the weft quasiparticle from the top position to the middle position after which U_{weave} will simply continue weaving this quasiparticle eventually ending with it in the middle position. (Note that by multiplying U_{weave} on the right by σ_2 , and not σ_2^{-1} , we are not requiring the initial elementary braid to be clockwise, since U_{weave} may have $n_1 = 0$ and $n_2 = -2$ or -4 so that the initial σ_2 is immediately multiplied by σ_2 to a negative power.) Similarly, the unitary operation produced by a weave in which the weft particle starts in the top position and ends in the bottom position can be written $\sigma_1 U_{\text{weave}}(\{n_i\})\sigma_2$, and so on.

To find a weave for which the corresponding unitary operation $U_{\text{weave}}(\{n_i\})$ approximates a particular desired unitary operation, the most straightforward approach is to simply perform a brute force search over all weaves, i.e. all sequences $\{n_i\}$ as described above, up to a certain length L , in order to find the $U_{\text{weave}}(\{n_i\})$ which is closest to the target operation. Here we will take as a measure of the distance between two operators U and V the operator norm distance $\epsilon(U, V) = \|U - V\|$ where $\|O\|$ is the operator norm, defined to be the square root of the highest eigenvalue of $O^\dagger O$. Again, if we are interested in fixing the relative phase of the total q -spin 1 and total q -spin 0 sectors then we would restrict the winding of the weaves so that the phases in 2.11 match those of the desired target gate.

For example, imagine our goal is to find a weave which approximates the unitary

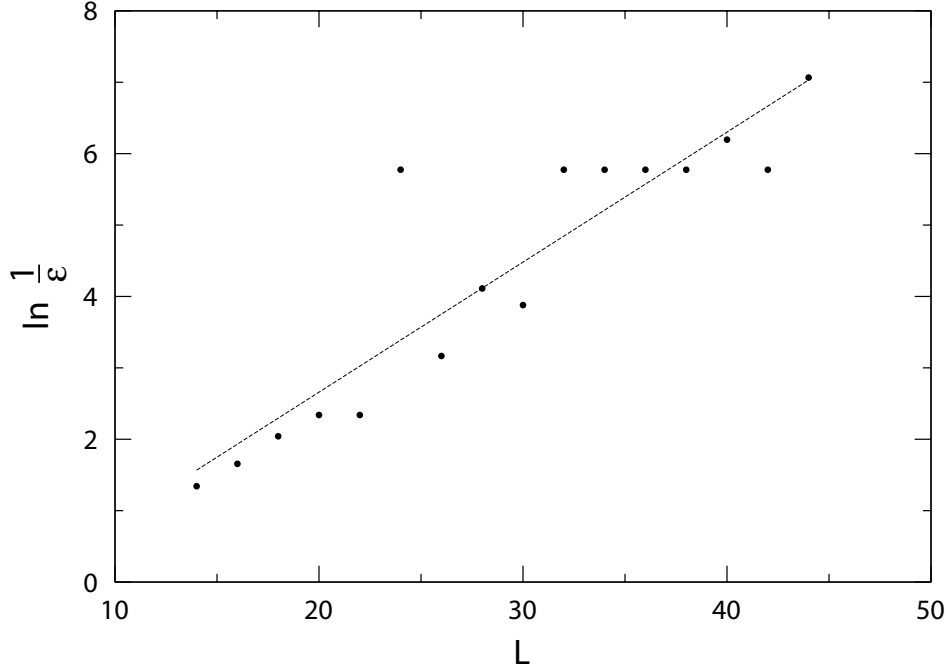


Figure 2.8: $\ln \frac{1}{\epsilon}$ vs. braid length L for weaves approximating the gate iX . Here ϵ is the distance (defined in terms of operator norm) between iX and the unitary transformation produced by a weave of length L which best approximates it. The line is a guide to the eye.

operation,

$$iX = \left(\begin{array}{cc|c} 0 & i & \\ i & 0 & \\ \hline & & 1 \end{array} \right). \quad (2.13)$$

If the resulting weave were to be used only for a single-qubit operation, then we would only require that the weave approximate the upper left 2×2 block of iX up to an overall phase and we would not care about the phase factor appearing in the lower right matrix element. There would then be no constraint on the winding of the braid. However, for this example we will assume that this weave will be used in a two-qubit gate construction, for which the overall phase and/or the phase difference between the total q -spin 1 and total q -spin 0 sectors will matter.

In this case, by comparing iX to [2.11](#), we see that the winding W of any weave

approximating iX must satisfy $e^{i3\pi W/5} = 1$ or $W = 0$ (modulo 10). Results of a brute force search over weaves satisfying this winding requirement which approximate iX are shown in Fig. 2.8. In this figure, $\ln \frac{1}{\epsilon}$ is plotted vs. braid length L , where ϵ is the minimum distance between U_{weave} and iX for weaves of length L . It is expected that, for any such brute force search for weaves approximating a generic target operation, the length should scale with distance according to $L \sim \log \frac{1}{\epsilon}$, because the number of braids grows exponentially with L . The results shown in Fig. 2.8 are consistent with such logarithmic scaling.

All the brute force searches used to find braids in this Chapter are straightforward sequential searches, meant mainly to demonstrate proof of principle. No doubt more sophisticated brute force search methods (e.g. bidirectional search) could be used to perform deeper searches resulting in longer and more accurate braids. Nevertheless, the exponential growth in the number of braids with L implies that finding optimal braids by any brute force search method will rapidly become infeasible as L increases. Fortunately one can still systematically improve a given braid to any desired accuracy by applying the Solovay-Kitaev algorithm [85, 86], which we now briefly review.

2.5.3 Implementation of the Solovay-Kitaev Algorithm for Braids

The general result of the Solovay-Kitaev theorem tells us that we can efficiently improve the accuracy of any given braid without the need to perform exhaustive brute force searches of ever improving accuracy [85, 86]. The essential ingredient in this procedure is an ϵ -net — a discrete set of operators which in the present case correspond to finite braids up to some given length, with the property that for any desired unitary operator there exists an element of the ϵ -net which is within some given distance ϵ_0 of that operator. Provided ϵ_0 is sufficiently small, the Solovay-Kitaev algorithm gives us a clever way to pick a finite number of braid segments out of the ϵ -net and sew them together so that the resulting gate will be an approximation to the desired gate with improved accuracy.

The implementation of the Solovay-Kitaev algorithm we use here follows closely that described in detail in Refs. [87] and [88]. The first step of this algorithm is to find a braid which approximates the desired gate, U , by performing a brute force search over the ϵ -net. Let U_0 denote the result of this search. Since we know that $\epsilon(U_0, U) \leq \epsilon_0$ it follows that $C = UU_0^{-1}$ is an operator which is within a distance ϵ_0 of the identity.

The next step is to decompose C as a group commutator. This means that we find two

unitary operators A and B for which $C = ABA^{-1}B^{-1}$. The unitary operators A and B are chosen so that their action on the computational qubit space corresponds to small rotations through the same angle but about perpendicular axes. For this choice, if A and B are then approximated by operators A_0 and B_0 in the ϵ -net, it can readily be shown that the operator $C_0 = A_0B_0A_0^{-1}B_0^{-1}$, will approximate C to a distance of order $\epsilon_0^{3/2}$. It follows that the operator $U_1 = A_0B_0A_0^{-1}B_0^{-1}U_0$ is an approximation to U within a distance $\epsilon_1 \simeq c\epsilon_0^{3/2}$, where c is a constant which determines the size of the ϵ -net needed to guarantee an improvement in accuracy.

What we have just described corresponds to one iteration of the Solovay-Kitaev algorithm. Subsequent iterations are carried out recursively. Thus, at the second level of approximation each search within the ϵ -net is replaced by the procedure described above, and so on, so that at the n^{th} level all approximations are made at the $(n-1)^{\text{st}}$ level. The result of this recursive process is a braid whose accuracy grows superexponentially in n , with the distance to the desired gate being of order $\epsilon_n \sim (c^2\epsilon_0)^{(3/2)^n}$ at the n^{th} level of recursion, while the braid length grows only exponentially in n , with $L \sim 5^n L_0$, where L_0 is a typical braid length in the initial ϵ -net. Thus, as the distance of the approximate gate from the desired target gate, ϵ , goes to zero, the braid length grows only polylogarithmically, with $L \sim \log^\alpha \frac{1}{\epsilon}$ where $\alpha = \ln 5 / \ln(3/2) \simeq 3.97$. While this scaling is, of course, worse than the logarithmic scaling for brute force searching, it is still only a polylogarithmic increase in braid length which is sufficient for quantum computation. Similar arguments [87, 88] can be used to show that the classical computer time t required to carry out the Solovay-Kitaev algorithm also only scales polylogarithmically in the desired accuracy, with $t \sim \log^\beta \frac{1}{\epsilon}$ where $\beta = \ln 3 / \ln(3/2) \simeq 2.71$.

It is worth noting that there is a particularly nice feature of this implementation of the Solovay-Kitaev algorithm when applied to compiling three-braids. Recall that when carrying out two-qubit gates it will be crucial to maintain the phase difference between the total q -spin 1 and total q -spin 0 sectors of the three-quasiparticle Hilbert space associated with a given three-braid, and, according to 2.11, this can be done by fixing the winding of the braid (modulo 10). Because of the group commutator structure of the Solovay-Kitaev algorithm, the winding of the n^{th} -level approximation U_n will be the same as that of the initial approximation U_0 . This is because all subsequent improvements involve multiplying this braid by group commutators of the form $A_nB_nA_n^{-1}B_n^{-1}$ which automatically have zero winding. The phase relationship between the total q -spin 1 and total q -spin 0 sectors is

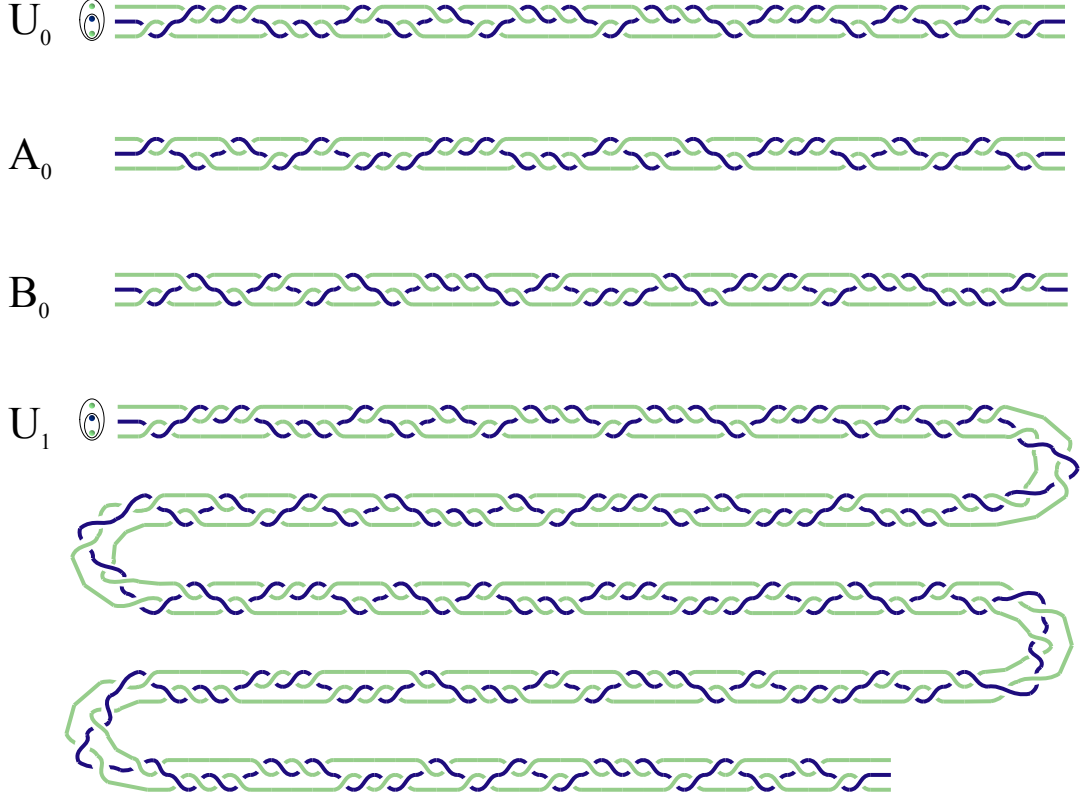


Figure 2.9: One iteration of the Solovay-Kitaev algorithm applied to finding a braid which approximates the operation $U = iX$. The braid U_0 is the result of a brute force search over weaves up to length 44 which best approximates the desired gate $U = iX$, with an operator norm distance between U and U_0 of $\epsilon \simeq 8.5 \times 10^{-4}$. The braids A_0 and B_0 are the results of similar brute force searches to approximate unitary operations A and B whose group commutator satisfies $ABA^{-1}B^{-1} = UU_0^{-1}$. The new braid $U_1 = A_0B_0A_0^{-1}B_0^{-1}U_0$ is then five times longer than U_0 , and the accuracy has improved so that the distance to the target gate is now $\epsilon_1 \simeq 4.2 \times 10^{-5}$. Given the group commutator structure of the $A_0B_0A_0^{-1}B_0^{-1}$ factor, the winding of the U_1 braid is the same as the U_0 braid. Note that when joining braids to form U_1 it is possible that elementary braid operations from one braid will multiply their own inverses in another braid, allowing the total braid to be shortened. Here we have left these “redundant” braids in U_1 , as the careful reader should be able to find.

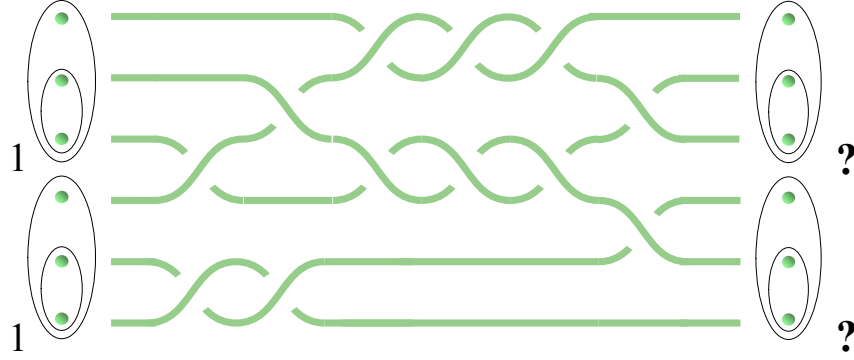


Figure 2.10: Two encoded qubits and a generic braid. Because quasiparticles are braided outside of their starting qubits these braids will generally lead to leakage out of the computational qubit space, i.e. the q -spin of each group of three quasiparticles forming these qubits will in general no longer be 1.

therefore preserved at every level of the construction.

Fig. 2.9 shows the application of one iteration of the Solovay-Kitaev algorithm applied to finding a braid which generates a unitary operation approximating iX . The braid labeled U_0 is the result of a brute force search with $L = 44$ corresponding to the best approximation shown in Fig. 2.8. (Note that although this braid is drawn as a sequence of elementary braid operations, it is topologically equivalent to a weave. In fact precisely this braid, drawn explicitly as a weave, is shown in Fig. 2.15.) The braids labeled A_0 and B_0 generate unitary operations which approximate operators A and B whose group commutator gives UU_0^{-1} where $U = iX$. Finally, the braid labeled U_1 is the new, more accurate, approximate weave.

2.6 Two-Qubit Gates

We have seen that single-qubit gates are “easy” in the sense that as long as we braid within an encoded qubit there will be no leakage errors (the overall q -spin of the group of three quasiparticles will remain 1). Furthermore, the space of unitary operators acting on the three-quasiparticle Hilbert space (essentially $SU(2)$) is small enough to find excellent approximate braids by performing brute force searches and subsequent improvement using the Solovay-Kitaev algorithm. We now turn to the significantly harder problem of finding braids which approximate entangling two-qubit gates.

Figure 2.10 depicts six quasiparticles encoding two qubits and a general braiding pattern. To entangle these qubits, quasiparticles from one qubit must be braided around quasiparticles from the other qubit, and this will inevitably lead to leakage out of the encoded qubit space, (i.e. the overall q -spin of the three quasiparticles constituting a qubit may no longer be 1). Furthermore, the space of all operators acting on the Hilbert space of six quasiparticles is much bigger than for three, making brute force searching extremely difficult. Here the unitary operations acting on this space are in $SU(5) \oplus SU(8)$, (up to winding dependent phase factors as in 2.11), which has 87 free parameters as opposed to 3 for the three quasiparticle case of $SU(2)$.

Still, as a matter of principle, it is possible to perform a brute force search of sufficient depth so that it corresponds to a fine enough ϵ -net to carry out the Solovay-Kitaev algorithm in this larger space [85]. This is essentially the program outlined in [25] as an “existence proof” that universal quantum computation is possible; however, it is not at all clear that, even if one could do this, it would be the most efficient procedure for compiling braids. For the same amount of classical computing power required to directly compile braids in $SU(5) \oplus SU(8)$, we believe one can find much more efficient (in the sense of having a more accurate computation with a shorter braid) braids by breaking the problem into smaller problems, each consisting of finding a specific three-braid embedded in the full six-braid space. As we’ve shown above, these three-braids can then be very efficiently compiled.

Here we present two classes of two-qubit gate constructions based on this “divide and conquer” approach. The first class consists of three constructions where the first two were originally introduced in [89]. These constructions are characterized by the weaving of a *pair* of quasiparticles from one qubit through the quasiparticles forming the second qubit. A new construction which only works for qubits encoded using four quasiparticles and can be carried out by weaving a pair of quasiparticles from one qubit around *pairs* of quasiparticles in the second qubit, is presented here for the first time. The second class which was first introduced in [90], can be carried out by weaving only a single quasiparticle from one qubit around one other quasiparticle from the same qubit, and two quasiparticles from the second qubit.

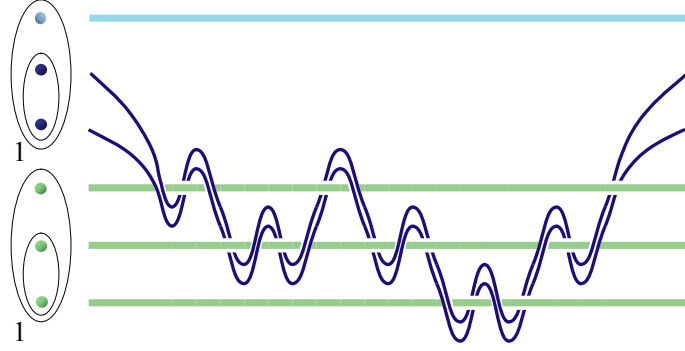


Figure 2.11: A two-qubit gate construction in which a pair of quasiparticles from the top (control) qubit is woven through the bottom (target) qubit. The mobile pair of quasiparticles is referred to as the control pair and has a total q -spin of 0 if the control qubit is in the state $|0_L\rangle$, and 1 if the control qubit is in the state $|1_L\rangle$. Since weaving an object with total q -spin 0 yields the identity operation, this construction is guaranteed to result in a transformation of the target qubit state only if the control qubit is in the state $|1_L\rangle$. Note that in this and subsequent figures world-lines of mobile quasiparticles will always be dark blue.

2.6.1 Two-Quasiparticle Weave Construction

We now review the two-qubit gate constructions first discussed in [89]. The basic idea behind these constructions is illustrated in Fig. 2.11. This figure shows two qubits and a braiding pattern in which a pair of quasiparticles from the top qubit (the control qubit) is woven through the quasiparticles forming the bottom qubit (the target qubit). Throughout this braiding the pair is treated as a single immutable object which, at the end of the braid, is returned to its original position.

If, as in Fig. 2.11, we choose the pair of weft quasiparticles to be the two quasiparticles whose total q -spin determines the logical state of the qubit, then we refer to this pair as the control pair. We can then immediately see why this construction naturally suggests itself. If the control qubit is in the state $|0_L\rangle$ the control pair will have total q -spin 0, and weaving this pair through the target qubit will have no effect. We are thus guaranteed that if the control qubit is in the state $|0_L\rangle$ the identity operation is performed on the target qubit.

The only non-trivial effect of this weaving pattern occurs when the control qubit is in the state $|1_L\rangle$. In this case, the control pair has total q -spin 1 and so behaves as a single Fibonacci anyon. The problem of constructing a two-qubit controlled gate then corresponds

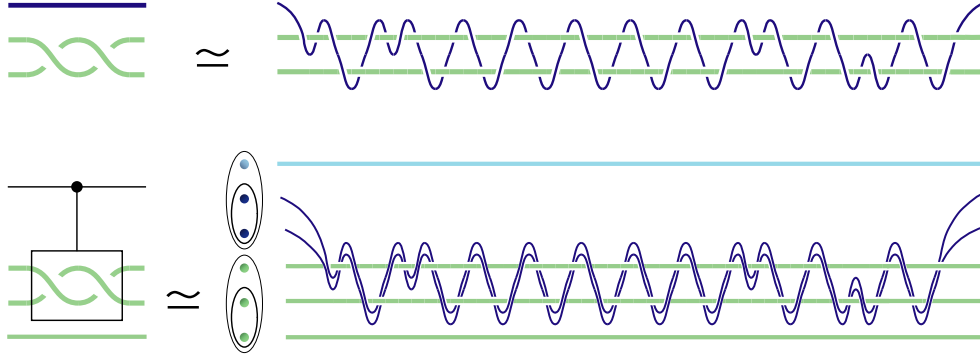


Figure 2.12: An effective braiding weave, and a two-qubit gate constructed using this weave. The effective braiding weave is a woven three-braid which produces a unitary operation which is a distance $\epsilon \simeq 2.3 \times 10^{-3}$ from that produced by simply interchanging the two target particles (σ_1^2). When the control pair is woven through the target qubit using this weave the resulting two-qubit gate approximates a controlled-(σ_2^2) gate to a distance $\epsilon \simeq 1.9 \times 10^{-3}$ or $\epsilon \simeq 1.6 \times 10^{-3}$ when the total q -spin of the two qubits is 0 or 1, respectively.

to finding a weaving pattern in which a single Fibonacci anyon weaves through the three quasiparticles of the target qubit, inducing a transition on this qubit without generating any leakage error out of the computational qubit space, or at least keeping such leakage as small as required for a particular computation. This reduces the problem of finding a two-qubit gate to that of finding a weaving pattern in which one Fibonacci anyon weaves around three others — a problem involving only four Fibonacci anyons. However, following our “divide and conquer” philosophy, we will further narrow our focus to weaving a single Fibonacci anyon through only two others at a time.

Effective Braiding

We define an “effective braiding” weave, to be a woven three-braid in which the weft quasiparticle starts at the top position, and returns to the top position at the end of the weave, with the requirement that the unitary transformation it generates be approximately equal to that produced by m clockwise interchanges of the two warp quasiparticles. To find such weaves we perform a brute force search, as outlined in Sec. 2.5, over sequences $\{n_i\}$ which approximately satisfy

$$\sigma_2 U_{\text{weave}}(\{n_i\}) \sigma_2 \simeq \sigma_1^m. \quad (2.14)$$

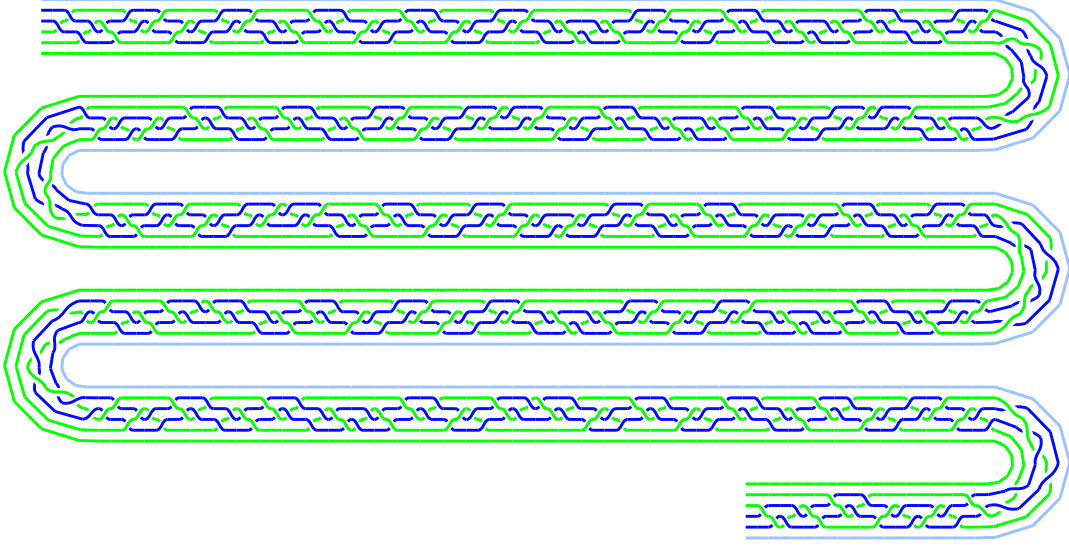


Figure 2.13: Solovay-Kitaev improved controlled- (σ_2^2) gate. This braid approximates a controlled- (σ_2^2) gate with an accuracy of $O(10^{-4})$.

If both sides of this equation are expressed using 2.11 it becomes evident that the winding of any effective braiding weave must satisfy $W = m$ (modulo 10). Since the weft particle starts and ends in the top position, W must be even, thus effective braiding weaves only exist for even m .

An example of an $m = 2$ effective braiding weave found through a brute force search is shown in Fig. 2.12. The corresponding unitary operation approximates that of interchanging the two warp quasiparticles twice to a distance $\epsilon \sim 10^{-3}$. (This is a typical distance for a woven three-braid of length $L \simeq 46$ which approximates a desired operation — precise distances of approximate weaves are given in the figure captions.) As for all approximate weaves considered here, the Solovay-Kitaev algorithm outlined in Sec. 2.5.3 can be used to improve the accuracy of this weave so that ϵ can be made as small as required with only a polylogarithmic increase in length.

The construction of a two-qubit gate using this effective braiding weave is also shown in Fig. 2.12. In this construction the control pair is woven through the top two quasiparticles of the target qubit using this weave. As described above, if the control qubit is in the state $|0_L\rangle$, the control pair has q -spin 0 and the target qubit is unchanged. But, if the control qubit is in

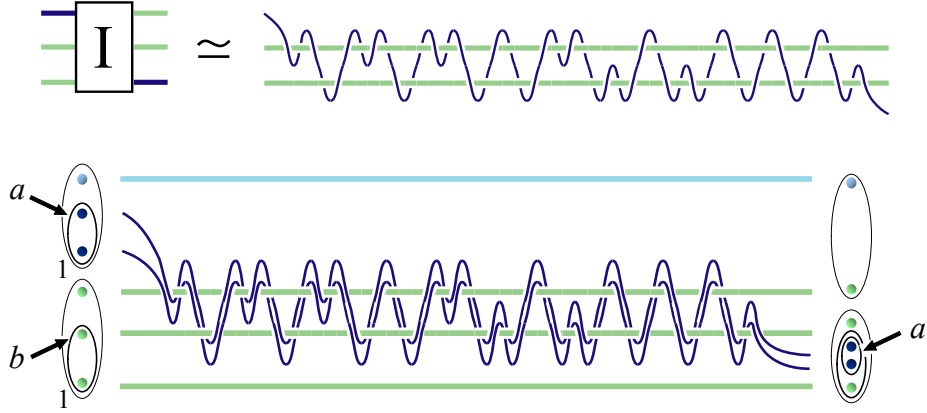


Figure 2.14: An injection weave, and step one in our injection based gate construction. The box labeled I represents an ideal (infinite) injection weave which is approximated by the weave shown to a distance $\epsilon \simeq 1.5 \times 10^{-3}$. In step one of our gate construction, this injection weave is used to weave the control pair into the target qubit. If the control qubit is in the state $|1_L\rangle$ then $a = 1$ and the result is to produce a target qubit with the same quantum numbers as the original, but with its middle quasiparticle replaced by the control pair.

the state $|1_L\rangle$, the control pair has q -spin 1 and the action on the target qubit is approximately equivalent to that of interchanging the top two quasiparticles twice, with the approximation becoming more accurate as the length of the effective braiding weave is increased, either by deeper brute force searching or by applying the Solovay-Kitaev algorithm. Because this effective braiding all occurs within an encoded qubit, leakage errors can be reduced to zero in the limit $\epsilon \rightarrow 0$. The resulting two-qubit gate is then a controlled- σ_2^2 gate which corresponds to controlled rotation of the target qubit through an angle of $6\pi/5$.

Unfortunately, due to the even m constraint, it is impossible to find an effective braiding gate which corresponds to a controlled π rotation of the target qubit. Such a gate would be equivalent to a controlled-NOT gate up to single-qubit rotations [86]. Nonetheless, it is known that any entangling two-qubit gate, when combined with the ability to carry out arbitrary single-qubit rotations, forms a universal set of quantum gates [14]. Thus, the efficient compilation of single-qubit operations described in Sec. 2.5 and the effective braiding construction just given provide direct procedures for compiling any quantum algorithm into a braid to any desired accuracy.

Although it can be used to form a universal set of gates, this effective braiding

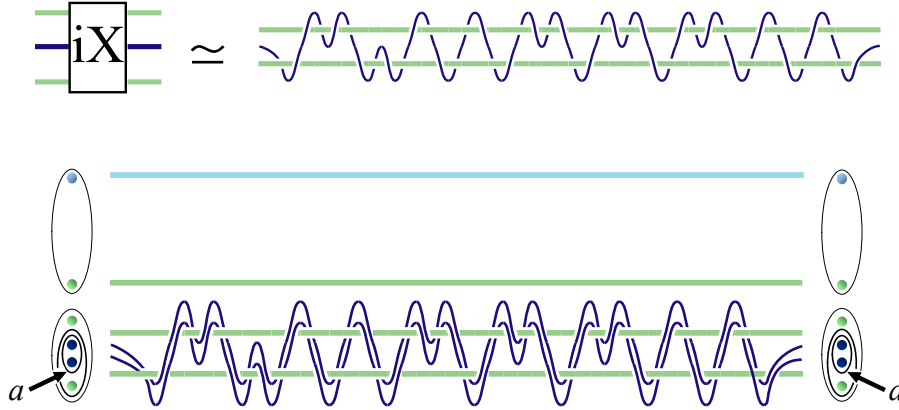


Figure 2.15: A weave which approximates iX (see Eq. 2.13), and step two in our injection based construction. The box labeled iX represents an ideal (infinite) iX weave which is approximated by the weave shown to a distance $\epsilon = 8.5 \times 10^{-4}$ (this is the same weave which appears at the top of Fig. 2.9). In step two of our gate construction the control pair is woven within the injected target qubit, following this weave, in order to carry out an approximate iX gate when $a = 1$, as shown.

construction is still rather restrictive. It is clearly desirable to be able to directly compile a controlled-NOT gate into a braid. In the next section, we give a construction which can be used to efficiently compile any arbitrary controlled rotation of the target qubit — including a controlled-NOT gate. This construction is based on a class of woven three-braids which we call “injection weaves”.

Injection-Based Controlled-Rotation Gates

In an injection weave the weft quasiparticle again starts at the top position but in this case ends at a different position. At the same time we require that the unitary operation generated by this weave approximate the identity. Thus the effect of an injection weave is to permute the quasiparticles involved without changing any of the underlying q -spin quantum numbers of the system.

Comparing the identity matrix to 2.11 we see that any three-braid approximating the identity must have winding $W = 0$ (modulo 10). The fact that this winding must be even implies that the final position of the weft particle must be at the bottom of the weave. Thus

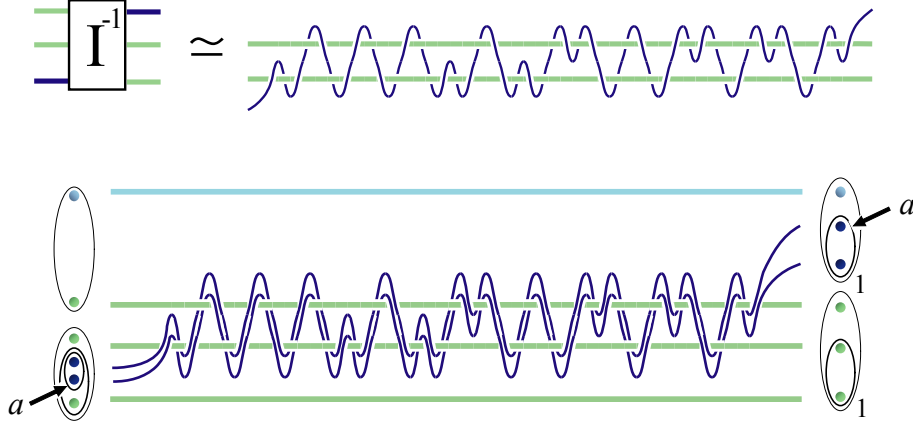


Figure 2.16: An inverse injection weave and step three in our injection based construction. The box labeled I^{-1} represents an ideal (infinite) inverse injection weave which is approximated by the inverse of the injection weave shown in Fig. 2.14, again to a distance $\epsilon \simeq 1.5 \times 10^{-3}$. This weave is used to extract the control pair out of the injected target qubit and return it to the control qubit, as shown.

injection weaves correspond to sequences $\{n_i\}$ which approximately satisfy the equation,

$$\sigma_1 U_{\text{weave}}(\{n_i\}) \sigma_2 \simeq \left(\begin{array}{cc|c} 1 & 0 & \\ 0 & 1 & \\ \hline & & 1 \end{array} \right). \quad (2.15)$$

An injection weave obtained through brute force search is shown in Fig. 2.14. The unitary operation produced by this weave approximates the identity operation to a distance $\epsilon \sim 10^{-3}$.

Our two-qubit gate construction based on injection weaving is carried out in three steps. In the first step, also shown in Fig. 2.14, the control pair is woven into the target qubit using the injection weave. If the control pair has total q -spin 1 (the only nontrivial case) the effect of this weave is merely to replace the middle quasiparticle of the target qubit with the control pair. Because the unitary operation approximated by the injection weave is the identity, in the $\epsilon \rightarrow 0$ limit this injection is accomplished *without changing any of the q -spin quantum numbers*. The injected target qubit is therefore (approximately) in the same quantum state as the original target qubit.

In the second step of our construction, illustrated in Fig. 2.15, we carry out an operation on the injected target qubit by simply weaving the control pair within the target. Because for $a = 1$ all of this weaving takes place within the injected target qubit, there will be no

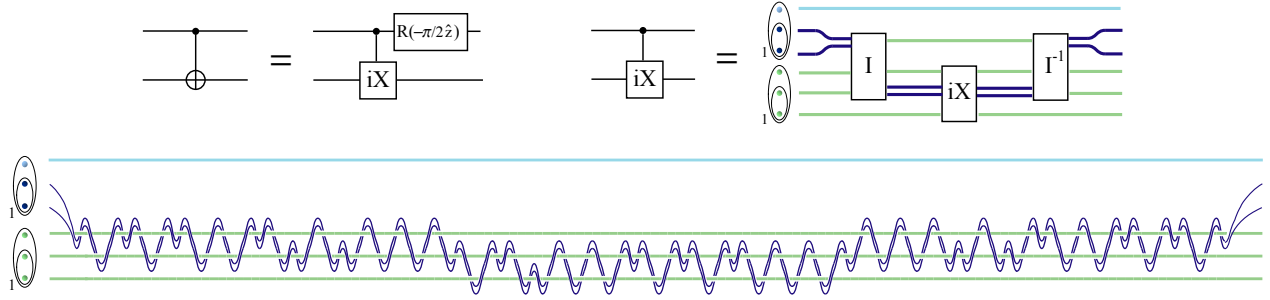


Figure 2.17: Injection-weave based compilation of a controlled-NOT gate into a braid. A controlled-NOT gate can be expressed as a controlled- (iX) gate and a single-qubit operation $R(-\pi/2 \hat{z}) = \exp(i\pi\sigma_z/4)$ acting on the control qubit. The single-qubit rotation can be compiled following the procedure outlined in Sec. 2.5, and the controlled- (iX) gate can be decomposed into ideal injection (I), iX , and inverse injection (I^{-1}) operations which can be similarly compiled. The full approximate controlled- (iX) braid obtained by replacing I , iX and I^{-1} with the weaves shown in the previous three figures is shown at bottom. The resulting gate approximates a controlled- (iX) to a distance $\epsilon \simeq 1.8 \times 10^{-3}$ and $\epsilon \simeq 1.2 \times 10^{-3}$ when the total q -spin of the two qubits is 0 or 1, respectively.

leakage error (again, strictly speaking, only in the limit of an exact injection weave). The only constraint on this weave is that the control pair must both start and end in the middle position, and so it must have even winding.

If our goal is to produce a gate which is equivalent to a controlled-NOT gate up to single-qubit rotations then we must apply a π rotation to the target qubit. Unfortunately, this cannot be accomplished by any finite weave with even winding, so we must again consider approximate weaves. Figure 2.15 shows the control pair being woven through the injected target qubit using a weave found by a brute force search which approximates a particular π rotation — the operator iX defined in 2.13 — to a distance $\epsilon \sim 10^{-3}$ (this is, in fact, the same weave shown at the top of Fig. 2.9).

The third step in our construction is the extraction of the control pair from the target qubit. This is accomplished, as shown in Fig. 2.16, by applying the inverse of the injection weave to the control pair. The effect of this extraction is to restore the control qubit to its original state, and replace the control pair inside the target qubit with the quasiparticle which originally occupied that position.

The full construction is summarized in Fig. 2.17, which provides a recipe for compiling

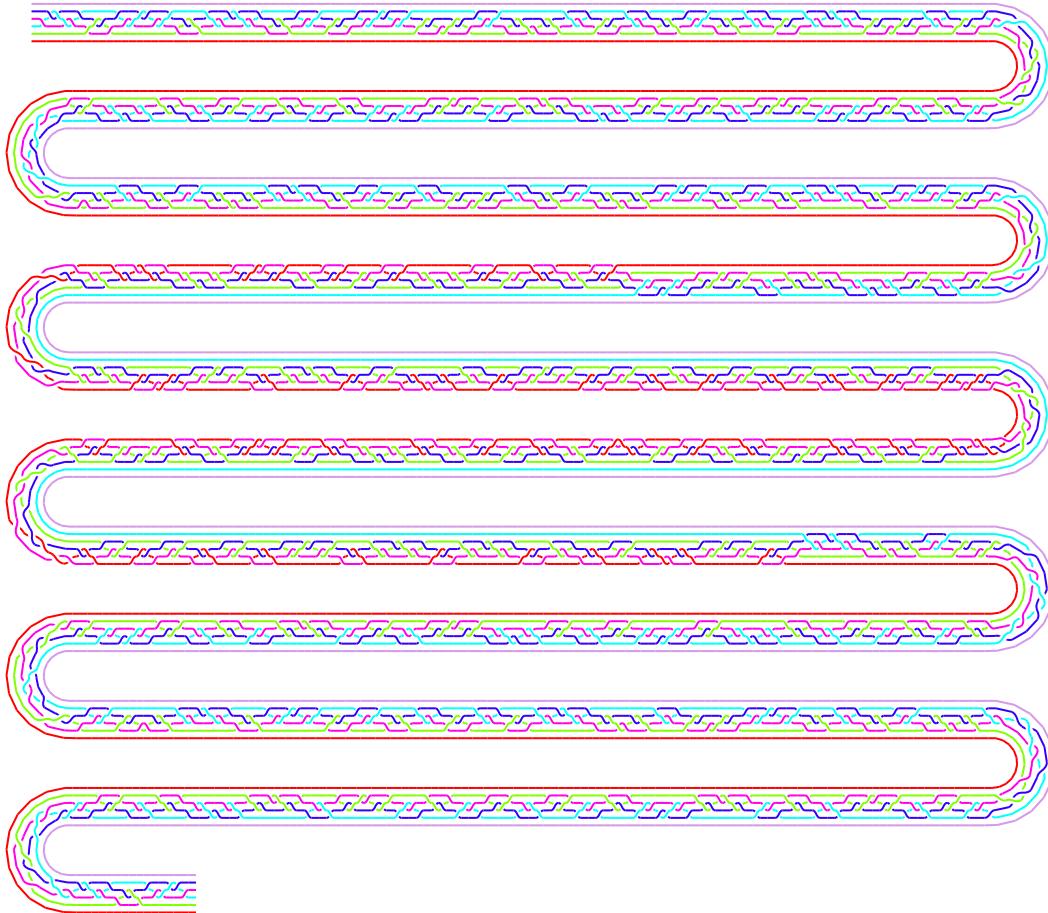


Figure 2.18: Solovay-Kitaev improved controlled- (iX) gate. This braid approximates a controlled- (iX) gate with an accuracy of $O(10^{-4})$.

a controlled-NOT gate into a two-quasiparticle weave. A quantum circuit showing that a controlled-NOT gate is equivalent to a controlled- (iX) gate and a single-qubit operation is shown in the top part of the figure. The single-qubit operation can be compiled to whatever accuracy is required following Sec. 2.5, and the controlled- (iX) gate can be decomposed into injection, iX , and inverse injection operations, as is also shown in the top part of the figure. These operations can then all be similarly compiled following Sec. 2.5.

The full braid shown at the bottom of Fig. 2.17 corresponds to using the approximate woven three-braids shown in Figs. 2.14-2.16 to carry out a controlled- (iX) gate. In this braid, if the control qubit is in the state $|0_L\rangle$ the control pair has total q -spin 0 and the resulting unitary transformation is exactly the identity. However, if the control qubit is in

the state $|1_L\rangle$ the control pair has total q -spin 1 and behaves like a single Fibonacci anyon. This pair is then woven into the target qubit using an injection weave, woven within the target in order to carry out the iX operation, and finally woven out of the target and back into the control qubit using the inverse of the injection weave. The resulting gate is therefore a controlled- (iX) gate.

By replacing the iX weave with an even winding weave which carries out an arbitrary operation U this construction will give a controlled- U gate. The only restriction on U is that its overall phase must be consistent with 2.11 with even winding W . However, this phase can be easily set to any desired value by applying the appropriate single-qubit rotation to the control qubit, as in Fig. 2.17.

Finally, note that at no point in either the effective braiding or injection weave constructions described above did we make reference to the total q -spin of the two qubits involved. It follows that, in the limit of exact effective braiding or injection weaves, the action of the corresponding two-qubit gates on the computational qubit space does not depend on the state of the external fusion space associated with the q -spin 1 quantum numbers of each qubit (see Sec. 2.4). These gates will therefore not entangle the computational qubit space with this external fusion space.

“Big Qubit” Controlled-Phase Gates

The construction of the two-qubit gates described above were based on the trick of weaving a pair of quasiparticles from one qubit around quasiparticles forming the second qubit, and using the fact that if the total q -spin of the weaving pair is 0, the resulting braids do not induce any transitions. When considering qubits encoded with four quasiparticles, it is possible to extend the use of this trick to the target qubit as well. Figure 2.19 (a) shows a braid in which, a pair of quasiparticles from the control qubit weaves around *pairs* of quasiparticles in the target qubit and returns to its original position in the control qubit. In this braid, when the control qubit is in the state $|0_L\rangle$, as before, the corresponding unitary operation is exactly the identity (see Fig. 2.19 (b)). Now consider the target qubit being in the state $|0_L\rangle$. This means the two pairs of quasiparticles in the target qubit each have total q -spin 0, therefore, as is illustrated in Fig. 2.19 (c), the braid is equivalent to weaving the control pair around objects with total q -spin 0, which again, does not induce a transition and the result is *exactly* the identity. Thus, the only nontrivial case is when both control

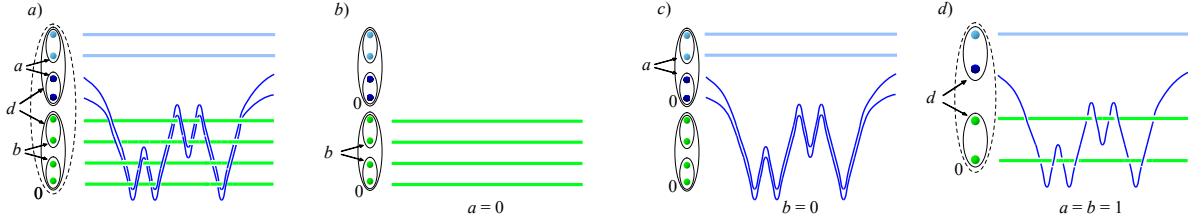


Figure 2.19: Constructing a controlled two-qubit gate. (a) The state of the control qubit (shown in blue) is labeled by a and the state of the target qubit (shown in green) is labeled by b . In this construction, a pair of quasiparticles from the control qubit weaves around pairs of quasiparticles in the target qubit. (b) When the control qubit is in the state zero ($a = 0$), weaving the control pair does not induce any (non-Abelian) transitions, therefore this braid is effectively the identity. (c) When the target qubit is in the state zero ($b = 0$), weaving the control pair around objects with q -spin zero, does not induce any transitions. Therefore, again, the result is exactly the identity. Note that the weaving pattern shown is topologically equivalent to two straight lines, i.e. the identity. (d) The only non-trivial case, when both control and target qubits are in state $|1_L\rangle$. In this case the original braid in (a) is effectively reduced to a three-braid corresponding to a single-qubit gate carried out in the “big qubit”. The two states of the big qubit can be determined by b .

and target qubits are in the state $|1_L\rangle$ ($a = b = 1$ as in Fig. 2.19 (d)).

In Sec. 2.1, it was pointed out that in the case of Fibonacci anyons (or more generally quasiparticles of $SU(2)_3$), braiding pairs of quasiparticles with total q -spin 1, induces *the same transitions* (up to overall Abelian phases) as braiding of single quasiparticles with q -spin 1 does. Therefore, when both control and target qubits are in the state $|1_L\rangle$ ($a = b = 1$ in Fig. 2.19), the problem of weaving a pair of quasiparticles around two other pairs, is equivalent to that of weaving a single quasiparticle around two other quasiparticles. Effectively, this is a single-qubit gate carried out on the “big qubit” shown in Fig. 2.19 (d).

Focusing on the case where both control and target qubits are in the state $|1_L\rangle$, to construct a two-qubit gate, we must find an effective single-qubit gate (acting on the “big qubit”). This effective single-qubit gate must be carried out in such way that it does not induce any leakage error on the original control and target qubits.

As is shown in Fig. 2.19 (d), the logical states of the big qubit are determined by the label d which can be either 0 or 1. Note however, that we are only interested in the phase acquired by the state corresponding to $d = 0$. This state corresponds to the two original

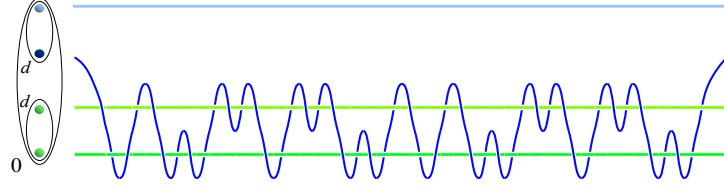


Figure 2.20: The “big qubit” and an effective single-qubit gate which approximates negative identity, with an accuracy of $O(10^{-3})$. The state of the big qubit is determined by d .

groups of four quasiparticles, each being in a qubit state (see Fig. 2.19 (a)). When $d = 1$, the groups of quasiparticles are no longer in qubit states, instead, they have leaked into the non-computational space. To ensure there are no transitions between the original qubit states and the non-computational states, the resulting unitary operation, in a basis that is labeled by $d = \{0, 1\}$ must be diagonal. If we focus in braids with zero total winding number^a, the resulting matrix must have the form

$$M = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix}. \quad (2.16)$$

Note that in this matrix $e^{i\phi}$ is the phase that we are interested in (it corresponds to the case where both qubits are in the state $|1_L\rangle$) (the other nonzero matrix element, $e^{-i\phi}$, is the phase acquired by a state in the non-computational space). (even winding)

We have not been able to find a finite braid which carries out M , exactly. However, as before, it is possible to find excellent approximations by carrying out a brute force search. As an example, Fig. 2.20 (adopted from [90]) shows the result of a brute force search that approximates a negative identity matrix with zero (modulo 10) total winding. If one follows this braiding pattern by weaving the control pair around pairs of quasiparticles in the target qubit (as is shown in Fig. 2.21), one can construct a controlled- Z gate which has the form,

$$U = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{array} \right). \quad (2.17)$$

^aNote that we can always set the overall Abelian phase (associated with weaving the control pair) to identity. This can be done by requiring the total winding of the braid to be zero (modulo 10). In addition, note that braids with zero total winding number, always result in unitary operations with determinant one.

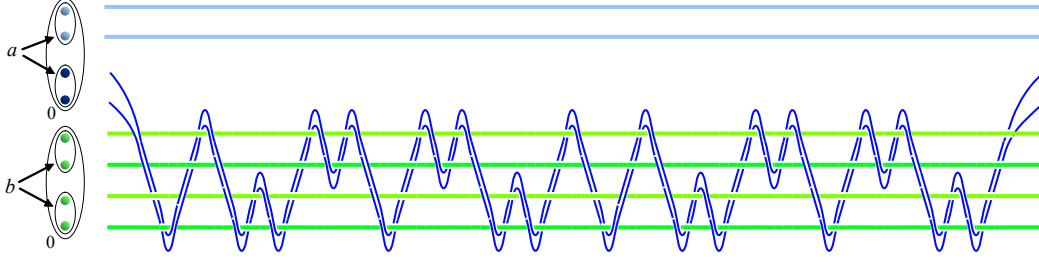


Figure 2.21: A braid that approximates a controlled- Z gate with an accuracy of $O(10^{-3})$. In this braid, a pair of quasiparticles from the control qubit weaves around pairs of quasiparticles in the second qubit.

A nice feature of this construction is that all the elements in the final gate are *exact*, except for -1 which is an approximation. The braid shown in Fig. 2.20 approximates this element with an accuracy of $O(10^{-3})$. Since M is effectively a single-qubit gate, as before, better approximations to this gate can be found, systematically, by applying the Solovay-Kitaev algorithm in $SU(2)$.

2.6.2 One-Quasiparticle Weave Constructions

We now show that two-qubit gates can be carried out with only a single mobile quasiparticle. This possibility follows from the general result of [84] that for any system of non-Abelian quasiparticles in which general braids are universal for quantum computation (such as Fibonacci anyons), single quasiparticle weaves are universal as well. However, the “proof of principle” weaves constructed in that work were extremely inefficient — involving a huge number of excess operations. Here we show how to efficiently construct a single-quasiparticle weave corresponding to a controlled-NOT gate (up to single-qubit rotations).

Our construction is based on a class of weaves which are similar to injection weaves in that they can be used to swap two q -spin 1 objects — where one object is a pair of Fibonacci anyons with total q -spin 1 and the other object is a single Fibonacci anyon — while acting effectively as the identity operation so that none of the other q -spin quantum numbers of the system are disturbed. However, unlike injection weaves, this new class of weaves accomplish this swap without moving the pair as a single object, and in fact can be carried out by moving just one quasiparticle.

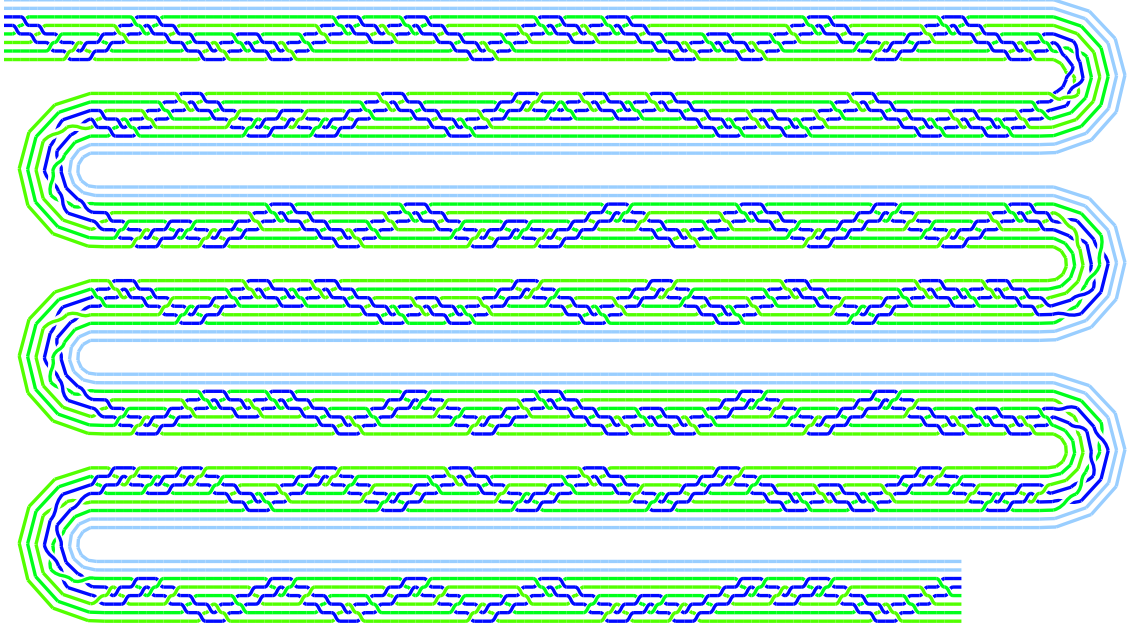


Figure 2.22: Solovay-Kitaev improved controlled-(Z) gate. This braid approximates a controlled-(Z) gate with an accuracy of $O(10^{-4})$.

The class of weaves we seek are those which approximate the transformation

$$U((\bullet, \bullet)_a, \bullet)_c = e^{i\phi}(\bullet, (\bullet, \bullet)_a)_c, \quad (2.18)$$

where ϕ is an overall (irrelevant) phase which does not depend on a or c . The relevant case for showing the similarity with injection is when $a = 1$, for which the initial and final states in 2.18 consist of two q -spin 1 objects — a single Fibonacci anyon and a pair of Fibonacci anyons with total q -spin 1. If both these objects are represented as single Fibonacci anyons then 2.18 can be written $U(\bullet, \bullet)_c = e^{i\phi}(\bullet, \bullet)_c$. In this representation U therefore acts effectively as the identity operation (times an irrelevant phase), similar to injection.

Using the F matrix 2.6 to expand the right hand side of 2.18 in the $((\bullet, \bullet), \bullet)$ basis yields

$$U((\bullet, \bullet)_a, \bullet)_c = e^{i\phi} \sum_b F_{ab}^c((\bullet, \bullet)_b, \bullet)_c. \quad (2.19)$$

Comparing this with the action of a unitary operation U with matrix representation

$$U = \left(\begin{array}{cc|c} U_{00}^1 & U_{01}^1 & \\ U_{10}^1 & U_{11}^1 & \\ \hline & & U_{11}^0 \end{array} \right), \quad (2.20)$$

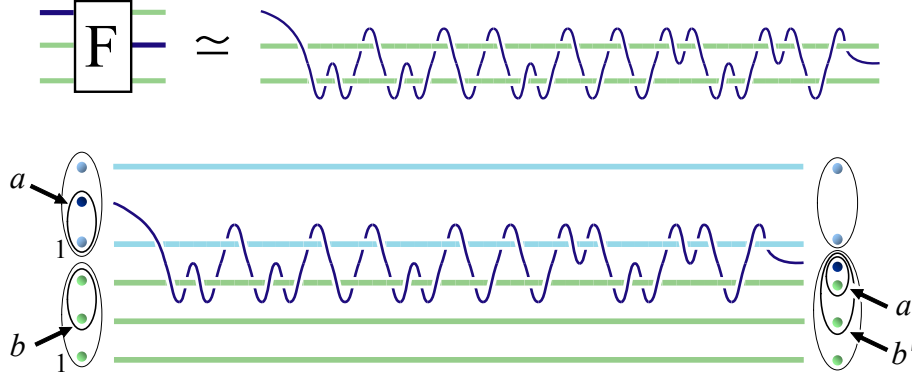


Figure 2.23: An F weave, and step one of our F weave based two-qubit gate construction. The box labeled F represents an ideal (infinite) F weave which is approximated by the weave shown to a distance $\epsilon \simeq 3.1 \times 10^{-3}$. Applying the F weave to the initial two-qubit state, as shown, produces an intermediate state with q -spins labeled a and b' which depend simply on a and b — the initial states of the two qubits (see Table I).

on the state $((\bullet, \bullet)_a, \bullet)_c$,

$$U((\bullet, \bullet)_a, \bullet)_c = \sum_b U_{ab}^c((\bullet, \bullet)_b, \bullet)_c, \quad (2.21)$$

we see that the matrix representation of the U we seek is precisely the F matrix (up to a phase): $U = e^{i\phi} F$. While the F matrix describes a “passive” operation, i.e. a change of basis, the operator U can be viewed as an “active” F operation which acts directly on the states of the Hilbert space. Note that, since $F = F^{-1}$, we also have

$$U(\bullet, (\bullet, \bullet)_a)_c = e^{i\phi}((\bullet, \bullet)_a, \bullet)_c. \quad (2.22)$$

We will refer to weaves which approximate the operation 2.18 (and thus also 2.22) as F weaves. As we have seen, the unitary operation U produced by an F weave need only approximate the F matrix (2.6) up to an overall irrelevant phase. To be consistent with 2.11 this phase must be -1 , as can be seen by writing the matrix $-F$ as

$$-F = \left(\begin{array}{c|c} \pm i \begin{pmatrix} \pm i\tau & \pm i\sqrt{\tau} \\ \pm i\sqrt{\tau} & \mp i\tau \end{pmatrix} & \\ \hline & -1 \end{array} \right), \quad (2.23)$$

where a factor of $\pm i$ has been pulled out of the upper left 2×2 block, leaving an $SU(2)$

matrix ($\det = \tau^2 + \tau = 1$). Comparing 2.23 with 2.11, it is also evident that any F weave must have winding $W = 5$ (modulo 10), which is necessarily odd.

The fact that F weaves must have an odd number of windings implies that if the weft quasiparticle starts at the top position of the weave it must end at the middle position. For this choice the F weave must then approximately satisfy the equation

$$U_{\text{weave}}(\{n_i\}) \sigma_2 \simeq -F. \quad (2.24)$$

The result of a brute force search for an F weave which approximates the operation $-F$ to a distance $\epsilon \sim 10^{-3}$ is shown in Fig. 2.23.

The first step in our single-quasiparticle weave construction is the application of an F weave to two qubits, also shown in Fig. 2.23. Note that in this figure for convenience we have made a change of basis on the bottom qubit, so that the pair which determines its state (the control pair) consists of the top two quasiparticles within it rather than the bottom two. There is no loss of generality in doing so since this just corresponds to a single-qubit rotation on the bottom qubit.

With this basis choice the initial state of the two qubits is determined by the q -spins of their respective control pairs which are indicated in Fig. 2.23 as a (top qubit) and b (bottom qubit). After carrying out the F weave, taking the middle quasiparticle of the top qubit as the weft quasiparticle and weaving it around both the bottom quasiparticle of the top qubit and the top quasiparticle of the bottom qubit, the resulting state (again, strictly speaking, only in the limit of an exact F weave) is shown at the end of the two-qubit weave in Fig. 2.23. From 2.22 it follows that the newly positioned weft quasiparticle and the quasiparticle beneath will have total q -spin a . When the quasiparticle beneath these two is also included, the three quasiparticles form what we will refer to as the intermediate state, $(\bullet, (\bullet, \bullet)_a)_{b'}$, where the total q -spin of all three quasiparticles, b' , has a well-defined value provided a and b are well defined, as we now show.

First consider the case $a = 1$. As described above, the effect of the F weave is then similar to that of the injection weave from the previous construction — it replaces the topmost quasiparticle in the bottom qubit with a pair of quasiparticles with q -spin 1, and the bottommost pair of quasiparticles in the top qubit (which also has total q -spin 1) with a single quasiparticle, without changing any of the other q -spin quantum numbers of the system. In the limit of an ideal F weave, this means that the b quantum number does not

change after this swap and so $b' = b$. The case $a = 0$ is simpler, since in this case the intermediate state is $(\bullet, (\bullet, \bullet)_0)_{b'}$ for which the fusion rules 2.3 imply $b' = 1$, regardless of the value of b . The resulting dependence of b' on a and b is summarized in Table 2.6.2.

Having used the F weave to create the intermediate state $(\bullet, (\bullet, \bullet)_a)_{b'}$, the next step in our construction is the application of a weave which performs an operation on this state which does not change a and b' but which does yield an a and b' dependent phase factor. After carrying out such a weave, which we will refer to as a phase weave, we can then apply the inverse of the F weave to restore the two qubits to their initial states a and b .

For any phase weave we will require that the weft quasiparticle both start and end in the top position so that when we join it to the F weave and its inverse there will be a single weft quasiparticle throughout the entire gate construction. The phase weave must therefore have even winding, and with no loss of generality we can consider the case for which the winding satisfies $W = 0$ (modulo 10). The unitary operation produced by such a phase weave must then approximately satisfy the equation

$$\sigma_2 U_{\text{weave}}(\{n_i\}) \sigma_2 \simeq F \left(\begin{array}{cc|c} e^{i\alpha} & 0 & \\ 0 & e^{-i\alpha} & \\ \hline & & 1 \end{array} \right) F^{-1}, \quad (2.25)$$

where the F matrices are needed to change the Hilbert space basis from that in which the operation produced by the phase braid must be diagonal, (the $(\bullet, (\bullet, \bullet))$ basis), to that in which the σ_1 and σ_2 matrices are defined, (the $((\bullet, \bullet), \bullet)$ basis).

We will see that a phase weave with $\alpha = \pi$ produces a two-qubit gate which is equivalent to a controlled-NOT gate up to single-qubit rotations. The result of a brute force search

Table 2.1: Values of b' for different values of a and b after applying the F weave as shown in Fig. 2.23, and the phase applied to the resulting state by a phase weave with zero winding. The value of b' is determined by the fact that $b' = 1$ when $a = 0$ and $b' = b$ when $a = 1$, as shown in the text.

a	b		b'	Phase Factor
0	0	$b' = 1$	1	$e^{i\alpha}$
0	1		1	$e^{i\alpha}$
1	0	$b' = b$	0	1
1	1		1	$e^{-i\alpha}$

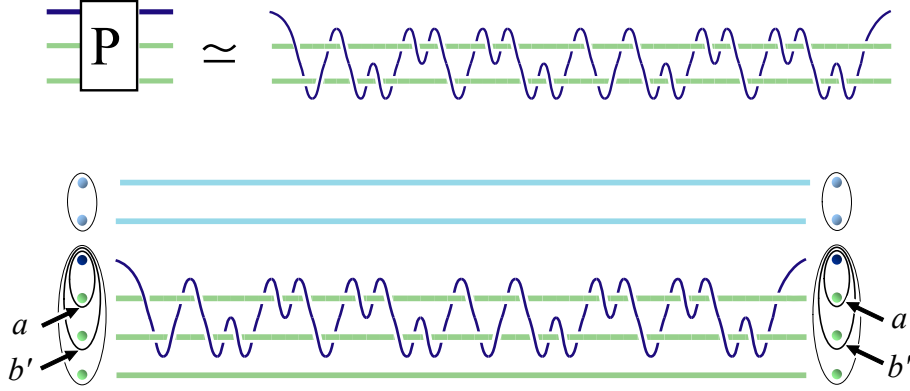


Figure 2.24: A phase weave with $\alpha = \pi$ (see text) which gives a π phase shift to the intermediate state when $b' = 1$, and step two of our F weave based construction. The box labeled P represents an ideal (infinite) $\alpha = \pi$ phase weave which is approximated by the weave shown to a distance $\epsilon \simeq 1.9 \times 10^{-3}$. Applying this phase weave to the intermediate state created by the F weave, as shown, results in a b' dependent π phase shift (see Table I with $\alpha = \pi$).

for such a phase weave which approximates the desired operation to a distance $\epsilon \sim 10^{-3}$ is shown in Fig. 2.24. This figure also shows the action of the phase weave on the intermediate state produced in Fig. 2.23. In this weave, the weft quasiparticle is now woven through the two quasiparticles beneath it, and returns to its original position. Because the phase weave produces a diagonal operation in the basis shown for the intermediate state, it does not change the values of a and b' . Its only effect is to give a phase factor of $e^{i\alpha}$ to the state with $a = 0$ (which necessarily has $b' = 1$) and $e^{-i\alpha}$ to the state with $a = 1$ and $b' = 1$. The state with $a = 1$ and $b' = 0$ is unchanged. These phase factors are also shown in Table 2.6.2.

The final step in this construction is to perform the inverse of the F weave to return the two qubits to their original states. This is shown in Fig. 2.25. In the limit of exact F and phase weaves, the resulting operation on the computational qubit space in the basis $ab = \{00, 01, 10, 11\}$ is then,

$$U = \begin{pmatrix} e^{i\alpha} & 0 & 0 & 0 \\ 0 & e^{i\alpha} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{-i\alpha} \end{pmatrix}. \quad (2.26)$$

If we take the top qubit to be the control qubit, and the bottom qubit to be the target qubit,

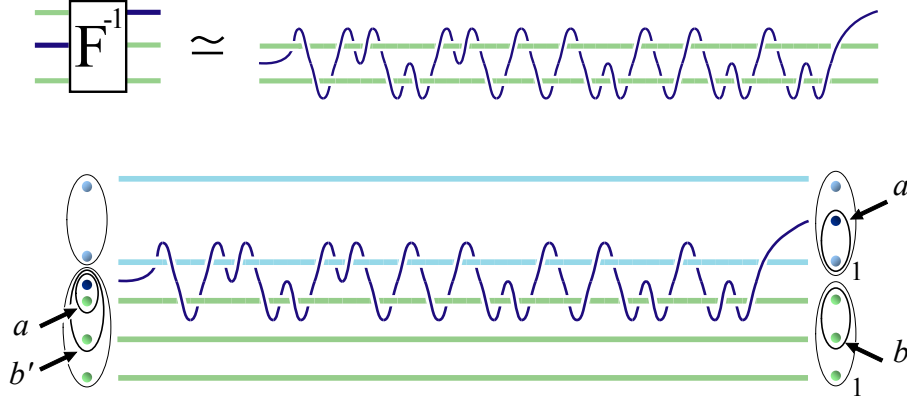


Figure 2.25: An inverse F weave and step three in our F weave construction. The box labeled F^{-1} is an ideal (infinite) inverse F weave which is approximated by the inverse of the F weave shown in Fig. 2.23, again to a distance $\epsilon \simeq 3.1 \times 10^{-3}$. By applying the inverse F weave to the state obtained after applying the phase weave, as shown, the two qubits are returned to their initial states, but now with an a and b dependent phase factor (see Table I).

then this gate corresponds, up to an irrelevant overall phase, to a controlled- $(e^{-i3\alpha/2}e^{i\alpha\sigma_z/2})$ operation. For the case $\alpha = \pi$ this is a controlled- $(-Z)$ gate (where $Z = \sigma_z$), i.e. a controlled-Phase gate, which, up to single-qubit rotations, is equivalent to a controlled-NOT gate.

The full F weave based gate construction is summarized in Fig. 2.26. A quantum circuit showing a controlled-NOT gate in terms of a controlled- $(-Z)$ gate and two single-qubit operations is shown in the top part of the figure. As in our injection based construction, the single-qubit operations can be compiled to whatever accuracy is required following the procedure outlined in Sec. 2.5. The controlled- $(-Z)$ gate can then be decomposed into ideal F , phase, and inverse F weaves as is also shown in the top part of the figure. Woven three-braids which approximate these operations can then be compiled to whatever accuracy is required, again following Sec. 2.5. The full controlled- $(-Z)$ weave corresponding to using the approximate F and phase weaves shown in Figs. 2.23-2.25 is shown in the bottom part of the figure.

Finally, in this construction, as for the constructions described in Sec. 2.6.B, we at no point made reference to the total q -spin of the two qubits involved. Thus, in the limit of exact

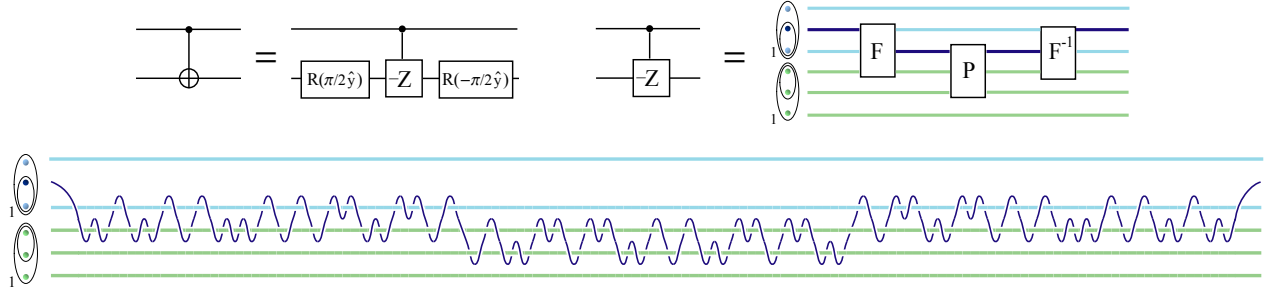


Figure 2.26: F weave based compilation of a controlled-NOT gate into a braid. A controlled-NOT gate is equivalent to a controlled- $(-Z)$ gate with the single-qubit operation $R(\pi/2 \hat{y}) = \exp(-i\pi\sigma_y/4)$ and its inverse applied to the target qubit before and after the controlled- $(-Z)$. Again, the single-qubit operations can be trivially compiled, and the controlled- $(-Z)$ gate decomposed into ideal F , phase (P), and inverse F (F^{-1}) weaves which can be similarly compiled. The full approximate controlled- $(-Z)$ weave obtained by replacing F , P and F^{-1} with the approximate weaves shown in the previous three figures is shown at bottom. The resulting gate approximates a controlled- $(-Z)$ to a distance $\epsilon \simeq 4.9 \times 10^{-3}$ and $\epsilon \simeq 3.2 \times 10^{-3}$ when the total q -spin of the two qubits is 0 or 1, respectively.

F and phase weaves, the action of the two-qubit gates constructed here will not entangle the computational qubit space with the external fusion space associated with the q -spin 1 quantum numbers of each qubit.

2.7 What's Special about $k = 3$?

All of the gate constructions discussed in this Chapter exploit the fact that the braiding and fusion properties of a pair of Fibonacci anyons are either trivial if their total q -spin is 0, or equivalent to those of a single Fibonacci anyon if their total q -spin is 1. The fact that these are the only two possibilities is a special property of the Fibonacci anyon model, and hence also the $SU(2)_3$ model, given their effective equivalence. It is then natural to ask to what extent our constructions can be generalized to $SU(2)_k$ theories for different values of the level parameter k .

Of course we know from the results of Freedman *et al.* [26] that the $SU(2)_k$ representations of the braid group are dense for $k = 3$ and $k > 4$. Thus, for example, braids which approximate controlled-NOT gates on encoded qubits exist and can, in principle, be found for all these k values. However, we will show below that things are somewhat simpler for the

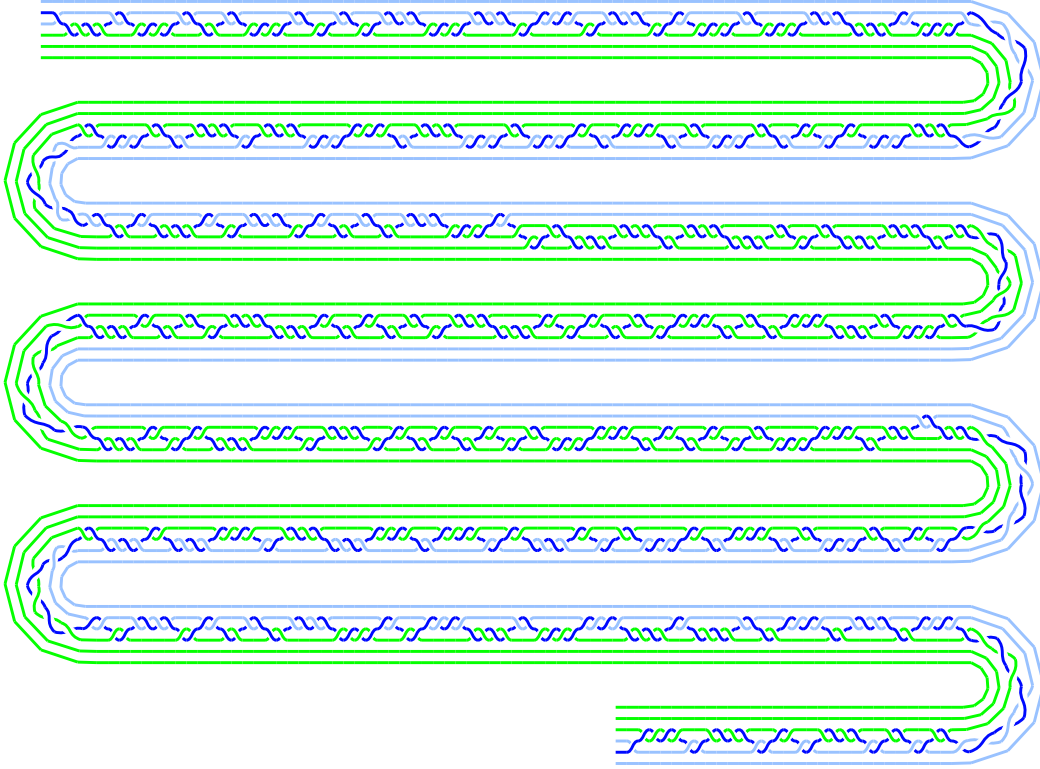


Figure 2.27: Solovay-Kitaev improved controlled- $(-Z)$ gate. This braid approximates a controlled- $(-Z)$ gate with an accuracy of $O(10^{-4})$.

case $k = 3$. Specifically we will show that for $k = 3$, and only $k = 3$, it is possible to carry out two-qubit entangling gates by braiding only four quasiparticles, as, for example, in our effective braiding and F weave constructions.

Consider a pair of $SU(2)_k$ four-quasiparticle qubits as shown in Fig. 2.28. Here each quasiparticle is assumed to have q -spin $1/2$ and the total q -spin of each qubit is required to be 0. The state of a given qubit is then determined by the q -spin of either the topmost or bottommost pair of quasiparticles within it, where, from the $SU(2)_k$ fusion rules 2.2, the q -spin of each pair must be the same for the total q -spin of the qubit to be 0. Thus, in Fig. 2.28, the state of the top qubit is determined by the q -spin labeled a and the state of the bottom qubit is determined by the q -spin labeled b , where, again from the fusion rules 2.2, a and b can be either 0 or 1.

If we are only allowed to braid the middle four quasiparticles, as shown in Fig. 2.28, then

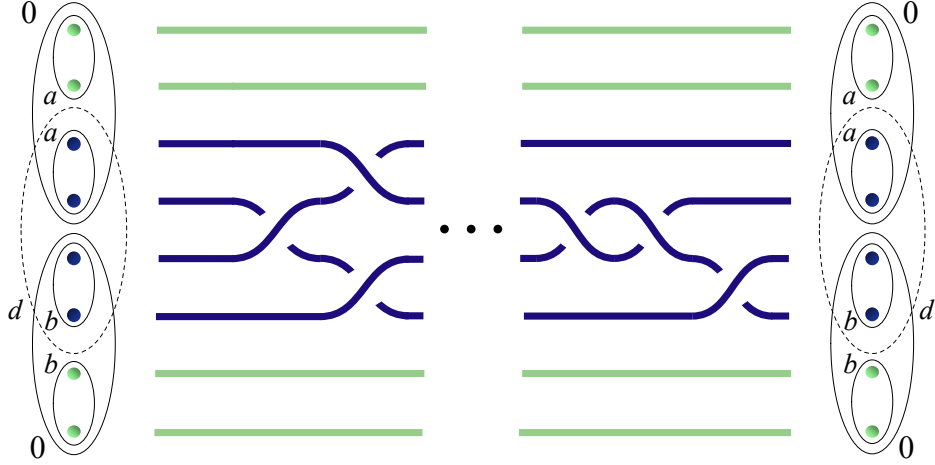


Figure 2.28: Two four-quasiparticle qubits and a braiding pattern in which only two quasiparticles from each qubit are braided. Here the quasiparticles are $SU(2)_k$ excitations with q -spin $1/2$. The state of the top qubit is determined by the total q -spin of the quasiparticle pairs labeled a and the state of the bottom qubit is determined by the total q -spin of the quasiparticle pairs labeled b . The overall q -spin of the four braided quasiparticles is d , (a dashed oval is used because when $a = b = 1$ these quasiparticles will not be in a q -spin eigenstate). For this braid to produce no leakage errors, the unitary operation it generates must be diagonal in a and b , though it can, of course, result in an a and b dependent phase factor. For $k > 3$, d can take the values 0, 1 or 2, while for $k = 3$ the only allowed values for d are 0 and 1. The existence of the $d = 2$ state for $k > 3$ makes it impossible to carry out an entangling two-qubit gate by braiding only four quasiparticles (see text).

the total q -spin of the two topmost quasiparticles of the top qubit and the two bottommost quasiparticles of the bottom qubit will remain, respectively, a and b . It follows that if the two qubits are to remain in their computational qubit spaces, the total q -spin of the two topmost and two bottommost quasiparticles that *are* being braided must also remain, respectively, a and b . (If this were not the case, the fusion rules (2.2) would imply that the total q -spin of the four quasiparticles forming each qubit would no longer be 0). Thus, in order for there to be no leakage errors after braiding these four quasiparticles, the resulting operation must be diagonal in a and b .

It is important to note that this result, and the results that follow, hold not just for four-quasiparticle qubits, but also for $SU(2)_k$ versions of the three-quasiparticle qubits used throughout this Chapter. This is because, as pointed out in Sec. 2.4, any gate acting on

a pair of three-quasiparticle qubits must result in an operation on the computational qubit space which is independent of the state of the external fusion space associated with the fact that each qubit has total q -spin $1/2$, (here the total q -spin of a three-quasiparticle qubit is $1/2$ rather than 1 because we are using $SU(2)_k$ quantum numbers and assuming each quasiparticle has q -spin $1/2$ — see Fig. 2.3 (b)). It is therefore sufficient to consider the special case when the state of two three-quasiparticle qubits corresponds to that of the two four-quasiparticle qubits shown in Fig. 2.28, but with the topmost and bottommost quasiparticles removed. The above arguments then imply any leakage free operation produced by braiding the four middle quasiparticles must be diagonal in a and b .

Now consider the four middle quasiparticles we are allowed to braid. A basis for the Hilbert space of these quasiparticles can be taken to be one labeled by the q -spin quantum numbers a and b , as well as the total q -spin of all four quasiparticles which we denote d (see Fig. 2.28). For $k > 3$ the fusion rules (2.2) imply this total q -spin d can be equal to 0 , 1 or 2 , while for $k = 3$ it can only be equal to 0 or 1 . We will see that this truncation of the $d = 2$ state is the crucial property of the $k = 3$ theory which makes our F weave and effective braiding constructions possible.

It is convenient at this stage to restrict ourselves to braids with zero total winding (i.e. equal numbers of clockwise and counterclockwise exchanges). For such braids, arguments similar to those used to derive 2.11 can be used to show the unitary operation enacted on the $d = 0, 1$ and 2 sectors must each have determinant 1 . There is no loss of generality in restricting ourselves to such braids, since a braid with arbitrary winding can always be turned into one with zero winding by adding the appropriate number of interchanges to either the two topmost or two bottommost of the braiding quasiparticles at either the beginning or end of the braid. These added interchanges will all be within encoded qubits and so correspond to single-qubit rotations which will not produce any entanglement between the two qubits.

If we restrict ourselves to braids with zero winding and insist that these braids approximate gates with zero leakage error — which, as shown above, implies the gate must be diagonal in the a and b quantum numbers — then in the $abd = \{000, 110, 011, 101, 111, 112\}$ basis the unitary transformation acting on the Hilbert space of the four braiding quasipar-

tics must have the form

$$U = \left(\begin{array}{cc|ccc|c} e^{i\alpha} & 0 & & & & \\ 0 & e^{-i\alpha} & & & & \\ \hline & & e^{i\beta} & 0 & 0 & \\ & & 0 & e^{i\gamma} & 0 & \\ & & 0 & 0 & e^{-i(\beta+\gamma)} & \\ \hline & & & & & 1 \end{array} \right), \quad (2.27)$$

where we have required that the $d = 0, 1$ and 2 blocks all have determinant 1, (in particular, the $d = 2$ block is simply 1).

Note that the case $a = b = 1$ has three entries in this matrix, corresponding to the three possible values for the total q -spin quantum number d . For this gate to produce no leakage error, the phase factors in all three of these sectors must be the same. To see this note that one can expand the relevant eight-quasiparticle state in terms of basis states with well-defined values of d as follows

$$\begin{aligned} & (((\bullet, \bullet)_1, (\bullet, \bullet)_1)_0, ((\bullet, \bullet)_1, (\bullet, \bullet)_1)_0)_0 \\ &= \sum_{d=0}^2 F_d ((\bullet, \bullet)_1, ((\bullet, \bullet)_1, (\bullet, \bullet)_1)_d, (\bullet, \bullet)_1)_0, \end{aligned} \quad (2.28)$$

where standard quantum group methods [73, 72] can be used to compute the coefficients F_d , with the result

$$F_0 = \frac{1}{[3]_q}, \quad F_1 = \frac{\sqrt{[3]_q}}{[3]_q}, \quad F_2 = \frac{\sqrt{[5]_q}}{[3]_q}. \quad (2.29)$$

Here we have introduced the q -integers $[m]_q \equiv (q^{m/2} - q^{-m/2})/(q^{1/2} - q^{-1/2})$, where $q = e^{i2\pi/(k+2)}$ is the deformation parameter.

For $k > 3$ all three F_d coefficients are nonzero. Thus, in order for the action of 2.27 on the $a = b = 1$ state to produce the same state back (up to a phase), the projection of this state in the three d sectors must all acquire the same phase. This implies that $\alpha = 0$ and $\beta = -\gamma$. The resulting unitary operation must therefore take the form

$$U = \left(\begin{array}{cc|ccc|c} 1 & 0 & & & & \\ 0 & 1 & & & & \\ \hline & & e^{i\beta} & 0 & 0 & \\ & & 0 & e^{-i\beta} & 0 & \\ & & 0 & 0 & 1 & \\ \hline & & & & & 1 \end{array} \right), \quad (2.30)$$

which corresponds to the following two-qubit gate in the $ab = \{00, 01, 10, 11\}$ basis,

$$U_{k>3}^{gate} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{i\beta} & 0 & 0 \\ 0 & 0 & e^{-i\beta} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.31)$$

This gate is simply the tensor product of two single-qubit rotations, $U_{k>3}^{gate} = e^{-i\beta\sigma_z^{(1)}/2} \otimes e^{i\beta\sigma_z^{(2)}/2}$. Thus we see that for $k > 3$ any two-qubit gate constructed by braiding only four quasiparticles for which there is no leakage error must necessarily also produce no entanglement.

For $k = 3$ this argument breaks down because the $d = 2$ sector of the braiding quasiparticles is not present. In this case, following the same argument as above, in the $abd = \{000, 110, 011, 101, 111\}$ basis the allowed leakage free unitary transformations which can be produced by braiding the four middle quasiparticles must be of the form (again taking the case of zero winding),

$$U = \left(\begin{array}{cc|ccc} e^{i\alpha} & 0 & & & & \\ 0 & e^{-i\alpha} & & & & \\ \hline & & e^{i\beta} & 0 & 0 & \\ & & 0 & e^{i(\alpha-\beta)} & 0 & \\ & & 0 & 0 & e^{-i\alpha} & \end{array} \right), \quad (2.32)$$

which corresponds to the following two-qubit gate in the $ab = \{00, 01, 10, 11\}$ basis,

$$U_{k=3}^{gate} = \begin{pmatrix} e^{i\alpha} & 0 & 0 & 0 \\ 0 & e^{i\beta} & 0 & 0 \\ 0 & 0 & e^{i(\alpha-\beta)} & 0 \\ 0 & 0 & 0 & e^{-i\alpha} \end{pmatrix}. \quad (2.33)$$

As for $U_{k>3}^{gate}$, the β dependence of $U_{k=3}^{gate}$ corresponds to a tensor product of single-qubit rotations. Gates of this form with fixed α but different values of β are thus equivalent up to single-qubit rotations. If we use this equivalence to set $\beta = \alpha$ we see that gates of the form $U_{k=3}^{gate}$ are equivalent to the gates produced by our F weave construction 2.26, and so, in particular, when $\alpha = \pi$ the resulting gate is equivalent to a controlled-NOT gate.

2.8 Summary

To summarize, we have shown how to construct both single-qubit and two-qubit gates for qubits encoded using non-Abelian quasiparticles described by $SU(2)_3$ theory, or, equivalently,

the $SO(3)_3$ theory (Fibonacci anyons). Qubits are encoded into triplets of quasiparticles and single-qubit gates are carried out by braiding quasiparticles within qubits. Two classes of two-qubit gate constructions were presented. In the first, a pair of quasiparticles from one qubit is woven through those forming the second qubit. In the second, a single quasiparticle is woven through three static quasiparticles (one from the same qubit as the mobile quasiparticle, the other two from the second qubit). A central theme in all of our two-qubit gate constructions is that of breaking the problem of compiling braids for the six quasiparticles used to encode two qubits into a series of braids involving only three objects at a time. While these constructions do not in general produce the optimal braid of a given length which approximates a desired two-qubit gate, we believe they do lead to the most accurate (or at least among the most accurate) two-qubit gates which can be obtained for a fixed amount of classical computing power. Finally, we proved a theorem which states that for the $SU(2)_k$ theory, two-qubit gates constructed by braiding only four quasiparticles (two from each qubit) can only lead to leakage free entangling two-qubit gates when $k = 3$.

CHAPTER 3

Compiling Braids for $SU(2)_k$ Anyons

In Chapter 2, an explicit recipe for carrying out universal quantum computation using Fibonacci anyons was presented. Specifically, it was shown how qubits can be encoded using Fibonacci anyons, and how a universal set of quantum gates (single-qubit rotations and CNOT gates) can be performed on the encoded qubits by braiding anyon world-lines.

As was pointed out in Chapter 2, the fact that Fibonacci anyons are, in principle, universal for quantum computation had already been proven mathematically by Freedman, Larsen and Wang [25, 26] prior to the work described in this Thesis. The main new contribution described in Chapter 2 was an *efficient* scheme for finding the necessary braids. This scheme was based on a “divide and conquer” approach which reduced all the required brute force searches to those involving braids of only three objects at a time. Furthermore, the corresponding unitary operations were all in $SU(2)$ — a Lie group for which the parameter space is just three-dimensional. Using this approach we were able to, for the first time, construct explicit braiding patterns corresponding to a universal set of quantum gates for Fibonacci anyons. Essentially what our work did was to provide a “compiler” for translating any high-level quantum algorithm (expressed as a sequence of single-qubit rotations and CNOT gates), into the “machine code” of a topological quantum computer (a braid).

In Chapter 2 it was also pointed out that the Fibonacci anyon model, which (for our purposes) is effectively equivalent to the $SU(2)_3$ anyon model, is one of a series of anyon models — the $SU(2)_k$ anyon models labeled by an index k (the so-called “level”) which can be any positive integer. These anyon models are particularly interesting since they are not just pure theory — there is evidence that it may be possible to realize them in fractional quantum Hall systems at filling fraction $\nu = k/(k + 2)$ [51, 73] and in rotating Bose gases

at filling fraction $\nu = k/2$ [28]. Furthermore, Freedman, Larsen and Wang have shown that $SU(2)_k$ anyons are universal for quantum computation, not just for $k = 3$ (corresponding to Fibonacci anyons), but also for all $k > 4$ [26].

This raises the natural question of whether the techniques described in Chapter 2 for Fibonacci anyons (or equivalently, quasiparticles of $SU(2)_3$ anyon model) can be generalized to all the $SU(2)_k$ models which are universal for quantum computation. The Fibonacci anyon gate constructions we have presented so far, have all relied on the special simplicity of this model, namely, the fact that there can only be *two* distinct q -spins of 0 and 1. This is no longer true for the general $SU(2)_k$ models. Nonetheless, in this Chapter I will give explicit constructions which, though more complex than those for Fibonacci anyons, can be used to efficiently find braids for $SU(2)_k$ particles. The key common feature between these constructions and those of Chapter 2 is that, again, we apply a “divide and conquer” approach, i.e. we break the problem of braiding many strands to a series of braids, each consisting of only three objects. This is done in such way that the corresponding unitary operations are all in $SU(2)$, therefore the braid sequences can be found efficiently.

In this Chapter I will briefly review the properties of the $SU(2)_k$ models with emphasis on those properties that are important from the point of view of quantum computing. Similar to the discussion in Chapter 2, I will describe how in these models qubits can be encoded, how single-qubit gates can be carried out and finally how two-qubit gates are constructed. I will explain why the methods introduced in Chapter 2 for Fibonacci anyons cannot be applied to quasiparticles of $SU(2)_k$ for $k > 3$ and introduce a new method for carrying out such gates.

3.1 $SU(2)_k$ Revisited

Although the properties of $SU(2)_k$ quasiparticles were reviewed in Sec. 2.1, the main focus there was on the equivalence of $SU(2)_3$ quasiparticles to Fibonacci anyons. Here, at the risk of some repetition, but for completeness, and to establish the notation for the rest of the Chapter, I will briefly review the properties of these quasiparticles which are crucial for understanding our gate constructions. (Note that detailed discussions of the mathematics of $SU(2)_k$ quasiparticles can be found in many textbooks on “quantum groups” — one excellent example is [72] — or in a 2001 paper by Slingerland and Bais [73].)

As described in Sec. 2.1, $SU(2)_k$ quasiparticles carry a quantum number which, in many

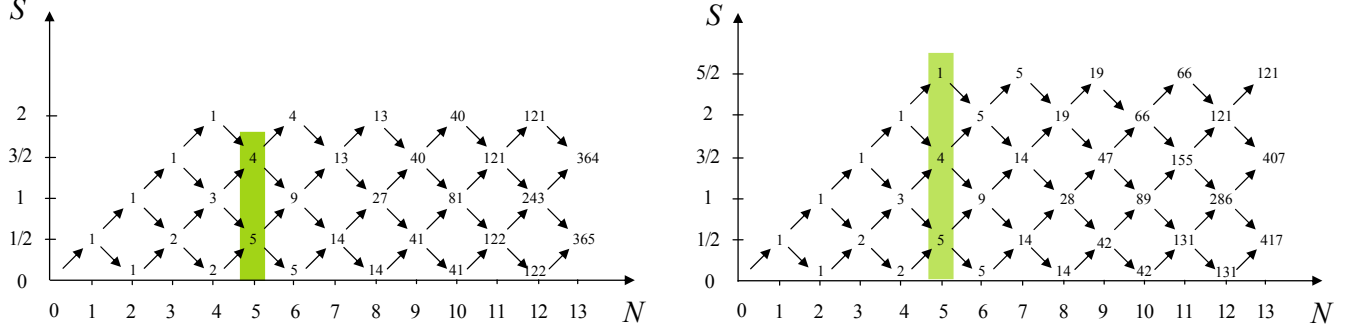


Figure 3.1: Bratteli diagrams for quasiparticles of $SU(2)_k$ for $k = 4$ and $k = 5$. N is the number of q -spin $1/2$ quasiparticles and S is the total q -spin. The highest possible total q -spin is $S = k/2$. The numbers written at each vertex (S, N) represent the dimensionality of the Hilbert space of N q -spin $1/2$ quasiparticles with total q -spin of S . For example, for four q -spin $1/2$ quasiparticles with $k = 4$ (marked with a green strip on the left diagram), the total q -spin can be $1/2$ or $3/2$ and the corresponding Hilbert space is 5 or 4 dimensional, respectively. When $k = 5$, the total q -spin of five q -spin $1/2$ quasiparticles can be $1/2$, $3/2$ or $5/2$ and the corresponding Hilbert space is 5, 4 or 1 dimensional, respectively.

ways, is analogous to ordinary spin and which we refer to as q -spin. For the level k theory the allowed values of this q -spin are,

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots, \frac{k}{2}. \quad (3.1)$$

The fusion rule which tells us what the allowed total q -spin values are when we combine two particles, is then a truncated version of the usual triangle rule for adding angular momentum,

$$s_1 \otimes s_2 = |s_1 - s_2| \oplus (|s_1 - s_2| + 1) \cdots \oplus \min(s_1 + s_2, k - (s_1 + s_2)). \quad (3.2)$$

These fusion rules can be used to construct the so-called Bratteli diagrams. As was mentioned in Sec. 1.3.3 and also in Sec. 2.1, Bratteli diagrams provide a convenient way to visualize the Hilbert space of non-Abelian particles. If we consider systems of quasiparticles with q -spin $1/2$, then Fig. 2.1 shows the Bratteli diagrams for the cases $k = 2$ and $k = 3$ and Fig. 3.1 shows these diagrams for $k = 4$ and $k = 5$. In these diagrams, the horizontal axis represents the number of quasiparticles with q -spin $1/2$ (N), and the vertical axis represents the total q -spin of the quasiparticles (S). Note that the maximum possible q -spin for quasiparticles of the $SU(2)_k$ theory is $k/2$. The numbers written in the diagram

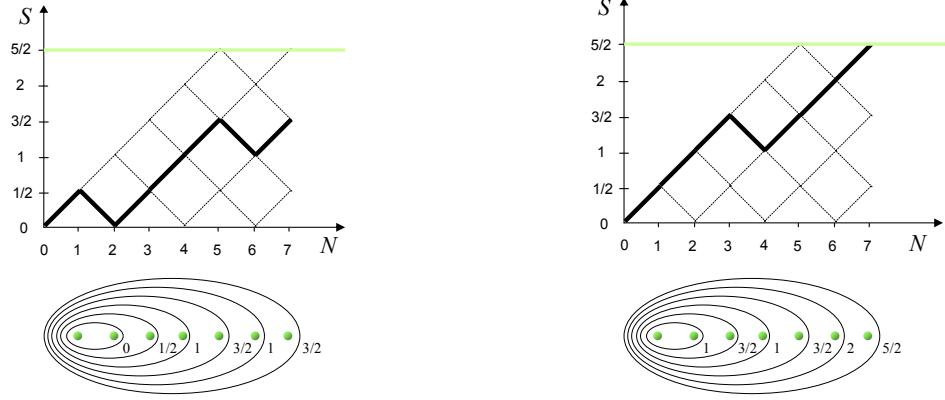


Figure 3.2: Bratteli diagram and the oval notation. Each path in the Bratteli diagram corresponds to a state in the Hilbert space of quasiparticles. The green lines represent the cutoff q -spin, $k/2$, which in this example is $5/2$. In the oval notation, each dot represents a q -spin $1/2$ quasiparticle and the numbers written next to each oval correspond to the total q -spin of the quasiparticles enclosed by the oval.

at each vertex (S, N) represent the dimensionality of the Hilbert space of N q -spin $1/2$ quasiparticles with total q -spin S . Asymptotically, the dimensionality of the Hilbert space of N quasiparticles with q -spin $1/2$ grows as $([2]_q)^N$ where $[m]_q$ is the q -deformed integer, defined as $[m]_q = (q^{m/2} - q^{-m/2}) / (q^{1/2} - q^{-1/2})$ and $q = e^{i2\pi/(k+2)}$ (see e.g. [75]). Note that the q -integer for most values of k is irrational, therefore, the Hilbert space of $SU(2)_k$ anyons cannot be decomposed as the tensor product of smaller subsystems. Consequently, to carry out quantum computation in the standard qubit model we must encode our qubits, using several quasiparticles. We postpone the definition of a qubit to the next Section.

Note that each “path” in the Bratteli diagram represents a state in the Hilbert space of $SU(2)_k$ quasiparticles. Figure 3.2 illustrates two examples of different states in the Hilbert space of seven $SU(2)_5$ quasiparticles. These states are shown as paths in the Bratteli diagram as well as ovals with numbers. In the figures throughout this Chapter, q -spin $1/2$ quasiparticles are denoted by dots, and ovals represent collections of quasiparticles. The number written next to each oval represents the total q -spin of the collection of quasiparticles enclosed by that oval. Note that in the text throughout this Chapter we represent q -spin $1/2$ quasiparticles with $\bullet$ and ovals with parenthesis. For example, in Fig. 3.2, on the left panel, the state of the first four quasiparticles can be represented by $(((\bullet, \bullet)_0, \bullet)_{1/2}, \bullet)_1$. In



Figure 3.3: R and F transformations for $SU(2)_k$. R is the unitary operation corresponding to the exchange of two q -spin $1/2$ quasiparticles in a clockwise manner. F represents a unitary transformation corresponding to a change of basis. The initial basis is shown on the left hand side of F in which, first, the two bottommost quasiparticles are fused and then the result is combined with the topmost quasiparticle. The final basis is shown on the right and in which first the two topmost quasiparticles are combined and then the result is fused with the bottommost quasiparticle.

this Chapter we also introduce a new notation, $\bigcirc$, which denotes a pair of quasiparticles with total q -spin of 1. In other words, we have $(\bullet, \bullet)_1 \equiv \bigcirc$.

Similar to the discussion of Fibonacci anyons in Chapter 2, $SU(2)_k$ anyon models have F and R structures which can be used to determine the effect of changing bases and interchanging quasiparticles. For the case of Fibonacci anyons these matrices were very simple due to the fact that there were only two possible q -spins, 0 and 1. In general, however, quasiparticles of $SU(2)_k$ models can have $k + 1$ different q -spins (see Eq. 3.1). Therefore F and R matrices for the $SU(2)_k$ models, in general, have more indices than for the Fibonacci anyon model. In this Thesis I will not give an exhaustive presentation of the properties of F and R matrices. However, the details of the F matrices of $SU(2)_k$, which we will need in our construction described in this Chapter, will be given in Appendix A. Here, for completeness, we present the simplest form of F and R matrices which are needed to construct single-qubit gates. These matrices are extracted from [73].

The simplest form of an R matrix provides the phase factors corresponding to the process of exchanging two q -spin $1/2$ quasiparticles with a certain sense. Figure 3.3 illustrates this exchange carried out in a clockwise manner which gives rise to,

$$R = \begin{pmatrix} e^{\frac{i(2k+1)\pi}{2(k+2)}} & 0 \\ 0 & e^{\frac{i\pi}{2(k+2)}} \end{pmatrix}. \quad (3.3)$$

Also, the simplest F matrix, corresponding to the following change of basis,

$$((\bullet, \bullet)_a, \bullet)_{1/2} = F_{ab}(\bullet, (\bullet, \bullet)_b)_{1/2}, \quad (3.4)$$

as shown in Fig. 3.3, has the form,

$$F = \begin{pmatrix} \frac{1}{[2]_q} & \frac{\sqrt{[3]_q}}{[2]_q} \\ \frac{\sqrt{[3]_q}}{[2]_q} & -\frac{1}{[2]_q} \end{pmatrix}. \quad (3.5)$$

Note that F and R matrices, as described above, are the only unitary operations that are needed for constructing the elementary braid matrices that are used for carrying out arbitrary single-qubit gates. The exact form of these elementary braid matrices will be given in the next Section.

3.2 Encoding Qubits and Single-Qubit Gates

It was shown in the previous Section that when quasiparticles of $SU(2)_k$ models are present, the corresponding Hilbert space does not have a simple tensor product form. Therefore, to impose a qubit structure on this Hilbert space we need to encode qubits. As before, the price we pay for this encoding is to introduce non-computational states. Transitions between the qubit states and the non-computational states are leakage errors which can potentially spoil the computation. In this Section I will briefly review the encoding scheme for quasiparticles of $SU(2)_k$ models and explain how single-qubit gates can be carried out.

Similar to Fibonacci anyons, for quasiparticles of $SU(2)_k$ models, qubits can be encoded using three or four quasiparticles, each carrying q -spin $1/2$, where the total q -spin of the entire collection of three or four quasiparticles is $1/2$ or 0 , respectively (see Fig. 2.4 and note that for three-quasiparticle qubits, in general, the overall q -spin is $1/2$ not 1). Note that as far as single-qubit gates are concerned, qubits encoded with three or four quasiparticles are essentially equivalent, however, when considering two-qubit gates, certain gate constructions can only be implemented on qubits encoded with four quasiparticles. An example is the construction of a controlled-Phase gate described in Sec. 2.6.1. The gate constructions described in this Chapter are also based on qubits that are encoded using four quasiparticles and they cannot be implemented on three-quasiparticle qubits. The implementation of a two-qubit gate using three-quasiparticle qubits for quasiparticles of $SU(2)_k$ anyon models with $k > 3$ is still an open question.

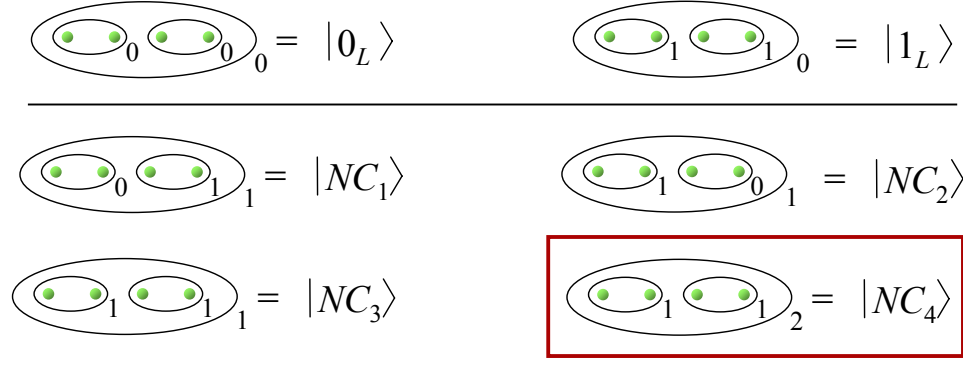


Figure 3.4: Four-quasiparticle qubit encoding. Top panel: Qubits can be encoded using four $SU(2)_k$ quasiparticles when the total q -spin of the group of four is 0. The logical states of the qubits can be determined by the total q -spin of either the two rightmost quasiparticles or the two leftmost quasiparticles (they must be the same). Bottom panel: The non-computational states of four quasiparticles. Note that the state $|NC_4\rangle$ (marked by a red box) was absent for Fibonacci anyons (Fig. 2.4) but is present for $SU(2)_k$ quasiparticles with $k > 3$.

As is shown in Fig. 3.4, for qubits encoded with four quasiparticles, the logical state of the qubit is determined by the total q -spin of either the two leftmost or the two rightmost quasiparticles (the two must have the same total q -spin to define a qubit). In general, the Hilbert space of four q -spin $1/2$ quasiparticles is six-dimensional (see Fig. 3.1), therefore, the remaining four states that do not correspond to qubit states are the non-computational states. Any transitions from the qubit space to this four-dimensional non-computational space is defined as a leakage error which can potentially spoil the computation.

Similar to Fibonacci anyons, for quasiparticles of $SU(2)_k$ anyon models, single-qubit gates can be carried out by braiding quasiparticles within qubits. It has been shown [25] that, in principle, for quasiparticles of $SU(2)_k$ for $k \geq 3$, $k \neq 4, 8$, all single-qubit gates can be approximated, to any desired accuracy, by braiding three quasiparticles with q -spin $1/2$ within a qubit. Similar to the discussion for Fibonacci anyons in Sec. 2.5.1, the elementary braid matrices that act on the Hilbert space of three $SU(2)_k$ quasiparticles with q -spin $1/2$, can be calculated from F and R matrices given in Eqs. 3.5 and 3.3.

Focusing on three q -spin $1/2$ quasiparticles as shown in Fig. 3.5, in a basis that is labeled as $((\bullet, \bullet)_a, \bullet)_1$, the unitary operation that corresponds to exchanging the two leftmost

quasiparticles (or the two bottommost quasiparticles in the figure) is simply the same as R ,

$$\sigma_1 = R. \quad (3.6)$$

Likewise, σ_2 which represents the unitary operation that results from exchanging the two rightmost quasiparticles (or the two topmost quasiparticles in Fig. 3.5) can be calculated by first changing the basis to which the two topmost quasiparticles have a well-defined q -spin using an F matrix, then exchanging the two topmost particles using an R matrix, and finally returning to the original basis using the inverse of an F matrix. Therefore,

$$\sigma_2 = F^{-1}RF. \quad (3.7)$$

σ_1 and σ_2 as described above, are the only braid matrices that are needed for carrying out single-qubit gates.

It has shown that any unitary operation in $SU(2)$ that can be approximated by braiding three strands, can also be approximated by *weaves* i.e. three-stranded braids in which, one strand weaves around the other two [84]. Following our strategy in Chapter 2, here we also focus on weaves. A single-qubit gate then in general can be written as,

$$U(\{n_i\}) = \prod_i \sigma_{m_i}^{n_i}. \quad (3.8)$$

In this expression $m_i \in \{1, 2\}$ and $n_i \in \{0, \pm 2, \pm 4, \dots, n\}$ where $n = k + 1$ if k is odd and $n = k + 2$ if k is even. The length of the sequence is $L(\{n_i\}) = \sum_i |n_i|$ and the winding number is simply $W(\{n_i\}) = \sum_i n_i$. Note that the winding number is defined modulo $(2(k + 2))$.

It should be noted that due to the results of Freedman *et al* [25], for quasiparticles of $SU(2)_k$ when $k \geq 3$ and $k \neq 4, 8$, in principle, all single-qubit gates can be approximated to any desired accuracy by performing a brute force search over the space of sequences of the form 3.8. In practice however, achieving arbitrary accuracy through brute force searching is not feasible. This is because the number of distinct braids of a fixed length grows exponentially with the length of the braid, so as the length of the braid increases, it takes exponentially longer to search through all the possibilities to find the best approximation. A practical way to obtain arbitrarily accurate braids is by applying the Solovay-Kitaev algorithm.



Figure 3.5: Elementary braid matrices. For three q -spin $1/2$ quasiparticles, σ_1 corresponds to exchanging the two bottommost quasiparticles in a clockwise sense and σ_2 corresponds to the exchange of the two topmost quasiparticles with the same sense. As is shown in the text, the two are related by an F matrix.

As was described in Chapter 2, roughly speaking, the Solovay-Kitaev algorithm provides a procedure for systematically improving the accuracy of a given braid without having to search through all possibilities. In this procedure one needs to generate a large library of braids up to a certain length, then based on the form of the desired unitary operation one will pick different segments of braids out of this library and sew them together to construct braids that are longer and systematically more accurate. Therefore, just like the case of Fibonacci anyons, both as a matter of principle and practice, single-qubits gates are easy to construct. Furthermore, since all the braiding takes place within the encoded qubits, one never needs to worry about leakage errors.

In Chapter 2, it was argued that two-qubit gates are significantly harder to construct, mainly due to the following two reasons. First, entangling two qubits requires braiding quasiparticles from one qubit around quasiparticles forming the second qubit and this will inevitably lead to leakage errors. Second, in general, the Hilbert space of eight quasiparticles is 14-dimensional and approximating any two-qubit gate that involves braiding eight quasiparticles requires a search in a 195-dimensional search space which is significantly larger than the 3-dimensional search space of a single-qubit gate. This large search space makes both the brute force searching and realization of the Solovay-Kitaev algorithm much harder (the Solovay-Kitaev algorithm must be implemented in $SU(14)$).

Following the recipe given in Chapter 2, our strategy to circumvent these problems is to restrict our search to those braids that involve at most three objects at a time and can be carried out by performing a search in $SU(2)$. This approach greatly reduces the computational difficulty of the problem but in general it leads to longer braids. This is

because we need to divide the two-qubit gate into several $SU(2)$ sectors (“divide and conquer” approach). In what follows, we describe the details of our construction for a two-qubit gate that can be carried out for quasiparticles of $SU(2)_k$ anyon models with $k \geq 3$, $k \neq 4$.

3.3 Two-Qubit Gates

All of our two-qubit gate constructions for Fibonacci anyons described in Chapter 2, take advantage of the fact that, for quasiparticles of $SU(2)_3$ anyon models, a pair of quasiparticles with q -spin 1 has the same braiding properties as a quasiparticle with q -spin $1/2$ (up to overall Abelian phases which are irrelevant for quantum computing). This, however, is not true in general. In particular, for quasiparticles of $SU(2)_k$ anyon models with $k \geq 4$, braiding a pair of quasiparticles with total q -spin 1 around other objects produces a different unitary operation from the one generated by braiding a q -spin $1/2$ quasiparticle around the same objects. Therefore, the methods introduced in Chapter 2, in general, cannot be applied to quasiparticles of $SU(2)_k$ models with $k \neq 3$.

In the following sections, I will describe a new method for constructing two-qubit gates, using quasiparticles of $SU(2)_k$ models. This construction is presented in two parts. In the first part, general ideas and the unitary operations corresponding to each step in the construction of the final gate are presented. In the second part, the details of how the braids can be found and all the information on braid matrices and fusion matrices that are necessary for building these gates are described in detail.

3.3.1 New Method for $k > 3$ – General Ideas

Similar to our two-qubit gate constructions described in Sec. 2.6.2, this new two-qubit gate is also carried out in three steps. In the first step, as before, the idea is to find a braid which, effectively, takes a pair of quasiparticles from the control qubit and “injects” them inside the target qubit. Carrying out this step will change the state of the system into an intermediate state in a controlled way (i.e. without introducing irreversible leakage errors to the final gate, in the limit of infinitely accurate braids^a). Then, in the second step a rotation is induced on the injected target qubit, and finally, in the third step all quasiparticles are

^aAs will become clear in the following, the intermediate state is, in fact, a mixture of qubit states and non-computational states, therefore strictly speaking, the first step does introduce leakage to the system however, the effects of these errors will be canceled in the third step before the final readout is performed.

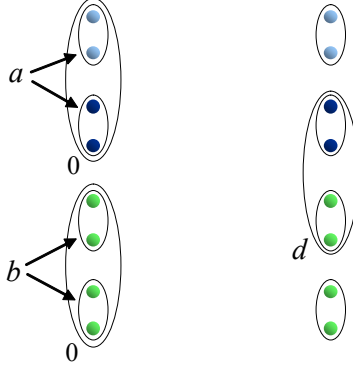


Figure 3.6: Left panel shows qubit basis and the right panel shows the d -basis.

returned to their original positions and the final state of the system can be read out.

To set the notation, consider the two qubits shown in Fig. 3.6. As before, the quasiparticles in the control qubit are shown in blue, and the quasiparticles in the target qubit are shown in green. The logical states of the control and target qubits are determined by the labels a and b , respectively. The two quasiparticles shown in dark blue form the *control pair* which is the only mobile part of the system i.e. all braids used in this construction consist of weaving the control pair around two other quasiparticles with q -spin $1/2$ which remain in their positions. As was mentioned above, when the overall q -spin of the control pair is 0 ($a = 0$), moving it around other objects does not induce any non-Abelian transition (note that the process of moving the control pair around other quasiparticles in the system does not change the total q -spin of the control pair). Therefore, as before, the only non-trivial case is when the total q -spin of the control pair is 1 ($a = 1$). In what follows, we mostly concentrate on this case, keeping in mind that weaving a pair of quasiparticles with total q -spin 0 can only produce an overall trivial phase.

In the first step of our two-qubit gate construction (see Fig. 3.7), the control pair starts from the bottom position in the control qubit, weaves around the two topmost quasiparticles in the target qubit, and ends underneath them in the target qubit. Considering the full Hilbert space of the four middle quasiparticles (see Fig. 3.7 (b)), and focusing on braids with zero total winding,^b the most general unitary operation that can be carried out in this way,

^bNote that we can restrict our braids to those with zero total winding without loss of generality.

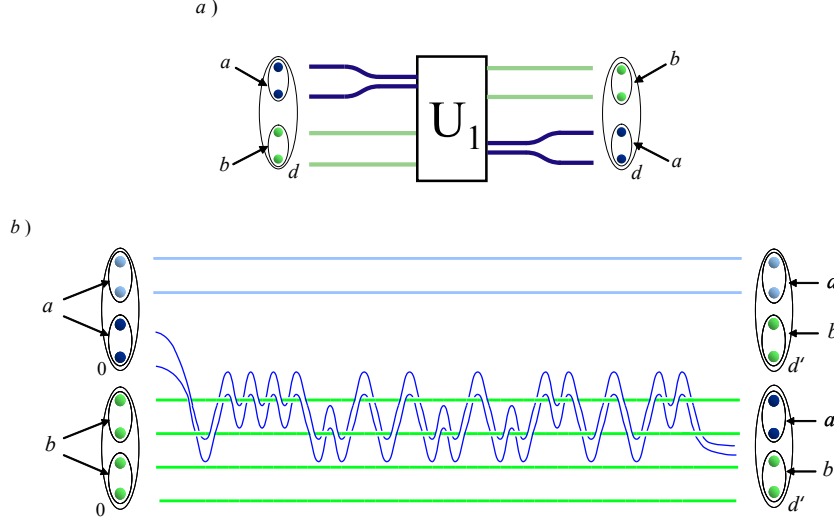


Figure 3.7: The first step in the construction of a controlled-Z gate. A pair of quasiparticles from the control qubit (the control pair) which is shown in dark blue is woven around single quasiparticles in the target qubit (green particles). This operation exchanges a from the control qubit with b from the target qubit without introducing leakage errors to the system. The box on the top panel represents an ideal (infinite) braid with a unitary operation that corresponds to $\mathcal{U}_1$. The braiding pattern in the bottom panel is the result of a brute force search which approximates $\mathcal{U}_1$ with an accuracy of $O(10^{-2})$, when $k = 5$.

in a basis which is labeled by $abd = \{000, 110, 011, 101, 111, 112\}$, can be shown to be

$$\mathcal{U}_1 = \left(\begin{array}{cc|ccc|c} \textcolor{green}{1} & 0 & & & & \\ 0 & \textcolor{red}{1} & & & & \\ \hline & & 0 & \gamma & \delta & \\ & & \textcolor{green}{1} & 0 & 0 & \\ & & 0 & \alpha & \beta & \\ \hline & & & & & \textcolor{red}{1} \end{array} \right). \quad (3.9)$$

where the 2×2 matrix

$$\mathcal{U}_1 = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad (3.10)$$

has determinant 1. Note that all the other matrix elements are completely fixed by considering the fact that weaving an object with total q -spin 0 does not induce any transitions (hence the matrix elements shown in green are both equal to 1), and the fact that for a braid with zero winding the determinant of each block must be 1.^c

^cSee e.g. Sec. 2.6 for an explanation.

This operation must be designed in such way that it effectively swaps the control pair with the topmost pair of quasiparticles in the target qubit (labeled by b). In other words, when focusing on the middle four quasiparticles (see Fig. 3.7 (a)), U_1 is a braid which starts from the state $|\psi_i\rangle = ((\bullet, \bullet)_a, (\bullet, \bullet)_b)_d$ and ends at the state $|\psi_f\rangle = ((\bullet, \bullet)_b, (\bullet, \bullet)_a)_d$.^d

Note that when a and b are both equal to 1, the above mentioned requirement indicates that $|\psi_i\rangle = |\psi_f\rangle$, i.e., the unitary operation corresponding to this braid must be equivalent to the identity. Therefore, the three matrix elements corresponding to $a = b = 1$ in the three blocks of $\mathcal{U}_1$ must all be equal to 1. From the form of $\mathcal{U}_1$ in 3.9, one can see that, by virtue of the zero winding requirement, two of these matrix elements are already equal to 1 (shown in red). Setting β to 1 will fix all the other matrix elements and yield,

$$\mathcal{U}_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (3.11)$$

Therefore, the goal in the first step of our construction is to find a braid U_1 , of the form shown in Fig. 3.7, that produces a unitary operation which approximates $\mathcal{U}_1$.

After the braid U_1 is applied, the system is in some intermediate state, labeled by a , b and d' , as shown in Fig. 3.7 (b). From this figure, it can easily be seen that when $a = 0$, we have $d' = b$, also when $a = 1$ and $b = 0$, then $d' = 1$. Note that when $a = 1$ and $b = 1$, U_1 is essentially equivalent to the identity operator (as argued above), therefore, in this case applying U_1 does not change the original state of the system and $d' = 0$. Thus, after applying U_1 , the state of the system in terms of a , b and d' is completely determined, as is summarized in Table 3.1.

Table 3.1: Values of d' as a function of a and b .

a	b	d'
0	0	0
0	1	1
1	0	1
1	1	0

^dNote that one might naively think that it might be possible to carry out this step by simply exchanging the control pair with the topmost pair of quasiparticles in the target qubit. However, as will be shown later in this section, this braid will lead to leakage errors in the final two-qubit gate therefore cannot be used as U_1 .

Having inserted the control pair inside the target qubit, we are now in the position to induce a rotation to the target qubit which is carried out in the second step of our construction. In this step, the control pair starts from the top position in the target qubit, weaves around the remaining two quasiparticles in the target qubit, and finally returns to its starting position on top of the target qubit (see Fig. 3.8). Focusing on the injected target qubit, in a basis that is labeled by $abd' = \{000, 110, 011, 101, 111, 112\}$, the most general unitary operation that is carried out in this way has the form

$$\mathcal{U}_2 = \left(\begin{array}{cc|ccc|c} 1 & 0 & & & & & \\ 0 & 1 & & & & & \\ \hline & & 1 & 0 & 0 & & \\ & & 0 & \alpha' & \beta' & & \\ & & 0 & \gamma' & \delta' & & \\ \hline & & & & & & 1 \end{array} \right), \quad (3.12)$$

where, again, the 2×2 matrix,

$$\mathcal{U}_2 = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}, \quad (3.13)$$

has determinant 1. Similar to the arguments presented for $\mathcal{U}_1$, all other matrix elements in $\mathcal{U}_2$ are fixed by the requirement that when the total q -spin of the control pair is zero, weaving it around other quasiparticles does not induce any transitions, and the fact that for a braid with zero total winding the determinant of each block must be 1.

To construct a two-qubit gate, a particular restriction must be imposed on this unitary operation. This restriction is that, there must not be any transitions between the states corresponding to $|abd'\rangle = |101\rangle$ and $|111\rangle$. As will become clear soon, this restriction is necessary to ensure that there are no leakage errors in the final two-qubit gate. Imposing this restriction implies that $\mathcal{U}_2$ must be a diagonal, unitary operation of the form

$$\mathcal{U}_2 = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix}. \quad (3.14)$$

In general, ϕ can be any phase and it determines the rotation induced on the target qubit. To summarize, in this step we must find a braid of the form shown in Fig. 3.8, with a unitary operation that approximates $\mathcal{U}_2$.

In the third and final step of this construction, the control pair must be returned to its original position in the control qubit. As before, this can be done by applying the inverse of U_1 . Note that applying this braid will also restore the original qubits, i.e., the two groups of

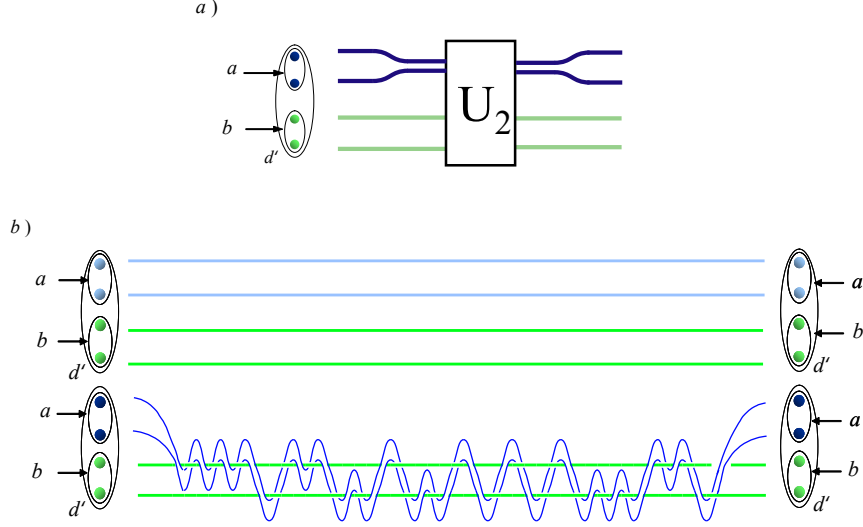


Figure 3.8: Second step in the construction of a controlled- Z gate for quasiparticles of $SU(2)_k$. In this braid the control pair (dark blue particles) weaves around two single particles in the target gate (green particles). The box on the top panel represents an ideal (infinite) braid with a unitary operation that corresponds to $\mathcal{U}_2$. The braiding pattern in the bottom panel is the result of a brute force search which approximates $\mathcal{U}_2$ with an accuracy of $O(10^{-2})$, when $k = 5$.

four quasiparticles, shown in the figures in blue and green, each have a total q -spin of 0 (see e.g. Fig. 3.9). This is possible because of the restriction imposed on $\mathcal{U}_2$ to be diagonal. The fact that applying U_2 did not mix the states corresponding to $|abd'\rangle = |101\rangle$ and $|111\rangle$ means applying the inverse of U_1 will take the system back to the state in which the overall q -spin of each set of four quasiparticles is equal to 0. As is illustrated in Fig. 3.9, this braid will return the control pair, as well as the pair of green quasiparticles with total q -spin b , back to their original positions in the control and target qubits, therefore restoring the original state of the control qubit.

Putting these three steps together, as is shown in Fig. 3.10, we end up with a braid which approximates a two-qubit gate of the form

$$M = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & e^{i\phi} & 0 \\ 0 & 0 & 0 & 1 \end{array} \right). \quad (3.15)$$

M represents a controlled-Phase gate in which, if the control qubit is in the state $|1_L\rangle$, the

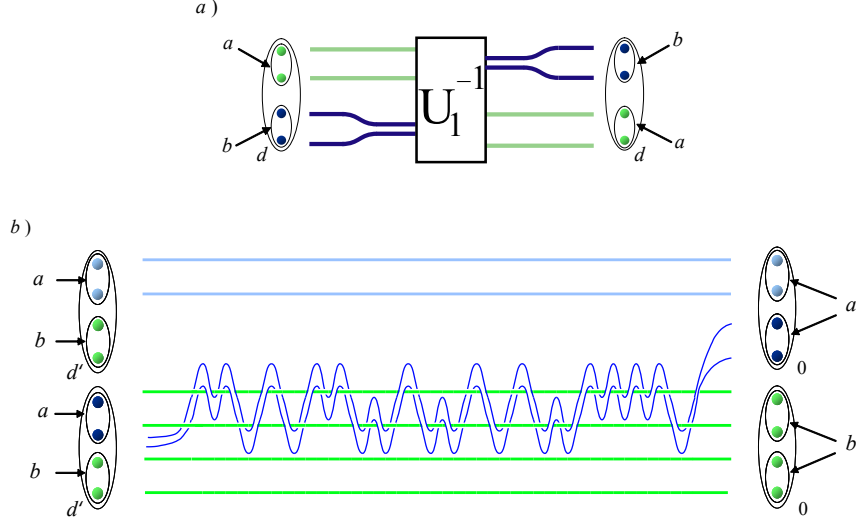


Figure 3.9: The third and last step in the construction of a controlled- Z gate. The braid corresponding to this step must return the control pair (shown in dark blue) to its original position in the control qubit and can be carried out by simply applying the inverse of U_1 , defined in Fig. 3.7.

target qubit is rotated around the $\hat{z}$ axis for an angle of ϕ . When $\phi = \pi$, this gate is a controlled- Z gate which is equivalent to a CNOT, up to single qubit rotations.

To summarize, in this Section, I presented an overview of the general ideas behind our construction of a two-qubit gate for quasiparticles of $SU(2)_k$ anyon models. I argued that these gates must be carried out in three steps, and presented the unitary operations corresponding to each step i.e. $\mathcal{U}_1$ and $\mathcal{U}_2$. In the next Section, I will describe in detail, how braids that correspond to these unitary operations can be found, by performing a search in $SU(2)$.

3.3.2 New Method for $k > 3$ – Details

In the previous Section we showed that in our construction of a two-qubit gate, braids corresponding to each step consist of four q -spin $1/2$ quasiparticles in which, a pair of quasiparticles (with total q -spin of 0 or 1) weaves around the two other quasiparticles (see Fig. 3.10). Knowing that braiding an object with total q -spin of 0 does not induce any non-Abelian transitions, the only nontrivial case is when the total q -spin of the moving pair

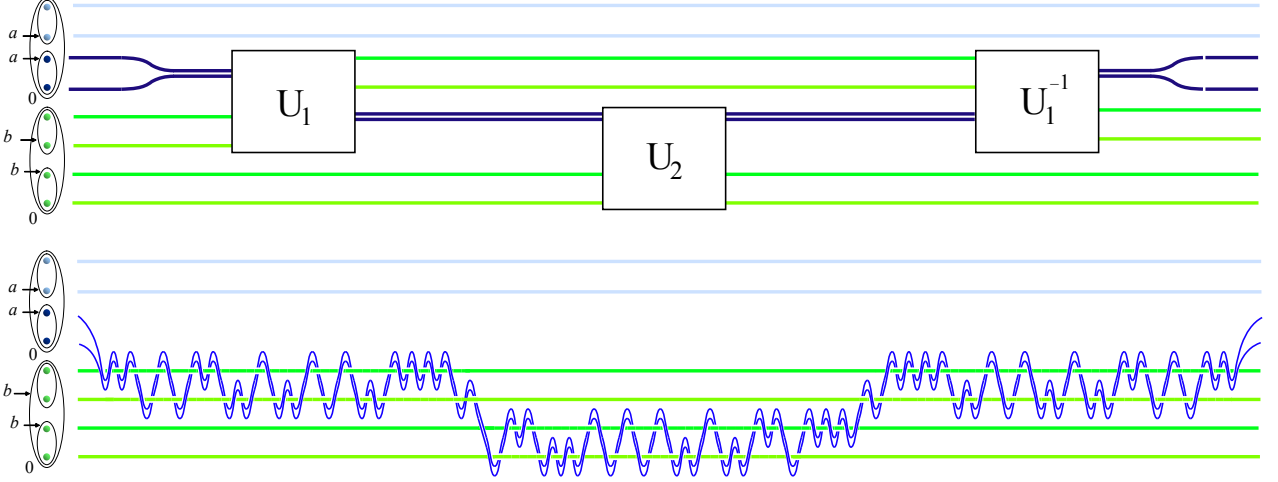


Figure 3.10: A controlled-Phase gate for quasiparticles of $SU(2)_5$. As before, the boxes labeled by U_1 , U_2 and U_1^{-1} represent ideal braids corresponding to unitary operations $\mathcal{U}_1$ and $\mathcal{U}_2$ in the text. The combination of these braids as shown in the top panel leads to a controlled-Phase gate. In the bottom panel, the result of a brute force search for braids that approximate $\mathcal{U}_1$ and $\mathcal{U}_2$ is illustrated. This braid approximates a controlled- Z gate with an accuracy of $O(10^{-2})$.

is 1.

In our constructions for Fibonacci anyons described in Chapter 2, we made extensive use of the fact that for quasiparticles of $SU(2)_3$, collections of quasiparticles with total q -spin of 1, when braided as a whole, produce the same non-Abelian operations as single q -spin $1/2$ quasiparticles (see Sec. 2.2). Consequently, the elementary braid matrices that corresponded to exchanging quasiparticles with q -spin $1/2$, could also be used for describing exchanges that involved pairs of quasiparticles, each with total q -spin of 1. However, for quasiparticles of $SU(2)_k$ when $k > 3$, this is no longer the case. In other words, a pair of quasiparticles with total q -spin 1 cannot be simply treated as a single quasiparticle with q -spin $1/2$. Therefore, braid matrices that involve braiding pairs of quasiparticles must be defined with extra care.

In particular, we must find braid matrices that correspond to weaving an object with total q -spin 1 around two other quasiparticles with q -spin $1/2$.^e Note that in what follows we will focus on a portion of the Hilbert space of four q -spin $1/2$ quasiparticles in which, two of

^eNote that all elementary braid matrices presented in this Section can be found in [73], as well as in standard text books on quantum groups, for example [72].

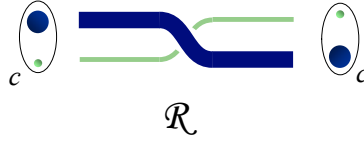


Figure 3.11: The “elementary” rotation. $\mathcal{R}$ is the rotation matrix corresponding to the exchange of a pair of quasiparticles with total q -spin 1 (shown in dark blue) with a q -spin 1/2 quasiparticle (shown in green).

the quasiparticles (the control pair) have total q -spin of 1 and the set of four quasiparticles also has total q -spin of 1 (in Fig. 3.7 (a), this corresponds to $a = 1$ and $d = 1$). Also, for simplicity, in the figures throughout this Section, we will denote a pair of quasiparticles with total q -spin of 1 with a big blue dot, and the corresponding world-line with a thick blue line. As before, the q -spin 1/2 quasiparticles which are stationary will be denoted by small green dots and the corresponding world-lines with thin green lines.

Figure 3.11 shows an elementary braid which corresponds to exchanging an object with total q -spin 1 (shown in the figure as a big blue dot), with a quasiparticle with q -spin 1/2 (small green dot). The corresponding unitary operation, in a basis that is labeled by $c = \{1/2, 3/2\}$, has the form

$$\mathcal{R} = \begin{pmatrix} e^{\frac{-i2\pi}{k+2}} & 0 \\ 0 & e^{\frac{i\pi}{k+2}} \end{pmatrix}. \quad (3.16)$$

This unitary operation is analogous to the rotation matrix, R , defined in 2.7 corresponding to the exchange of two Fibonacci anyons. Similar to the braid matrices of Fibonacci anyons discussed in Chapter 2, the exact form of all braid matrices which are used in this new construction can be calculated using $\mathcal{R}$ and a set of unitary operations that can be used to carry out arbitrary change of bases — the F matrices. The main difference between the calculation of braid matrices for quasiparticles of $SU(2)_k$ models and those of Fibonacci anyons introduced in Chapter 2 is that, for the latter we had only one non-trivial particle type, i.e., all objects had total q -spin of either 0 or 1. In general, however, groups of quasiparticles can have $k + 1$ different total q -spins (see Eq. 3.1), and several different F and R matrices need to be defined to account for different kinds of bases change and interchanges

of different objects. As was mentioned above, in our construction we do not need a complete set of F and R matrices. All of our braids consist of weaving an object with total q -spin 1 around quasiparticles carrying a q -spin of $1/2$. The corresponding F matrices will be presented in the following.

Figure 3.12 illustrates the three F matrices used in the two-qubit gate construction presented in this Chapter. The subscripts 1, 2 and 4 refer to the first, second and fourth F matrix in a set of five matrices that satisfy the pentagon consistency equation (see Sec. 2.3). This set of matrices can be calculated using equations given in [72] and [73] for example. Using these equations, a full set of matrices that satisfy the pentagon equation, for four quasiparticles with q -spin $1/2$ is presented in Appendix A. For simplicity, in this Chapter we focus on a truncated version of these F matrices and the reader is referred to the Appendix for more details.

As shown in Fig. 3.12, $\mathcal{F}_1$ corresponds to the following change of basis,

$$((\bigcirc, \bullet)_c, \bullet)_1 = \sum_b F_{cb}(\bigcirc, (\bullet, \bullet)_b)_1, \quad (3.17)$$

where $\bigcirc$ denotes a pair of quasiparticles with total q -spin of 1. Note that in this expression $c \in \{1/2, 3/2\}$ and $b \in \{0, 1\}$. This unitary operation has the form,

$$\mathcal{F}_1 = \begin{pmatrix} -\sqrt{\frac{[4]_q}{[2]_q[3]_q}} & \frac{1}{\sqrt{[3]_q}} \\ \frac{1}{\sqrt{[3]_q}} & \sqrt{\frac{[4]_q}{[2]_q[3]_q}} \end{pmatrix}. \quad (3.18)$$

Likewise, $\mathcal{F}_2$ corresponds to the following change of basis,

$$((\bullet, \bullet)_b, \bigcirc)_1 = \sum_{c'} F_{bc'}(\bullet, (\bullet, \bigcirc)_{c'})_1, \quad (3.19)$$

where, $c' \in \{1/2, 3/2\}$ and it has the form,

$$\mathcal{F}_2 = \begin{pmatrix} \frac{1}{\sqrt{[3]_q}} & \sqrt{\frac{[4]_q}{[2]_q[3]_q}} \\ -\sqrt{\frac{[4]_q}{[2]_q[3]_q}} & \frac{1}{\sqrt{[3]_q}} \end{pmatrix}. \quad (3.20)$$

Finally, $\mathcal{F}_4$ corresponds to

$$((\bullet, \bigcirc)_c, \bullet)_1 = \sum_{c'} F_{cc'}(\bullet, (\bigcirc, \bullet)_{c'})_1, \quad (3.21)$$

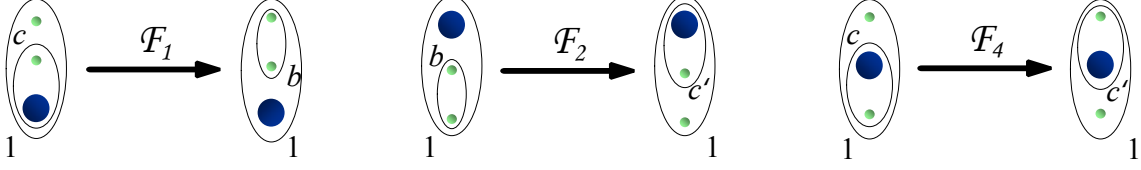


Figure 3.12: Change of bases transformations. Following our notation throughout this Chapter, the big blue dot corresponds to a pair of quasiparticles with total q -spin 1 and the small green dots represent q -spin 1/2 quasiparticles. Each symbol $\mathcal{F}$, represents a unitary operation that changes the basis from the one shown on the left to the basis on the right of each $\mathcal{F}$.

and it has the form

$$\mathcal{F}_4 = \begin{pmatrix} -\frac{1}{[3]_q} & \frac{\sqrt{[4]_q[2]_q}}{[3]_q} \\ \frac{\sqrt{[4]_q[2]_q}}{[3]_q} & \frac{1}{[3]_q} \end{pmatrix}. \quad (3.22)$$

Having set the notation, we can now proceed with the actual construction. Consider the four quasiparticles shown in Fig. 3.7 (a). As was mentioned above, we focus on the $a = 1$ and $d = 1$ sector of the Hilbert space, and we consider braids in which the control pair starts from the top position, weaves around the two green quasiparticles and ends underneath them in the bottom position. The desired braid must produce a unitary operation which approximates $\mathcal{U}_1$ (as is given in 3.11). Following a notation similar to that presented in Chapter 2, we consider the following Equation,

$$\sigma_{out_1} U(\{n_i\}) \sigma_{in} = \mathcal{U}_1. \quad (3.23)$$

As shown in Fig. 3.13, σ_{in} is a unitary operation that corresponds to exchanging the control pair, with the q -spin 1/2 quasiparticle underneath it in a clockwise manner, hence, moving the control pair from the top position to the middle position. Likewise, σ_{out_1} is a unitary operation that corresponds to moving the control pair from the middle position to the bottom position by exchanging it with the remaining q -spin 1/2 underneath it.

How do these unitary operations look like? As shown in Fig. 3.13, σ_{in} is defined in two different basis: $(\bigcirc, (\bullet, \bullet)_b)_1$ on the left and $((\bullet, \bigcirc)_c, \bullet)$ on the right. To find this unitary operation in terms of $\mathcal{R}$, we must change the bases on both sides in such way that the

overall q -spin of the control pair and the single quasiparticle it exchanges positions with, is well defined. This can be done by using the F matrices given in Fig. 3.12, in the following way,

$$\sigma_{in} = \mathcal{F}_4 \mathcal{R} \mathcal{F}_2. \quad (3.24)$$

Similarly, σ_{out_1} is also defined in two different bases: $((\bullet, \bigcirc)_{c'}, \bullet)$ on the left and $((\bullet, \bullet)_b, \bigcirc)_1$ on the right. Note that in the basis on the left, the total q -spin of the control pair and the quasiparticle it switches with is well defined (labeled by c' in the Fig. 3.13), therefore we only need to change the basis on the right. This can be done by using $\mathcal{F}_1$ (see Fig. 3.12). We have,

$$\sigma_{out_1} = \mathcal{F}_1 \mathcal{R}. \quad (3.25)$$

Thus, in Eq. 3.23, σ_{in} , σ_{out_1} and $\mathcal{U}_1$ are known. $U(\{n_i\})$ is then, the unitary operation corresponding to a braid in which, the control pair starts from the middle position, weaves around the two q -spin 1/2 quasiparticles on its top and bottom, and ends in the middle position. A brute force search should be carried out over sequences $\{n_i\}$, such that

$$U(\{n_i\}) = \prod_i (\sigma_{m_i}^d)^{n_i}, \quad (3.26)$$

where $m_i \in \{1, 2\}$ and $n_i \in \mathbb{Z}$ (modulo $(k+2)$).^f Note that σ_1^d and σ_2^d are the elementary *weaving* matrices which correspond to taking the control pair from the middle position, moving it around either the top or bottom quasiparticle, and returning it to its original position (as shown in Fig. 3.14). The subscript ‘ d ’, refers to *double* and is used as a reminder that σ_1^d and σ_2^d correspond to moving the control pair *one complete round* around a single quasiparticle.

In Fig. 3.14, σ_1^d is shown to be the unitary operation which corresponds to taking the control pair around the quasiparticle underneath it and returning it to its original position, in a clockwise manner. In the basis that is shown in Fig. 3.14, the overall q -spin of the control pair and the quasiparticle it encircles, is well defined (c in the figure), therefore σ_1^d can be simply defined as,

$$\sigma_1^d = \mathcal{R}^2. \quad (3.27)$$

^fNote that the definition of $U(\{n_i\})$ given in 3.26 is different from the definition of $U_{\text{weave}}(\{n_i\})$ given in 2.12. In particular, here σ_1^d and σ_2^d are *double* weave generators as will be described later, therefore n_i do not need to be even (which was the requirement for n_i in $U_{\text{weave}}(\{n_i\})$).

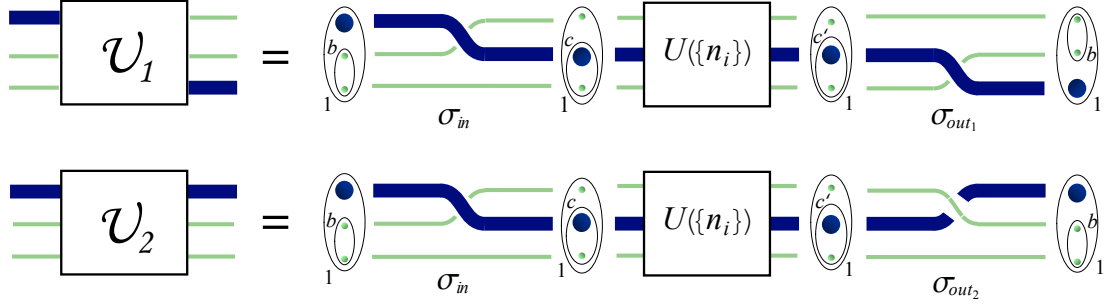


Figure 3.13: Braid in and braid out. The two boxes on the left represent ideal (infinite) braids that correspond to $\mathcal{U}_1$ and $\mathcal{U}_2$ given in the text in Eqs. 3.11, 3.14, respectively. On the right, the inner structures of the braids, i.e. graphical equivalent of Eqs. 3.23 and 3.29 are illustrated. σ_{in} is the unitary operation corresponding to the exchange of the control pair with total q -spin 1 (shown in blue) and a single q -spin 1/2 quasiparticle (shown in green). This braid will place the control pair at the middle position. $U\{n_i\}$ is the unitary operation corresponding to a braid in which the control pair starts from the middle position and ends in the middle position, and is found through brute force searching. In the top panel, σ_{out_1} is the unitary operation corresponding to a braid which takes the control pair to the bottom position by exchanging it with the quasiparticle under neath it. In the bottom panel, σ_{out_2} is the unitary operation which returns the control pair to its original top position by exchanging it with the quasiparticle above it. Note that σ_{in} , σ_{out_1} and σ_{out_2} are defined in two different bases on the right and on the left and this must be taken into consideration when the matrix representation of these operations are worked out. Also note that $U\{n_i\}$ is just a notation and the two boxes labeled by $U\{n_i\}$ on the top and bottom panels correspond to different braids.

Likewise, σ_2^d is the unitary operation that corresponds to taking the control pair around the quasiparticle in the top position (in a clockwise manner) and returning it to its original position. To find this matrix, as before, we must use the appropriate F matrix to change the basis to which the total q -spin of the control pair and the quasiparticle it encircles is well defined. In that basis then σ_2^d is simply the same as σ_1^d . From Fig. 3.12 one can see that this change of basis can be carried out by using $\mathcal{F}_4$, as defined in 3.22. Therefore, we have,

$$\sigma_2^d = \mathcal{F}_4^{-1} \sigma_1^d \mathcal{F}_4. \quad (3.28)$$

Using σ_1^d and σ_2^d , together with their inverses, one can carry out a brute force search over sequences of the form $U(\{n_i\})$ (as in 3.26) to find a unitary operation that satisfies Eq. 3.23. Note that all the matrices are 2×2 and, as before, the brute force search can be carried



Figure 3.14: “Double” braid matrices. σ_1^d is a unitary operation which corresponds to taking the control pair with total q -spin 1 (shown in dark blue) one complete round around the q -spin 1/2 quasiparticle underneath it (shown in green), in a clockwise manner. Similarly, σ_2^d is the unitary operation which corresponds to taking the control pair one complete round around the quasiparticle on top of it with the same sense.

out in $SU(2)$. In Fig. 3.7 (b), the result of a brute force search for the unitary operation that approximated $\mathcal{U}_1$ is illustrated.^g The unitary operation corresponding to this braid approximates $\mathcal{U}_1$ with an accuracy of $O(10^{-2})$. Note that similar to our constructions in Chapter 2, this approximation can be systematically improved by applying the Solovay-Kitaev algorithm in $SU(2)$.

To carry out the second step of our construction, consider the following Equation,

$$\mathcal{U}_2 = \sigma_{out_2} U(\{n_i\}) \sigma_{in}. \quad (3.29)$$

As is shown in Fig. 3.13, this Equation corresponds to a braid in which the control pair starts from the top position, weaves around the quasiparticles on the top and bottom, and finally returns to its original position on the top. As was described above, σ_{in} is the unitary operation that corresponds to exchanging the control pair with the quasiparticle underneath it and it is given in Eq. 3.24.

As before, $U(\{n_i\})$ is the unitary operation corresponding to a sequence $\{n_i\}$ which consists of σ_1^d , σ_2^d and their inverses, as given in Eq. 3.26. To take the control pair back to its original position, we need another elementary braid which describes the exchange of the control pair with the quasiparticle in the top position, as shown in Fig. 3.13. Once again, note that the corresponding unitary operation, σ_{out_2} is defined between two bases: $((\bullet, \bigcirc)_{c'}, \bullet)_1$ on the left and $((\bullet, \bullet)_b, \bigcirc)_1$ on the right. To change the basis to which the total q -spin of the control pair and the quasiparticle it exchanges with is well defined, we

^gIt should be noted that this braid does not exactly correspond to Eq. 3.23, instead, it corresponds to a unitary operation of the form $\widetilde{\sigma_{out_1}} U(\{n_i\}) \sigma_{in} = \mathcal{U}_1$ where $\widetilde{\sigma_{out_1}} = \mathcal{F}_1 \mathcal{R}^{-1}$. In the text, for simplicity, we focus on σ_{out_1} instead.

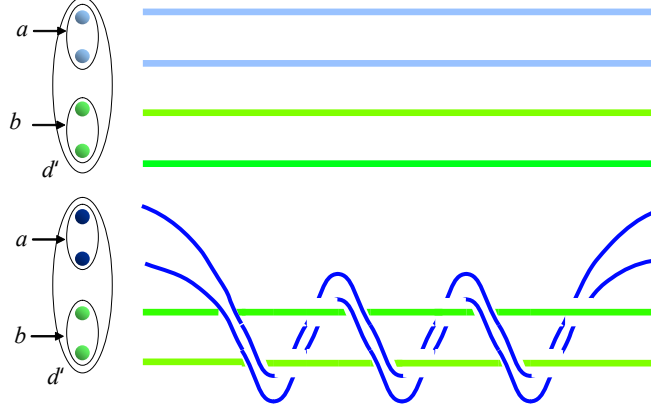


Figure 3.15: A finite braid that produces negative identity when $k = 22$. Here $W = 3$.

need to use the appropriate F matrices. From Fig. 3.12, one can see that σ_{out_2} can be defined as,

$$\sigma_{out_2} = \mathcal{F}_2^{-1} \mathcal{R} \mathcal{F}_4. \quad (3.30)$$

Knowing σ_{in} , σ_{out_2} and $\mathcal{U}_2$, we can carry out a brute force search over sequences $\{n_i\}$ to find $U(\{n_i\})$ such that Eq. 3.29 is satisfied. Figure 3.8 (b), illustrates a braid which is found through brute force searching.^h This braid approximates $\mathcal{U}_2$ with an accuracy of $O(10^{-2})$, where in this case,

$$\mathcal{U}_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.31)$$

Again, since the search for this braid is carried out in $SU(2)$, the accuracy of our approximations can be systematically improved by applying the Solovay-Kitaev algorithm in $SU(2)$.

The third step is easy, since we only need to reverse the braid found for $\mathcal{U}_1$. This reversed braid, takes the control pair back to its original position in the control qubit and the corresponding unitary operation approximates $\mathcal{U}_1^{-1}$ again with an accuracy of $O(10^{-2})$.

Putting the three braid segments, U_1 , U_2 and U_1^{-1} together (as shown in Fig. 3.10), we have a weave in which, the control qubit starts from the bottom position in the control qubit,

^hAgain, note that this braid does not exactly correspond to Eq. 3.29, instead in this braid $\widetilde{\sigma_{out_2}} = \mathcal{F}_2^{-1} \mathcal{R}^{-1} \mathcal{F}_4$.

weaves, first, around the two quasiparticles on top of the target qubit, and, second, around the two bottommost quasiparticles in the target qubit, and, finally, weaves back around the two topmost quasiparticles in the target qubit and returns to its original position in the control qubit. The result is an approximation to a controlled- Z gate of the form,

$$M = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{array} \right). \quad (3.32)$$

The braid shown in Fig. 3.10 approximates this gate with an accuracy of $O(10^{-2})$. Note that similar to our constructions for Fibonacci anyons described in Sec. 2.6.1, it is only the lower 2×2 corner of this matrix that is an approximation and all other matrix elements are exact.

It is interesting to note that if we loosen the zero winding constraint, certain $\mathcal{U}_2$ unitary operations can be carried out exactly. As is illustrated in Fig. 3.15, this can be done by weaving the control pair, an even number of times, around the pair of green quasiparticles with total q -spin of b . As before, for this braid, if $a = 0$ or $b = 0$, the corresponding unitary operation is exactly the identity. When $a = 1$ and $b = 1$, in a basis that is labeled by $d' = \{0, 1, 2\}$, this braid will produce the following unitary operation

$$U = \begin{pmatrix} e^{i\alpha} & 0 & 0 \\ 0 & e^{i\beta} & 0 \\ 0 & 0 & e^{i\gamma} \end{pmatrix}. \quad (3.33)$$

The matrix element relevant for our construction is $e^{i\alpha}$ and the resulting two-qubit gate will have the form

$$M = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\alpha} \end{array} \right). \quad (3.34)$$

where $\alpha = -8\pi W/(k+2)$ and W is the number of times the control pair (labeled by a in Fig. 3.15) weaves a complete round around the pair of quasiparticles labeled by b (see Fig. 3.15)ⁱ; for example in the figure, $W = 3$. Note that for certain values of k , it is possible to find an integer W such that $\alpha = -1$ which can be used to construct a controlled- Z gate. One can easily check to see this is possible for $k = 8W - 2$, i.e. $k = 6, 14, 22, \dots$.

ⁱNote that W as defined here is twice the winding number defined in Sec. 2.5.1.

Note that when the total q -spin of the control pair is 1, the Hilbert space of the system of four quasiparticles is in the form $U(1) \oplus U(2) \oplus U(1)$, corresponding to d , the overall q -spin of the four quasiparticles, to be 0, 1 or 2. In our discussion throughout this Chapter, we mostly focused on the $U(2)$ part. This was because the two $U(1)$ phases can always be fixed by restricting the total winding number of the braids. The actual brute force search is carried out by using σ_1^d and σ_2^d which are 2×2 matrices. Just like our two-qubit gate constructions for Fibonacci anyons, this brute force search can be carried out efficiently and the accuracy of the resulting braids can be improved systematically by applying the Solovay-Kitaev algorithm in $SU(2)$.

3.4 Summary

In this Chapter we provided efficient methods for carrying out single-qubit gates and two-qubit gates using quasiparticles of $SU(2)_k$ theory. Similar to the results of Chapter 2, the main idea used in this construction is to divide the hard problem of searching for braids that consist of many strands, into smaller sections each involving only three strands, in particular, weaving a pair of quasiparticles from one qubit around quasiparticles of the second qubit. This divide and conquer approach allows us to find braids that approximate two-qubit gates with excellent accuracy, by performing a search in $SU(2)$. The accuracy of the resulting braids can then be further improved, to any desired accuracy, by applying the Solovay-Kitaev procedure in $SU(2)$.

APPENDIX A

The Pentagon Equation

In this appendix we present the change of bases matrices which were used in Chapter 3. As was mentioned in Chapter 3, in general, when dealing with braiding properties of quasiparticles of $SU(2)_k$ models, it is important to have a consistent set of unitary operators that transform different bases to each other. The matrices which perform these transformations are the F matrices which we have already introduced for the case of Fibonacci anyons in Chapter 2. Because the Fibonacci anyon case is particularly simple, the F matrix given in Eq. 2.6, which is essentially just a 2×2 matrix, is sufficient to perform any basis change. For the general $SU(2)_k$ models this is no longer the case, and the F matrix must be treated as a full, rank 6 tensor, and is defined by the following equation,^a

$$(s_1, (s_2, s_3)_{s_{23}})_{s_{tot}} = \sum_{s_{12}} F_{s_1, s_2, s_3}^{s_{tot}, s_{12}, s_{23}} ((s_1, s_2)_{s_{12}}, s_3)_{s_{tot}}. \quad (\text{A.1})$$

Here the sum is over all values of s_{12} which are consistent with the relevant $SU(2)_k$ fusion rules. The F matrices for $SU(2)_k$ are in fact known, and tabulated (see, for example Eq. 78 in [73], which can be used to obtain the matrices given in Eqs. A.2-A.6).

Fortunately, for our gate constructions we will not need the full F matrices. Instead, we only need those F matrices which are relevant when changing basis between states made up of four q -spin $1/2$ quasiparticles. Fig. A.1 illustrates five possible ways to fuse four quasiparticles with q -spin $1/2$. This figure corresponds to the *pentagon equation*, which was mentioned in passing in Chapter 2. In this figure, each of the five vertices of the pentagon represents a different basis for four quasiparticles and each of the five F_i ($i = 1, \dots, 5$) matrices corresponds

^aNote that in this equation we are using a hopefully obvious generalization for our notation in which objects with q -spin s are represented by the symbol s — thus the equation describes three particles with q -spins s_1 , s_2 and s_3 , and two different bases in one of which the q -spin of s_1 and s_2 are combined first, and in the other the q -spin of s_2 and s_3 are combined first.

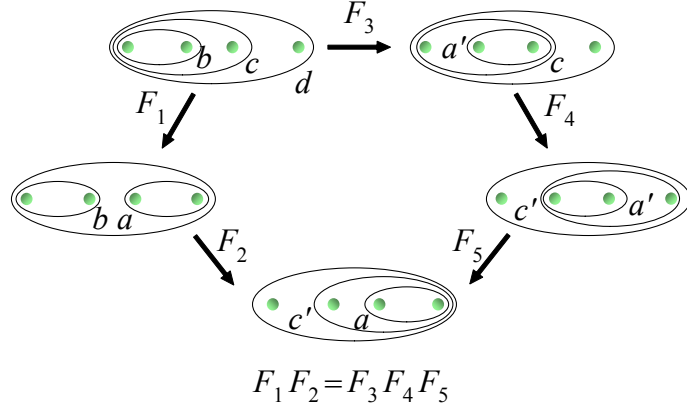


Figure A.1: Pentagon equation shows how to change basis from Bratteli basis (upperleft corner of the figure) to anti-Bratteli basis (lowerright corner). The basis labeled with a and b (obtained after applying F_1) is the qubit basis.

to a unitary operator that transforms one basis into another. To set the nomenclature, we call the basis in the upper left corner of Fig. A.1, which is in the form $(((\bullet, \bullet)_a, \bullet)_c, \bullet)_d$, the *Bratteli* basis and the basis in the bottom of the figure, the *anti-Bratteli* basis. Also, we call the basis in on the left, labeled by a and b , the *qubit* basis, for obvious reasons.

As is shown in Fig. A.1, F_1 is the matrix that takes us from the Bratteli basis labeled by $bcd = \{0\frac{1}{2}0, 1\frac{1}{2}0, 0\frac{1}{2}1, 1\frac{1}{2}1, 1\frac{3}{2}1, 1\frac{3}{2}2\}$ to the qubit basis labeled by $bad = \{000, 110, 011, 111, 101, 112\}$ and has the form,

$$F_1 = \left(\begin{array}{cc|ccc|c} 1 & 0 & & & & \\ 0 & 1 & & & & \\ \hline & & 1 & 0 & 0 & \\ & & 0 & -\sqrt{\frac{[4]_q}{[2]_q[3]_q}} & \frac{1}{\sqrt{[3]_q}} & \\ & & 0 & \frac{1}{\sqrt{[3]_q}} & \sqrt{\frac{[4]_q}{[2]_q[3]_q}} & \\ \hline & & & & & 1 \end{array} \right). \quad (\text{A.2})$$

F_2 is the unitary transformation that takes us from the qubit basis, labeled by $bad = \{000, 110, 011, 111, 101, 112\}$ to the anti-Bratteli basis, labeled by $ac'd = \{0\frac{1}{2}0, 1\frac{1}{2}0, 0\frac{1}{2}1, 1\frac{1}{2}1, 1\frac{3}{2}1, 1\frac{3}{2}2\}$,

and has the form,

$$F_2 = \left(\begin{array}{cc|ccc|c} 1 & 0 & & & & & \\ 0 & 1 & & & & & \\ \hline & & 0 & & 0 & & 1 \\ & & \frac{1}{\sqrt{[3]_q}} & & -\sqrt{\frac{[4]_q}{[2]_q[3]_q}} & & \\ & & \sqrt{\frac{[4]_q}{[2]_q[3]_q}} & & \frac{1}{\sqrt{[3]_q}} & & \\ \hline & & & & & & 1 \end{array} \right). \quad (\text{A.3})$$

Likewise, F_3 changes bases from the Bratteli basis, labeled by $bcd = \{0\frac{1}{2}0, 1\frac{1}{2}0, 0\frac{1}{2}1, 1\frac{1}{2}1, 1\frac{3}{2}1, 1\frac{3}{2}2\}$ to the basis labeled by $a'cd = \{0\frac{1}{2}0, 1\frac{1}{2}0, 0\frac{1}{2}1, 1\frac{1}{2}1, 1\frac{3}{2}1, 1\frac{3}{2}2\}$, and has the form,

$$F_3 = \left(\begin{array}{cc|ccc|c} \frac{1}{[2]_q} & \frac{\sqrt{[3]_q}}{[2]_q} & & & & & \\ \sqrt{[3]_q} & -\frac{1}{[2]_q} & & & & & \\ \hline & & \frac{1}{[2]_q} & \frac{\sqrt{[3]_q}}{[2]_q} & 0 & & \\ & & \sqrt{[3]_q} & -\frac{1}{[2]_q} & 0 & & \\ & & 0 & 0 & 1 & & \\ \hline & & & & & & 1 \end{array} \right). \quad (\text{A.4})$$

Similarly, F_4 represents change of bases from the basis labeled by $a'cd = \{0\frac{1}{2}0, 1\frac{1}{2}0, 0\frac{1}{2}1, 1\frac{1}{2}1, 1\frac{3}{2}1, 1\frac{3}{2}2\}$ to the basis labeled by $a'c'd = \{0\frac{1}{2}0, 1\frac{1}{2}0, 0\frac{1}{2}1, 1\frac{1}{2}1, 1\frac{3}{2}1, 1\frac{3}{2}2\}$ and it has the form,

$$F_4 = \left(\begin{array}{cc|ccc|c} 1 & 0 & & & & & \\ 0 & 1 & & & & & \\ \hline & & 1 & 0 & 0 & & \\ & & 0 & -\frac{1}{[3]_q} & \frac{\sqrt{[4]_q[2]_q}}{[3]_q} & & \\ & & 0 & \frac{\sqrt{[4]_q[2]_q}}{[3]_q} & \frac{1}{[3]_q} & & \\ \hline & & & & & & 1 \end{array} \right). \quad (\text{A.5})$$

Finally, F_5 which takes us from the basis labeled by $a'c'd = \{0\frac{1}{2}0, 1\frac{1}{2}0, 0\frac{1}{2}1, 1\frac{1}{2}1, 1\frac{3}{2}1, 1\frac{3}{2}2\}$ back to the anti-Bratteli basis, labeled by $ac'd = \{0\frac{1}{2}0, 1\frac{1}{2}0, 0\frac{1}{2}1, 1\frac{1}{2}1, 1\frac{3}{2}1, 1\frac{3}{2}2\}$, has exactly the same form as F_3 ,

$$F_5 = F_3. \quad (\text{A.6})$$

The consistency requirement that the F matrices satisfy the pentagon equation, guarantees that the two ways to transform the Bratteli basis to anti-Bratteli basis are essentially equivalent. More explicitly,

$$F_2 F_1 = F_5 F_4 F_3 \quad (\text{A.7})$$

as can be explicitly checked.

Note that the change of basis matrices, $\mathcal{F}_1$, $\mathcal{F}_2$ and $\mathcal{F}_4$, introduced in Chapter 3, correspond to the $d = 1$ sector of F_1 when $b = 1$, F_2 when $a = 1$ and F_4 when $a' = 1$, respectively. Also note that the F matrix introduced in 3.5, corresponds to the $c = 1/2$ sector of F_3 as defined above.

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Phys. Rev. Lett. **95**, 140503 (2005)

CONFERENCE PRESENTATIONS

Topological Quantum Compiling

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Aspen Center for Physics

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Topological Quantum Compiling

Invited Talk, APS March Meeting

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CNOT for Fibonacci Anyons with Only One Mobile Quasiparticle

Chalk Talk given at the Program on Topological Phases and Quantum Computation

Kavli Institute for theoretical physics, Santa Barbara, CA (2/2006 – 5/2006)

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CNOT for Fibonacci Anyons with Only One Mobile Quasiparticle

Contributed Talk, APS March Meeting

Baltimore, MD (2006)

Braid Topologies for Quantum Computation

Contributed Talk, Quantum Computing and Many-Body Systems (QCMBS)

Key West, FL (1/2006)

Braid Topologies for Quantum Computation

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Braid Topologies for Quantum Computation

Poster Session, Physical Phenomena at High Magnetic Fields (PPHMF-V)

Tallahassee, FL (8/2005)

Topological Quantum Compiling

Contributed Talk, APS March Meeting

Los Angeles, CA (2005)

Designer Entanglement in Heisenberg Spin Chains

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Program on Topological Phases and Quantum Computation

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Aspen Center for Physics

Program on Topological Phases and Applications to Quantum Information Processing

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